CHMENG 250

Transport Processes

Fluid Mechanics, Heat and Mass Transport

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Preface

These notes accompany the graduate course *Transport Processes*, offered to first-year graduate students in the Department of Chemical & Biomolecular Engineering at the University of California, Berkeley. They are intended as a self-contained reference covering the fundamentals of momentum, energy, and mass transport, with an emphasis on rigorous derivation and physical intuition. Corrections and suggestions are welcome.

Although trained as a chemical engineer, I have spent the main part of my research career so far in computational biology and neuroscience, and have not been, in a matter of speaking, "cradled" in transport phenomena. While I have always been interested in the subject, I only became a serious student of it in the last three years, through the study of electrochemical transport near lipid-membrane interfaces in the context of modelling the phenomena underlying bioelectricity.

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Recommended Reading

These notes were inspired by the following sources:

[1] R. B. Bird, W. E. Stewart, and E. N. Lightfoot, Transport Phenomena—the author's first course in the subject, taken as a sophomore at IIT Bombay in 2004 (taught by Prof. Uday V. Shenoy).

[2] D. J. Acheson, Elementary Fluid Dynamics—a clear, physically grounded introduction to fluid mechanics and an ideal first book on the subject.

[3] W. M. Deen, Analysis of Transport Phenomena—based on a graduate course taught by Deen at MIT and a model of rigor and clarity in a chemical-engineering context.

[4] K. K. Mandadapu, together with his doctoral students A. M. Alkadri, J. B. Fernandes, and J. Farhadi, *CBE 250 Lecture Notes*—developed at the University of California, Berkeley, over successive offerings of the course from 2022 to 2025. Ongoing discussions and a scientific collaboration with K. K. Mandadapu have strongly shaped the author's thinking, especially the continuum-mechanical and non-equilibrium-thermodynamic viewpoints emphasized here.

[5] D. Tong, Lectures on Fluid Mechanics—a physicist's viewpoint presented in exceptionally clear, approachable prose that is fun to read.

[6] R. P. Feynman, R. B. Leighton, and M. Sands, The Feynman Lectures on Physics—neither conventional nor comprehensive, but its chapters on vectors and tensors are extremely insightful, as is the chapter on kinetic theory. The two chapters on fluid mechanics—"The Flow of Dry Water" and "The Flow of Wet Water"—are thoroughly entertaining.

[7] L. D. Landau and E. M. Lifshitz, Fluid Mechanics—a classic containing deep physics drawn economically from first principles, though terse and demanding.

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1
Introduction and Preliminaries

Transport processes describe the movement of mass, momentum, and energy through matter. Place any finite quantity of material in a box, shake it up, and its subsequent evolution will be governed by the laws of transport. These laws predict how physical quantities redistribute in space and time through the interplay of microscopic molecular motion and macroscopic constraints. Remarkably, mass, momentum, and energy transport share a common mathematical structure. And that structure works extraordinarily well in explaining a remarkable diversity of commonplace phenomena, and predicting uncommon ones testable by new experiments!

Although the foundations of transport processes have been developed across many disciplines, a modern synthesis emerged within chemical engineering from the recognition that fluid flow, heat transfer, and mass transfer are not separate collections of empirical rules, but manifestations of a common set of underlying principles. This unified viewpoint, championed most influentially by R. B. Bird, W. E. Stewart, and E. N. Lightfoot,[1] transformed chemical engineering from the study of particular unit operations into a general science of transport across scales and systems with both explanatory and predictive power. The same principles have since been applied to a wide range of problems in physics, chemistry, biology, and engineering.

At its foundation lie conservation laws, symmetry, and thermodynamic constraints. Mass is conserved, motion obeys Newton's laws, and energy is transferred through work and heat in accordance with the laws of thermodynamics. These balances are expressed locally in terms of fields such as density $\rho(\boldsymbol{x}, t)$, velocity $\boldsymbol{v}(\boldsymbol{x}, t)$, and temperature $T(\boldsymbol{x}, t)$, where $\boldsymbol{x}$ denotes spatial location and t denotes time. Transport theory connects these macroscopic fields to microscopic mechanisms (though not explicitly at the molecular level as we shall shortly see). Molecular motion produces diffusion, collisions produce viscous momentum transport, and thermal agitation produces heat conduction. These effects enter through *constitutive relations*, characterized by transport coefficients such as diffusivity, viscosity,

[1] R. B. Bird, W. E. Stewart, and E. N. Lightfoot, *Transport Phenomena*, 2nd ed. (John Wiley & Sons, New York, 2002); first edition 1960.

and thermal conductivity.

Transport phenomena is inherently *field-theoretic*: physical quantities are represented by *fields*, which are continuous functions over space and time and whose dynamical evolution is governed by local balance laws. The same language underlies classical field theories such as electromagnetism and relativity. This universality demands a compact, coordinate-independent way to manipulate scalars, vectors, tensors, and their spatial derivatives, motivating the formalism developed in this chapter.

Important | What this chapter does

This preliminary chapter develops the mathematical vocabulary used throughout these notes. We introduce the continuum hypothesis; classify physical quantities as scalars, vectors, and tensors; and develop the two notations used throughout: compact *vectorial* (Gibbs) notation and explicit *indicial* (Einstein) notation. We conclude with vector calculus, using indices to establish the differential identities that recur in later chapters.

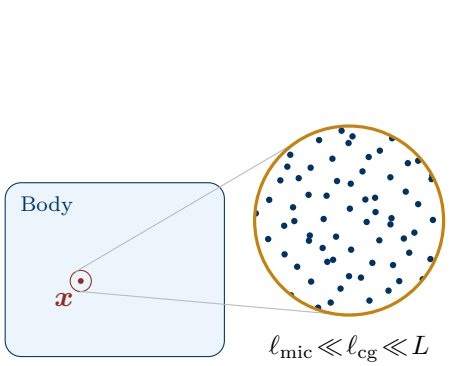
1.1 The Continuum Hypothesis

The continuum hypothesis asserts that matter can be treated as continuously distributed in space, so that physical properties are described by smooth fields. Quantities such as density ρ, temperature T, and velocity $\boldsymbol{v}$ are then defined at every point $\boldsymbol{x}$ and time t.

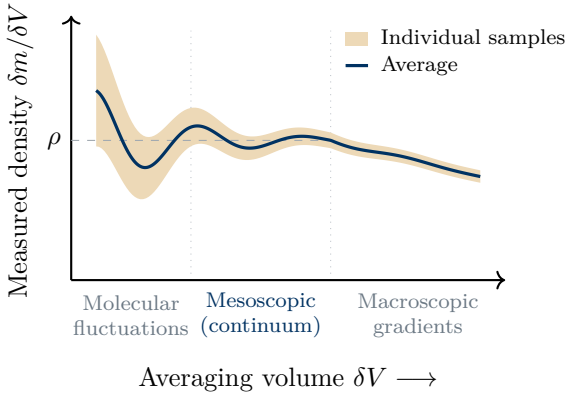
Strictly speaking, this is a fiction! You already learned in middle school that matter is molecular, not continuous. A fluid is not actually defined at a mathematical point, and there is no literal value of "the density at a point" if the point is taken all the way down to zero volume. The British would call this "codswallop". The continuum hypothesis is therefore an *approximation*, a remarkably successful lie, in which the microscopic discreteness of matter is replaced by smooth macroscopic fields. A great number of predictions of fluid mechanics, heat transfer, and mass transfer are based on this approximation, and they are borne out by experiment to astonishing accuracy.

Let me repeat for the sake of clarity. The so-called continuum limit is *not* one in which the volume element shrinks until it contains no molecules. Rather, one imagines a mesoscopic volume δV small enough that macroscopic fields vary only weakly across it, yet large enough to contain many molecules so that molecular fluctuations average out. The continuum "point" is really this small averaging volume (Fig. 1.1(a)), and throughout these notes we call it a *fluid particle* (or *fluid parcel*): not a molecule, but a mesoscopic blob of many many molecules that moves with the flow.

Historically, this viewpoint sits at the intersection of eighteenth- and nineteenth-century continuum mechanics and kinetic theory. L. Euler and later C.-L. Navier and G. G. Stokes formulated fluid motion in terms of smooth fields and local balance laws, long



(a) A point zooms into a volume of many molecules.



(b) Samples (ribbon) and their average.

Figure 1.1: The continuum hypothesis. **(a)** The "point" $\boldsymbol{x}$ of a continuum field, when zoomed in, is really a mesoscopic averaging volume δV that, though vanishingly small on the macroscopic scale, still contains a great many molecules (the coarse-graining volume $\delta V \sim \ell_{\mathrm{cg}}^3$ obeys $\ell_{\mathrm{mic}} \ll \ell_{\mathrm{cg}} \ll L$; in a dilute gas ℓ_{mic} is the mean free path λ). **(b)** A local field such as the density, measured as $\delta m/\delta V$: individual samples are irregular (the ribbon shows their spread), while their average oscillates when δV is molecular, settles onto a well-defined plateau over an intermediate range of averaging volumes, and finally drifts once δV grows large enough to feel macroscopic gradients. The plateau value is what we call the density $\rho(\boldsymbol{x})$ in the continuum approximation.

before the molecular basis of matter was universally accepted. L. Boltzmann and J. C. Maxwell then showed how such continuum equations could emerge statistically from molecular motion[2]. In this sense, continuum mechanics is not a denial of molecular structure, but a *coarse-grained* description of it.

1.1.1 When Is the Continuum Description Valid?

Its validity rests on the *separation of scales* illustrated in the plateau of Fig. 1.1(b), which we can now make quantitative. The coarse-graining volume $\delta V \sim \ell_{\mathrm{cg}}^3$ must fit inside a hierarchy of length scales

$$\ell_{\mathrm{mic}} \;\ll\; \ell_{\mathrm{cg}} \;\ll\; L,$$

where ℓ_{mic} is a microscopic correlation length (the scale below which the medium looks discrete or structured), ℓ_{cg} the coarse-graining scale of the averaging volume, and L the distance over which the macroscopic fields vary. Crucially, ℓ_{mic} is not universal but system-dependent. In a dilute gas the relevant ℓ_{mic} is the mean free path λ, while in dense liquids and electrochemical systems ℓ_{mic} is instead a molecular correlation or the Debye screening length rather than a mean free path. Thus, the fundamental requirement is:

(i) **Coarse-grained fields exist.** Molecular degrees of freedom can be averaged into

[2]We won't have to cover this, but if you'd like textbook recommendations, ask me during office hours.

reproducible fields that vary smoothly on the scale of interest. When no intermediate averaging scale exists, because the system contains too few molecules, or because the macroscopic variation occurs over only a few molecular dimensions, the continuum description may cease to be useful. This assumption alone, the existence of smooth coarse-grained fields, *is* the continuum hypothesis. Cool?

For the development of non-equilibrium thermodynamics seen later in these notes (**??**), a second assumption is also required:

(ii) **Local equilibrium hypothesis.** Each sufficiently small fluid particle may be assigned thermodynamic variables such as temperature, pressure, and chemical potential, even though the body as a whole is out of equilibrium[3]. The equilibrium equations of state may then be applied *locally*. This assumes that the microscopic relaxation time must be short compared with the observation time,

$$\tau_{\text{mic}} \ll \tau_{\text{obs}},$$

so that each fluid particle relaxes to equilibrium before its macroscopic state changes appreciably.

These two assumptions are related but distinct. Smooth continuum fields may remain meaningful even when local thermodynamic equilibrium becomes inaccurate. Conversely, when molecular fluctuations or interfacial structure cannot be averaged over, the fields themselves may lose their usual macroscopic meaning.

In dense liquids, the continuum description is remarkably robust and often remains useful down to nanometer scales. At such scales, however, molecular structure may first appear through modified transport coefficients (e.g., conductivity or viscosity), interfacial properties (e.g., surface tension), or boundary conditions rather than through an immediate failure of the bulk balance laws. This is why continuum models such as Poisson–Nernst–Planck theory can describe many electrochemical systems at the scale of Debye layers and nanopores, while still requiring care near molecularly structured interfaces.

Important | What "continuum" assumes

The essential continuum assumption is that molecular matter can be coarse-grained into smooth fields. For the classical thermodynamics of continuous media, we additionally

[3] Recall from undergraduate thermodynamics that *equilibrium* is a state with uniform, time-independent properties and no net fluxes, attained by an isolated system with no external driving. This is distinct from a steady state, where properties are likewise time-independent but nonzero fluxes are sustained by external forcing, as in conduction through a slab held at two different temperatures.

assume that each fluid particle is in local thermodynamic equilibrium:

$$\left.\begin{array}{l}\text{coarse-grained fields exist,}\\ \text{local equilibrium holds.}\end{array}\right\}$$

These assumptions should be assessed at the length and time scales relevant to the problem.

1.2 Scalars, Vectors, and Tensors

Physical quantities in transport phenomena are classified by their *tensorial rank*: a scalar (rank 0) carries no directional information, a vector (rank 1) carries a single direction, and a second-rank tensor is a linear map from vectors to vectors. We introduce the three in turn, fixing along the way the two notations, symbolic and indicial, used throughout these notes. We will worry about higher-rank tensors (rank 2 and above) when they arise in the context of constitutive relations.

Scalars

A **scalar** (rank 0) is a single number, the same in every coordinate system: it is unchanged by a rotation of the axes. Density ρ, temperature T, and pressure p are scalars. A *scalar field* $\phi(\boldsymbol{x}, t)$ assigns such a number to every point $\boldsymbol{x}$ and time t.

Vectors

A **vector** (rank 1) has a magnitude and a direction. In symbolic (*Gibbs*) notation[4, 5] we denote it by a single boldface letter, $\boldsymbol{v}$, with no reference to any coordinate system; an identity written this way holds in every basis. The vectors we consider live in a *Euclidean vector space* $\mathcal{E}^3$ over the real numbers $\mathbb{R}$: a real vector space of dimension three, equipped with an inner product. Two of its properties will be used constantly. The space is *closed* under the two vector operations,

$$\boldsymbol{u}, \boldsymbol{v} \in \mathcal{E}^3, \quad \alpha \in \mathbb{R} \qquad \Longrightarrow \qquad \boldsymbol{u} + \boldsymbol{v} \in \mathcal{E}^3, \qquad \alpha\, \boldsymbol{u} \in \mathcal{E}^3,$$

[4] The symbolic calculus of vectors was developed by the great American physicist J. W. Gibbs, regarded as the main architect of modern thermodynamics, and particularly of chemical thermodynamics, in his Yale lectures of the 1880s and circulated as the privately printed *Elements of Vector Analysis* (New Haven, 1881–1884). It was shaped into a textbook by his student E. B. Wilson, *Vector Analysis* (Yale University Press, 1901). O. Heaviside arrived at an almost identical system independently in *Electromagnetic Theory* (London, 1893). Their streamlined notation eventually displaced W. R. Hamilton's quaternions after a spirited late-nineteenth-century debate now remembered as the "vector wars."

[5] An early edition of Wilson's *Vector Analysis*, once belonging to the great Indian-American theoretical physicist S. Chandrasekhar, is in the possession of Prof. K. K. Mandadapu.

so that adding two velocities, or doubling one, cannot carry you out of the space. And a set $\{\boldsymbol{u}_1, \ldots, \boldsymbol{u}_k\}$ is *linearly independent* when the only combination of its members that vanishes is the trivial one,

$$\sum_{i=1}^{k} \alpha_i \, \boldsymbol{u}_i = \boldsymbol{0} \qquad \Longrightarrow \qquad \alpha_1 = \cdots = \alpha_k = 0,$$

that is, when no member can be assembled from the others. To say that $\mathcal{E}^3$ has dimension three is to say that three is the largest number of linearly independent vectors it admits; any such three therefore span the space and constitute a *basis*. To compute with components we adopt a right-handed orthonormal Cartesian basis, $\{\boldsymbol{e}_1, \boldsymbol{e}_2, \boldsymbol{e}_3\}$. The labels $1, 2, 3$ stand for the Cartesian axes x, y, z, so that $\boldsymbol{e}_1 = \boldsymbol{e}_x$, $\boldsymbol{e}_2 = \boldsymbol{e}_y$, and $\boldsymbol{e}_3 = \boldsymbol{e}_z$. Any vector is expanded on this basis as

$$\boldsymbol{v} = v_1 \boldsymbol{e}_1 + v_2 \boldsymbol{e}_2 + v_3 \boldsymbol{e}_3 = \sum_{i=1}^{3} v_i \, \boldsymbol{e}_i, \qquad v_i = \boldsymbol{v} \cdot \boldsymbol{e}_i .$$

The scalar (dot) product measures the overlap of two vectors,

$$\boldsymbol{v} \cdot \boldsymbol{w} = v_1 w_1 + v_2 w_2 + v_3 w_3 = \sum_{i=1}^{3} v_i w_i = |\boldsymbol{v}| \, |\boldsymbol{w}| \cos\theta,$$

and returns a scalar; here θ is the angle between $\boldsymbol{v}$ and $\boldsymbol{w}$. Stated as an operation, it takes a pair of vectors to a real number,

$$\mathcal{E}^3 \times \mathcal{E}^3 \longrightarrow \mathbb{R},$$

symmetric in its two arguments, linear in each, and positive definite, in that $\boldsymbol{v} \cdot \boldsymbol{v} > 0$ for every $\boldsymbol{v} \neq \boldsymbol{0}$. That last property is what lets the inner product measure length. The *norm*, or magnitude, of a vector is

$$|\boldsymbol{v}| = \sqrt{\boldsymbol{v} \cdot \boldsymbol{v}} = \sqrt{\sum_{i=1}^{3} v_i^2} \, ,$$

a vector of unit norm is a *unit vector*, and two vectors are *orthogonal* when $\boldsymbol{v} \cdot \boldsymbol{w} = 0$, which by the formula above is exactly $\theta = \pi/2$. That the basis is orthonormal is recorded by the *Kronecker delta*, named for the German mathematician L. Kronecker.[6]

[6] L. Kronecker (1823–1891), known for his work in number theory and algebra and for the aphorism "God made the integers; all else is the work of man."

Definition 1.1 | Kronecker delta

The Kronecker delta records the orthonormality of the basis,

$$\boldsymbol{e}_i \cdot \boldsymbol{e}_j = \delta_{ij} = \begin{cases} 1, & i = j, \\ 0, & i \neq j. \end{cases}$$

The vector (cross) product is an operation of a different kind, taking a pair of vectors to a third vector,

$$\mathcal{E}^3 \times \mathcal{E}^3 \longrightarrow \mathcal{E}^3,$$

linear in each argument and *antisymmetric*,

$$\boldsymbol{u} \times \boldsymbol{v} = -\,\boldsymbol{v} \times \boldsymbol{u},$$

from which $\boldsymbol{u} \times \boldsymbol{u} = \boldsymbol{0}$ follows immediately. Combined with the dot product it produces the *box product*, or scalar triple product,

$$[\boldsymbol{u}, \boldsymbol{v}, \boldsymbol{w}] \coloneqq \boldsymbol{u} \cdot (\boldsymbol{v} \times \boldsymbol{w}),$$

which is unchanged by a cyclic permutation of its arguments,

$$[\boldsymbol{u}, \boldsymbol{v}, \boldsymbol{w}] = [\boldsymbol{v}, \boldsymbol{w}, \boldsymbol{u}] = [\boldsymbol{w}, \boldsymbol{u}, \boldsymbol{v}].$$

That one property settles the geometry, with no appeal to a picture. Taking the first two arguments equal and cycling,

$$\boldsymbol{u} \cdot (\boldsymbol{u} \times \boldsymbol{v}) = [\boldsymbol{u}, \boldsymbol{u}, \boldsymbol{v}] = [\boldsymbol{v}, \boldsymbol{u}, \boldsymbol{u}] = \boldsymbol{v} \cdot (\boldsymbol{u} \times \boldsymbol{u}) = 0,$$

and the same step applied to $[\boldsymbol{v}, \boldsymbol{u}, \boldsymbol{v}] = [\boldsymbol{u}, \boldsymbol{v}, \boldsymbol{v}]$ gives $\boldsymbol{v} \cdot (\boldsymbol{u} \times \boldsymbol{v}) = 0$. The product $\boldsymbol{u} \times \boldsymbol{v}$ is therefore orthogonal to both factors, and so lies along the normal to the plane they span. Which of the two normal directions it takes is a convention, not a theorem: we adopt the *right-hand rule*, and with it the magnitude $|\boldsymbol{u} \times \boldsymbol{v}| = |\boldsymbol{u}|\,|\boldsymbol{v}| \sin\theta$, the area of the parallelogram the two vectors span. On the right-handed basis $\boldsymbol{e}_1 \times \boldsymbol{e}_2 = \boldsymbol{e}_3$ together with its cyclic partners; these relations are collected by the *Levi-Civita symbol*, named for the Italian mathematician T. Levi-Civita.[7]

[7] G. Ricci and T. Levi-Civita, "Méthodes de calcul différentiel absolu et leurs applications," *Mathematische Annalen* **54** (1900), 125–201. It was precisely this tensor calculus that later furnished A. Einstein with the mathematical language of general relativity.

Definition 1.2 | Levi-Civita (permutation) symbol

The Levi-Civita symbol is the totally antisymmetric array

$$\epsilon_{ijk} = \begin{cases} +1, & (i,j,k) \text{ an even permutation of } (1,2,3), \\ -1, & (i,j,k) \text{ an odd permutation of } (1,2,3), \\ 0, & \text{any index repeated}, \end{cases}$$

so that $\boldsymbol{e}_i \times \boldsymbol{e}_j = \sum_k \epsilon_{ijk}\, \boldsymbol{e}_k$.

The cross product then has the compact component form

$$(\boldsymbol{v} \times \boldsymbol{w})_i = \sum_{j,k} \epsilon_{ijk}\, v_j w_k, \tag{1.1}$$

which the reader should check reproduces the familiar determinant expansion. A *vector field* $\boldsymbol{v}(\boldsymbol{x}, t)$ assigns a vector to every point and time.

Tensors

A scalar carries no direction and a vector carries one. Many of the quantities we shall need carry *two*. The traction on an internal surface depends on the direction of the force and, separately, on the orientation of the surface it acts across; the flux of momentum depends on which component of momentum is being carried and, separately, on the direction it is carried in. An object holding two directions at once is a second-rank tensor, and the way to make that precise is to let it consume one direction and return the other.

A **second-rank tensor** (rank 2) is a linear map that sends a vector to a vector: $\boldsymbol{T} \cdot \boldsymbol{v}$ is a vector, linear in $\boldsymbol{v}$, by which we mean

$$\boldsymbol{T} \cdot (\alpha \boldsymbol{u} + \beta \boldsymbol{v}) = \alpha\, \boldsymbol{T} \cdot \boldsymbol{u} + \beta\, \boldsymbol{T} \cdot \boldsymbol{v} \qquad \text{for all } \boldsymbol{u}, \boldsymbol{v} \in \mathcal{E}^3 \text{ and } \alpha, \beta \in \mathbb{R}.$$

Handed the surface normal, the stress tensor returns the traction; that is the whole content of the definition. We write a generic tensor as a boldface capital, $\boldsymbol{T}$. Such maps form a vector space in their own right, ~~$\mathrm{Lin}(\mathbb{R}^3)$~~$\mathrm{Lin}(\mathcal{E}^3, \mathcal{E}^3)$, of dimension nine. The simplest tensors are built from two vectors by the *dyadic* (or tensor) product.

Definition 1.3 | Dyadic product

The dyadic product $\boldsymbol{v} \otimes \boldsymbol{w}$ of two vectors is the second-rank tensor defined by its action

on an arbitrary vector $\boldsymbol{c}$,

$$(\boldsymbol{v} \otimes \boldsymbol{w}) \cdot \boldsymbol{c} = (\boldsymbol{w} \cdot \boldsymbol{c})\, \boldsymbol{v}, \qquad \text{with components} \qquad (\boldsymbol{v} \otimes \boldsymbol{w})_{ij} = v_i w_j.$$

It is common, and we shall often follow the practice, to write a dyad simply by juxtaposition, $\boldsymbol{v}\boldsymbol{w}$, with no symbol between the two vectors, meaning exactly $\boldsymbol{v} \otimes \boldsymbol{w}$. This is what makes the tensor character of a quantity such as the momentum flux $\rho\, \boldsymbol{v}\boldsymbol{v} = \rho\, \boldsymbol{v} \otimes \boldsymbol{v}$ transparent. The nine dyads $\boldsymbol{e}_i \otimes \boldsymbol{e}_j$ (likewise written $\boldsymbol{e}_i\boldsymbol{e}_j$) are a basis of this nine-dimensional space, so any tensor expands as

$$\boldsymbol{T} = \sum_{i,j} T_{ij}\, \boldsymbol{e}_i \otimes \boldsymbol{e}_j, \qquad T_{ij} = \boldsymbol{e}_i \cdot \boldsymbol{T} \cdot \boldsymbol{e}_j,$$

and acts on a vector componentwise, $(\boldsymbol{T} \cdot \boldsymbol{v})_i = \sum_j T_{ij}\, v_j$.[8] The simplest tensor of all is the one that does nothing to what it is handed.

Definition 1.4 | Identity tensor

The *identity tensor* $\boldsymbol{I}$ is the linear map that returns every vector unchanged,

$$\boldsymbol{I} \cdot \boldsymbol{v} = \boldsymbol{v} \qquad \text{for every } \boldsymbol{v} \in \mathcal{E}^3.$$

Its components in an orthonormal basis are precisely the Kronecker delta,

$$I_{ij} = \boldsymbol{e}_i \cdot \boldsymbol{I} \cdot \boldsymbol{e}_j = \boldsymbol{e}_i \cdot \boldsymbol{e}_j = \delta_{ij}, \qquad \text{so that} \qquad \boldsymbol{I} = \sum_{i,j} \delta_{ij}\, \boldsymbol{e}_i \otimes \boldsymbol{e}_j = \sum_i \boldsymbol{e}_i \otimes \boldsymbol{e}_i.$$

The stress tensor, the rate-of-strain tensor $\boldsymbol{D}$, and the identity tensor $\boldsymbol{I}$ are all second-rank tensors; a *tensor field* $\boldsymbol{T}(\boldsymbol{x}, t)$ assigns one to every point and time.

Two symbols we have already met are tensors in exactly this sense. The Kronecker delta δ_{ij} is a **rank-2** tensor, precisely the components of the identity tensor $\boldsymbol{I}$, and as a linear map it is the identity, returning any vector unchanged, so that for example $\delta_{ij}\, v_j = v_i$. The Levi-Civita symbol ϵ_{ijk} is a **rank-3** tensor. Just as a rank-2 tensor is a linear map that turns a vector into a vector, a rank-3 tensor is a linear map that turns a vector into a rank-2 tensor, or, contracted with two vectors, into a single vector: $\epsilon_{ijk}\, v_j w_k = (\boldsymbol{v} \times \boldsymbol{w})_i$, so ϵ_{ijk} is exactly the map that sends a pair of vectors to their cross product. Contracted with all three vectors it returns the scalar triple product $\epsilon_{ijk}\, a_i b_j c_k = \boldsymbol{a} \cdot (\boldsymbol{b} \times \boldsymbol{c})$, the signed volume of the parallelepiped they span.

[8] The dot contracts the *second* (nearer) index of $\boldsymbol{T}$ with $\boldsymbol{v}$, leaving i free to label the component of the resulting vector, exactly as a matrix multiplies a column vector row by row. Summing the *first* index instead, $T_{ij}\, v_i$, would give a *different* vector, $(\boldsymbol{v} \cdot \boldsymbol{T})_j = (\boldsymbol{T}^{\top} \cdot \boldsymbol{v})_j$, which coincides with $\boldsymbol{T} \cdot \boldsymbol{v}$ only when $\boldsymbol{T}$ is symmetric.

Rank and the Transformation Law

The three products just introduced—dot, cross, and dyadic—lower, preserve, and raise the rank of their operands, respectively[9]. What ultimately makes an array of components a scalar, vector, or tensor is how it transforms when the axes are rotated. Under a rotation described by an orthogonal matrix Q_{ij} (with $\sum_k Q_{ik}Q_{jk} = \delta_{ij}$), a scalar is unchanged, a vector transforms as $v'_i = \sum_j Q_{ij}\, v_j$, and a second-rank tensor transforms on *each* index, $T'_{ij} = \sum_{k,l} Q_{ik}Q_{jl}\, T_{kl}$. Any physical law must equate objects of the same rank, so that both sides transform identically, the reason a scalar equation can never equal a vector, and why the stress that resists a shear must be a tensor rather than a mere vector.

Remark 1.5 | Components are not the object

A vector is *not* an array of three numbers, and a second-rank tensor is *not* a 3×3 matrix. The array (v_1, v_2, v_3) says nothing at all until one states the basis it was measured against; the vector is the geometric object, and its components are only the shadow it casts on one particular set of axes. Rotate those axes and every component changes, $v'_i = \sum_j Q_{ij} v_j$, while the arrow itself has not moved. What survives a change of basis is the object, and with it any quantity assembled from dot products and contractions alone, such as $|\boldsymbol{v}|$, $\boldsymbol{v} \cdot \boldsymbol{w}$, or $\operatorname{tr} \boldsymbol{T}$. This is why the balance laws are written symbolically first: a statement such as $\boldsymbol{\nabla} \cdot \boldsymbol{T} + \rho \boldsymbol{b} = \rho \frac{\mathrm{D}\boldsymbol{v}}{\mathrm{D}t}$ is a claim about the objects and holds in every basis, whereas its component form holds only in the basis it was written for.

Remark 1.6 | Polar vectors, axial vectors, and observers

The law above was written for a proper rotation ($\det Q = +1$, where det is the determinant). For a general *orthogonal* Q, which may also include a reflection ($\det Q = -1$), one must distinguish two kinds of vector. A *polar* (or true) vector transforms as

$$v'_i = Q_{ij}\, v_j,$$

whereas an *axial* (or pseudo) vector carries an extra sign,

$$a'_i = (\det Q)\, Q_{ij}\, a_j.$$

Cross products, the angular velocity, and the vorticity $\boldsymbol{\omega} = \boldsymbol{\nabla} \times \boldsymbol{v}$ are axial vectors, because the Levi-Civita symbol ϵ_{ijk} is a *pseudotensor*: under an improper transformation its array is reproduced only up to the factor $\det Q$. Under proper rotations the

[9] This subsection stores important concepts, but I am being purposefully brief here, so if you feel like you don't fully understand what this is all about, just park your worries, and proceed to the next subsection!

distinction is invisible and an axial vector behaves exactly like a polar one; it surfaces only under reflections.

This transformation of *components* under a change of coordinates, a passive relabeling of one and the same physical event, should not be confused with a change of *observer*, which may entail a genuinely different and possibly moving frame. We reserve the terms *objective* or *frame-indifferent* for quantities that transform properly under *time-dependent* rigid changes of observer; that notion and the associated calculation is developed in Ch. 2.

1.3 Einstein's Indicial Notation

Symbolic notation is elegant but grows opaque once products pile up. Indicial (index) notation is the answer to your worries. It may seem formidable at first, but it provides an elegant framework in which every identity reduces to bookkeeping of subscripts. Its cornerstone is a convention introduced by A. Einstein that drops the explicit summation signs we have written out up to this point.[10]

Definition 1.7 | Einstein summation convention

An index that appears exactly *twice* in a single term is summed over its range $1, 2, 3$, and the summation sign is dropped:

$$v_i\, \boldsymbol{e}_i \;\equiv\; \sum_{i=1}^{3} v_i\, \boldsymbol{e}_i, \qquad v_i w_i \;\equiv\; \sum_{i=1}^{3} v_i w_i.$$

Every explicit sum written in the previous section, the component expansion, the dot and cross products, the action of a tensor on a vector, the rotation law, is from here on abbreviated by simply repeating the index. The Kronecker delta and the Levi-Civita symbol, also introduced there, supply the rest of the machinery.

Three rules govern every expression written in this notation, and between them they catch almost every error one can make.

Rule 1: the dummy index rule. An index repeated exactly twice in a term is a *dummy* index: it is summed over $1, 2, 3$ and vanishes from the result. Its name is therefore

[10] A. Einstein introduced the convention of summing over a repeated index in his foundational paper on general relativity, "Die Grundlage der allgemeinen Relativitätstheorie," *Annalen der Physik* **49** (1916), 769–822. He later remarked, only half in jest, that he had made a great discovery in mathematics: the suppression of the summation sign whenever an index is repeated.

arbitrary and may be changed at will,

$$v_i w_i = v_j w_j = v_m w_m,$$

a freedom used constantly to avoid a clash when two expressions are multiplied together. In $A_{ij}B_{jk}$ the index j is summed away; in $v_i w_i$ both indices are, leaving a scalar.

Rule 2: the free index rule. An index appearing exactly once in a term is *free*. It is not summed, it fixes the rank of the expression, and it must appear, unrepeated, in *every* term of an equation. In $A_{ij}u_j = v_i$ the index j is dummy and i is free, so each side is a vector and the single line stands for three equations, one per value of i. By the same test $A_{ij}u_j = v_j$ is meaningless, since the free index on the left is i and on the right is j. Matching free indices across an equals sign is the quickest check available.

Rule 3: no index appears more than twice. A term such as $A_{ij}B_{jk}u_j$ is not defined: with j occurring three times there is no way to say which pair is to be summed. When such a collision appears in the middle of a calculation it is a signal to rename a dummy index by Rule 1, not to press on.

Thus $v_i w_i$ has no free index (a scalar), $\epsilon_{ijk}v_j w_k$ has the single free index i (a vector), and T_{ij} has two (a tensor).

Example 1.8 | A first proof in indices

For any tensor $\boldsymbol{T}$ and any vectors $\boldsymbol{u}$ and $\boldsymbol{v}$,

$$\boldsymbol{u} \cdot (\boldsymbol{T} \cdot \boldsymbol{v}) = \boldsymbol{v} \cdot (\boldsymbol{T}^{\top} \cdot \boldsymbol{u}),$$

where the transpose is defined componentwise by $(\boldsymbol{T}^{\top})_{ij} = T_{ji}$.

Write each side out. On the left,

$$\boldsymbol{u} \cdot (\boldsymbol{T} \cdot \boldsymbol{v}) = u_i \, (\boldsymbol{T} \cdot \boldsymbol{v})_i = u_i \, T_{ij} \, v_j,$$

in which i and j are both dummy indices, so nothing is free and the result is a scalar, as it must be. On the right,

$$\boldsymbol{v} \cdot (\boldsymbol{T}^{\top} \cdot \boldsymbol{u}) = v_j \, (\boldsymbol{T}^{\top})_{jk} \, u_k = v_j \, T_{kj} \, u_k.$$

By Rule 1 the dummy index k may be renamed i, which turns the right-hand side into $u_i \, T_{ij} \, v_j$, the left-hand side exactly. The two are the same scalar. The entire proof is a change in the name of a summed index, and this is the flavour of nearly every manipulation that follows.

The Kronecker delta acts as a *substitution operator*: summed against an indexed quantity, it renames the dummy index,

$$\delta_{ij}\, v_j = v_i, \qquad \delta_{ij}\, T_{jk} = T_{ik}, \qquad \delta_{ii} = 3. \tag{1.2}$$

It is also what makes the dot product so simple in components, $\boldsymbol{v}\cdot\boldsymbol{w} = v_i\boldsymbol{e}_i\cdot w_j\boldsymbol{e}_j = v_i w_j\,\delta_{ij} = v_i w_i$, while the Levi-Civita symbol gives the cross product its component form Eq. (1.1).

The single most useful relation between the two symbols is the *epsilon–delta identity*, obtained by contracting two Levi-Civita symbols on a shared first index:

$$\epsilon_{ijk}\,\epsilon_{ilm} = \delta_{jl}\,\delta_{km} - \delta_{jm}\,\delta_{kl}. \tag{1.3}$$

Remark 1.9 | How to remember Eq. (1.3)

The summed index is the first slot of each symbol. Of the remaining free indices, the "outer" pair in matching order (j,l) and (k,m) enters with a $+$ sign, and the "crossed" pair with a $-$ sign. Two further contractions follow immediately by setting indices equal in Eq. (1.3):

$$\epsilon_{ijk}\,\epsilon_{ijm} = 2\,\delta_{km}, \qquad \epsilon_{ijk}\,\epsilon_{ijk} = 6.$$

1.4 Proving Identities with Indices

We now put the machinery to work and demonstrate that, once you get the hang of it, the indicial notation makes vector algebra and calculus deliciously simple. The strategy is always the same: convert each vector operation to components using Eq. (1.1), manipulate the resulting δ's and ϵ's, above all via the epsilon–delta identity Eq. (1.3), and translate the surviving components back to symbolic form. Although it may seem like idle algebra at first, these identities will recover in the physics of later chapters, as noted below. We work only two representative examples here; the remaining identities are left as exercises at the end of the chapter. You will also see them in your homework.

Example 1.10 | Magnitude of a cross product

Let us prove $|\boldsymbol{v}\times\boldsymbol{w}|^2 = |\boldsymbol{v}|^2\,|\boldsymbol{w}|^2 - (\boldsymbol{v}\cdot\boldsymbol{w})^2$. Writing the squared magnitude as a dot product of Eq. (1.1) with itself,

$$\begin{aligned} |\boldsymbol{v}\times\boldsymbol{w}|^2 &= (\epsilon_{ijk}v_j w_k)(\epsilon_{ilm}v_l w_m) \\ &= \epsilon_{ijk}\epsilon_{ilm}\, v_j w_k v_l w_m. \end{aligned}$$

Apply the epsilon–delta identity Eq. (1.3):

$$\begin{aligned} &= (\delta_{jl}\delta_{km} - \delta_{jm}\delta_{kl})\, v_j w_k v_l w_m \\ &= v_j v_j\, w_k w_k - v_j w_j\, v_l w_l \\ &= |\boldsymbol{v}|^2\, |\boldsymbol{w}|^2 - (\boldsymbol{v}\cdot\boldsymbol{w})^2. \end{aligned}$$

Setting $\boldsymbol{v}\cdot\boldsymbol{w} = |\boldsymbol{v}|\,|\boldsymbol{w}|\cos\theta$ recovers $|\boldsymbol{v}\times\boldsymbol{w}| = |\boldsymbol{v}|\,|\boldsymbol{w}|\sin\theta$. This magnitude is the area of the parallelogram spanned by $\boldsymbol{v}$ and $\boldsymbol{w}$; the same cross product gives the moment (torque) $\boldsymbol{r}\times\boldsymbol{F}$ and the angular momentum $\boldsymbol{r}\times\boldsymbol{p}$ that govern the balance of angular momentum in a continuum.

Theorem 1.11 | The BAC–CAB rule

For any three vectors,

$$\boldsymbol{a}\times(\boldsymbol{b}\times\boldsymbol{c}) = \boldsymbol{b}\,(\boldsymbol{a}\cdot\boldsymbol{c}) - \boldsymbol{c}\,(\boldsymbol{a}\cdot\boldsymbol{b}).$$

Proof. Work with the ith component and apply Eq. (1.1) twice, once for the outer product and once for the inner:

$$\begin{aligned} \big[\boldsymbol{a}\times(\boldsymbol{b}\times\boldsymbol{c})\big]_i &= \epsilon_{ijk}\, a_j\, (\boldsymbol{b}\times\boldsymbol{c})_k \\ &= \epsilon_{ijk}\, a_j\, \epsilon_{klm}\, b_l c_m. \end{aligned}$$

Cycle the second symbol so that the summed index k sits first, $\epsilon_{ijk}\to\epsilon_{kij}$, so that $\epsilon_{ijk}\epsilon_{klm} = \epsilon_{kij}\epsilon_{klm}$. Now, apply the epsilon–delta identity Eq. (1.3) on the shared index k:

$$\epsilon_{kij}\epsilon_{klm} = \delta_{il}\delta_{jm} - \delta_{im}\delta_{jl}.$$

Therefore

$$\begin{aligned} \big[\boldsymbol{a}\times(\boldsymbol{b}\times\boldsymbol{c})\big]_i &= (\delta_{il}\delta_{jm} - \delta_{im}\delta_{jl})\, a_j b_l c_m \\ &= b_i\,(a_j c_j) - c_i\,(a_j b_j) \\ &= b_i\,(\boldsymbol{a}\cdot\boldsymbol{c}) - c_i\,(\boldsymbol{a}\cdot\boldsymbol{b}). \end{aligned}$$

Since this holds for each free index i, it is the ith component of the claimed vector identity. ■

Remark 1.12 | Where the BAC–CAB rule appears

This identity (Thm. 1.11) is a workhorse of momentum transport. Applied to the velocity field, it recasts the convective acceleration as

$$(\boldsymbol{\nabla}\boldsymbol{v})\,\boldsymbol{v} = \boldsymbol{\nabla}\Big(\tfrac{1}{2}\,|\boldsymbol{v}|^2\Big) - \boldsymbol{v}\times\boldsymbol{\omega}, \qquad \boldsymbol{\omega} = \boldsymbol{\nabla}\times\boldsymbol{v},$$

where $\boldsymbol{\omega}$ is the vorticity, the form of the inertial term that underlies Bernoulli's theorem (see Exercise). The same rule produces the centripetal acceleration $\boldsymbol{\Omega}\times(\boldsymbol{\Omega}\times\boldsymbol{r})$ that appears when the momentum balance is written in a rotating frame.

1.5 Vector Calculus

Everything from here on is a *field*: a quantity carrying a value at every point of space and every instant of time. Three kinds will be needed, distinguished only by what they return,

$$\begin{aligned}\phi(\boldsymbol{x},t) &: \mathcal{E}^3\times\mathbb{R}\longrightarrow\mathbb{R},\\ \boldsymbol{v}(\boldsymbol{x},t) &: \mathcal{E}^3\times\mathbb{R}\longrightarrow\mathcal{E}^3,\\ \boldsymbol{T}(\boldsymbol{x},t) &: \mathcal{E}^3\times\mathbb{R}\longrightarrow \mathrm{Lin}(\mathcal{E}^3,\mathcal{E}^3),\end{aligned}$$

a *scalar field* such as the temperature or the pressure, a *vector field* such as the velocity, and a *tensor field* such as the stress. The operators of this section measure how each of them varies from one point to the next.

Spatial variations of fields are described by the vector differential operator $\boldsymbol{\nabla}$ ("del"), whose Cartesian components are the partial derivatives

$$\boldsymbol{\nabla} = \boldsymbol{e}_i\,\partial_i, \qquad \partial_i \;\equiv\; \frac{\partial}{\partial x_i}.$$

In indicial notation, differentiation simply introduces a derivative index, and, crucially, partial derivatives commute, $\partial_i\partial_j = \partial_j\partial_i$. This symmetry, played against the antisymmetry of ϵ_{ijk}, is the engine behind the identities below.

Definition 1.13 | Gradient, divergence, curl, Laplacian

Let $\phi(\boldsymbol{x})$ be a scalar field and $\boldsymbol{v}(\boldsymbol{x})$ a vector field.

$$(\boldsymbol{\nabla}\phi)_i = \partial_i\phi, \qquad \boldsymbol{\nabla}\cdot\boldsymbol{v} = \partial_i v_i, \qquad (\boldsymbol{\nabla}\times\boldsymbol{v})_i = \epsilon_{ijk}\,\partial_j v_k, \qquad \nabla^2\phi = \partial_i\partial_i\phi.$$

The gradient raises rank ($0\to1$), the divergence lowers it ($1\to0$), and the curl preserves it ($1\to1$).

Each of the three answers a different question about how a field varies, and each repays a closer look.

The gradient. The whole meaning of $\boldsymbol{\nabla}\phi$ is contained in the change of ϕ between two neighbouring points separated by $\mathrm{d}\boldsymbol{x}$,

$$\mathrm{d}\phi = \boldsymbol{\nabla}\phi \cdot \mathrm{d}\boldsymbol{x} = \partial_i \phi \, \mathrm{d}x_i.$$

Read as a statement about direction, this says that for a step of fixed small length the change in ϕ is greatest when $\mathrm{d}\boldsymbol{x}$ is parallel to $\boldsymbol{\nabla}\phi$ and vanishes when the step is perpendicular to it. So $\boldsymbol{\nabla}\phi$ points along the steepest ascent of ϕ, its magnitude is the rate of that ascent per unit length, and the level surfaces $\phi =$ constant are everywhere normal to it. This is the content of $\boldsymbol{q} = -k\boldsymbol{\nabla}T$, which sends heat down the temperature gradient, and of the pressure force $-\boldsymbol{\nabla}p$, which drives fluid from high pressure to low.

Applied to a *vector* field the same operation returns a tensor rather than a vector, since each of three components now carries three derivatives. The velocity gradient $\boldsymbol{\nabla}\boldsymbol{v}$, with components $\partial_j v_i$, is the object that holds every piece of local information about a flow, and Ch. 2 is largely the study of it. The gradient, in short, *raises* rank by one.

The divergence. The divergence contracts the derivative index against the field's own index, $\boldsymbol{\nabla}\cdot\boldsymbol{v} = \partial_i v_i$, and therefore *lowers* rank by one: a vector field yields a scalar, and a tensor field a vector, $(\boldsymbol{\nabla}\cdot\boldsymbol{T})_i = \partial_j T_{ij}$. What it measures is local expansion. A small material volume carried along by the flow changes at the fractional rate $\boldsymbol{\nabla}\cdot\boldsymbol{v}$, so a positive value marks a point out of which the field spreads, a negative value one into which it converges, and $\boldsymbol{\nabla}\cdot\boldsymbol{v} = 0$ a flow that preserves the volume of every element it carries. That condition is *incompressibility*; it is proved and put to work in Ch. 2.

The curl. The curl extracts the part of the spatial variation that is a rotation. Alone among the three it *preserves* rank, taking a vector field to a vector field, $(\boldsymbol{\nabla}\times\boldsymbol{v})_i = \epsilon_{ijk}\,\partial_j v_k$, and the antisymmetry of ϵ_{ijk} is precisely what discards the symmetric, stretching part of $\boldsymbol{\nabla}\boldsymbol{v}$ and retains the antisymmetric, spinning part. Applied to a velocity field it returns the vorticity $\boldsymbol{\omega} = \boldsymbol{\nabla}\times\boldsymbol{v}$, twice the local angular velocity of a fluid element; a field with $\boldsymbol{\nabla}\times\boldsymbol{v} = \boldsymbol{0}$ everywhere is called *irrotational*.

1.5.1 Derivatives of Vector and Tensor Fields

The operators just defined differentiate a scalar or a vector, but the balance laws to come also differentiate *vectors* into tensors and *tensors* into vectors, the velocity gradient $\boldsymbol{\nabla}\boldsymbol{v}$, the stress divergence $\boldsymbol{\nabla}\cdot\boldsymbol{T}$, the viscous term $\nabla^2\boldsymbol{v}$, and the stress power $\boldsymbol{T}:\boldsymbol{\nabla}\boldsymbol{v}$. Each is defined componentwise, by letting ∂_i act in the natural way.

Definition 1.14 | Derivatives of vector and tensor fields

For a vector field $\boldsymbol{v}(\boldsymbol{x})$ and a second-rank tensor field $\boldsymbol{T}(\boldsymbol{x})$,

$$(\boldsymbol{\nabla v})_{ij} = \partial_j v_i, \qquad (\boldsymbol{\nabla}\cdot\boldsymbol{T})_i = \partial_j T_{ij}, \qquad (\nabla^2\boldsymbol{v})_i = \partial_j\partial_j v_i, \qquad \boldsymbol{T}:\boldsymbol{\nabla v} = T_{ij}\,\partial_j v_i.$$

The *gradient of a vector* is a second-rank tensor whose first index labels the vector component and whose second labels the direction of differentiation; the *divergence of a tensor* is a vector; the *vector Laplacian* acts componentwise; and the double contraction $\boldsymbol{T}:\boldsymbol{\nabla v}$ is a scalar.

The convention for $\boldsymbol{\nabla}\cdot\boldsymbol{T}$ is not arbitrary, although it may seem like that. Contracting the *second* index is precisely what makes $(\boldsymbol{\nabla}\cdot\boldsymbol{T})_i = \partial_j T_{ij}$ the ith component of the force per unit volume in the momentum balance (Ch. 3); the double contraction $\boldsymbol{T}:\boldsymbol{\nabla v} = T_{ij}\partial_j v_i$, in the same way, returns the local rate at which stresses do work, which underlies viscous dissipation. These derivatives obey the ordinary Leibniz rule of calculus on each index. The one identity we shall lean on most is the divergence of $\boldsymbol{T}^{\top}\cdot\boldsymbol{v}$, the combination that arises when a surface working $(\boldsymbol{T}\cdot\boldsymbol{n})\cdot\boldsymbol{v}$ is converted to a volume integral. For *any* second-rank tensor $\boldsymbol{T}$,

$$\boldsymbol{\nabla}\cdot(\boldsymbol{T}^{\top}\cdot\boldsymbol{v}) = (\boldsymbol{\nabla}\cdot\boldsymbol{T})\cdot\boldsymbol{v} + \boldsymbol{T}:\boldsymbol{\nabla v}, \qquad \partial_i(T_{ji}v_j) = (\partial_i T_{ji})\,v_j + T_{ji}\,\partial_i v_j. \tag{1.4}$$

The transpose on the left is exactly what lets the two named operators on the right appear without one of their own; for a *symmetric* tensor such as the stress, $\boldsymbol{T}^{\top} = \boldsymbol{T}$ and it may be dropped. It is this identity that later separates the divergence of the mechanical-energy flux from the local work done by the stress (Sec. 3.8).

Operation	Symbol	Components	Rank
Scalar gradient	$\boldsymbol{\nabla}\phi$	$\partial_i\phi$	1
Vector gradient	$\boldsymbol{\nabla v}$	$\partial_j v_i$	2
Vector divergence	$\boldsymbol{\nabla}\cdot\boldsymbol{v}$	$\partial_i v_i$	0
Tensor divergence	$\boldsymbol{\nabla}\cdot\boldsymbol{T}$	$\partial_j T_{ij}$	1
Vector Laplacian	$\nabla^2\boldsymbol{v}$	$\partial_j\partial_j v_i$	1
Double contraction	$\boldsymbol{T}:\boldsymbol{\nabla v}$	$T_{ij}\,\partial_j v_i$	0

Theorem 1.15 | Curl of a gradient vanishes

For any twice-differentiable scalar field, $\boldsymbol{\nabla}\times(\boldsymbol{\nabla}\phi) = \boldsymbol{0}$.

Proof. Taking the ith component,

$$\begin{aligned}[\boldsymbol{\nabla}\times(\boldsymbol{\nabla}\phi)]_i &= \epsilon_{ijk}\,\partial_j(\partial_k\phi)\\ &= \epsilon_{ijk}\,\partial_j\partial_k\phi.\end{aligned}$$

The derivative factor $\partial_j\partial_k\phi$ is symmetric under $j \leftrightarrow k$, while ϵ_{ijk} is antisymmetric. The contraction of a symmetric object with an antisymmetric one over the same pair of indices vanishes identically, so each component is zero. Physically, any gradient field is irrotational; conversely, an irrotational velocity field ($\boldsymbol{\nabla}\times\boldsymbol{v}=\boldsymbol{0}$) may be written locally as the gradient of a velocity potential, $\boldsymbol{v}=\boldsymbol{\nabla}\varphi$, the starting point of potential-flow theory. This representation is guaranteed *globally* only on a *simply connected domain*, the content of the Poincaré lemma, which we will not deal with any further. On a multiply connected region a single-valued potential need not exist. The point vortex of Ch. 2, for example, is irrotational everywhere away from its core, yet possesses no globally single-valued velocity potential, precisely because the punctured domain it lives on is not simply connected. The companion identity $\boldsymbol{\nabla}\cdot(\boldsymbol{\nabla}\times\boldsymbol{v})=0$ (Exercise 1.4), the divergence of any curl vanishes, guarantees in turn that the vorticity field $\boldsymbol{\omega}=\boldsymbol{\nabla}\times\boldsymbol{v}$ is itself divergence-free; the converse representation of a divergence-free field as a curl, $\boldsymbol{v}=\boldsymbol{\nabla}\times\boldsymbol{A}$, likewise holds locally and, globally, only under appropriate topological assumptions on the domain. ∎

Theorem 1.16 | Curl of a curl

For any twice-differentiable vector field,

$$\boldsymbol{\nabla}\times(\boldsymbol{\nabla}\times\boldsymbol{v}) = \boldsymbol{\nabla}(\boldsymbol{\nabla}\cdot\boldsymbol{v}) - \nabla^2\boldsymbol{v}.$$

Proof. Apply the curl component twice and use the epsilon–delta identity Eq. (1.3) on the shared index i:

$$\begin{aligned}[\boldsymbol{\nabla}\times(\boldsymbol{\nabla}\times\boldsymbol{v})]_p &= \epsilon_{pqi}\,\partial_q\,(\boldsymbol{\nabla}\times\boldsymbol{v})_i\\ &= \epsilon_{pqi}\,\partial_q\,\epsilon_{ijk}\,\partial_j v_k\\ &= \epsilon_{ipq}\,\epsilon_{ijk}\,\partial_q\partial_j v_k.\end{aligned}$$

With $\epsilon_{ipq}\epsilon_{ijk}=\delta_{pj}\delta_{qk}-\delta_{pk}\delta_{qj}$,

$$\begin{aligned}[\boldsymbol{\nabla}\times(\boldsymbol{\nabla}\times\boldsymbol{v})]_p &= (\delta_{pj}\delta_{qk}-\delta_{pk}\delta_{qj})\,\partial_q\partial_j v_k\\ &= \partial_p\partial_k v_k - \partial_q\partial_q v_p\\ &= \partial_p(\boldsymbol{\nabla}\cdot\boldsymbol{v}) - \nabla^2 v_p,\end{aligned}$$

which is the pth component of the right-hand side. ■

Remark 1.17 | The viscous term

For an incompressible flow the velocity is divergence-free, $\boldsymbol{\nabla}\cdot\boldsymbol{v}=0$, and the identity (Thm. 1.16) collapses to $\boldsymbol{\nabla}\times(\boldsymbol{\nabla}\times\boldsymbol{v})=-\nabla^2\boldsymbol{v}$. The viscous force per unit volume in the Navier–Stokes equations, $\mu\nabla^2\boldsymbol{v}$, can therefore be written as $-\mu\,\boldsymbol{\nabla}\times\boldsymbol{\omega}$: momentum diffuses down gradients of vorticity. The many product rules that arise when differentiating compound fields, $\boldsymbol{\nabla}\cdot(\phi\boldsymbol{v})$, $\boldsymbol{\nabla}\times(\phi\boldsymbol{v})$, $\boldsymbol{\nabla}\cdot(\boldsymbol{v}\times\boldsymbol{w})$, and the like, reduce to the ordinary Leibniz rule for ∂_i and are collected in Exercise 1.3.

Remark 1.18 | Integral theorems

Two integral identities convert local (differential) statements into global (control-volume) ones and are used repeatedly when deriving balance laws. For a volume V with closed boundary S and outward unit normal $\boldsymbol{n}$, the *divergence theorem* states

$$\int_V \boldsymbol{\nabla}\cdot\boldsymbol{v}\,\mathrm{d}V=\oint_S \boldsymbol{v}\cdot\boldsymbol{n}\,\mathrm{d}S,$$

and for a surface S bounded by the closed curve C, *Stokes' theorem* states

$$\int_S(\boldsymbol{\nabla}\times\boldsymbol{v})\cdot\boldsymbol{n}\,\mathrm{d}S=\oint_C \boldsymbol{v}\cdot\mathrm{d}\boldsymbol{\ell}.$$

Both are indispensable in Ch. 3, where the balance laws convert control-volume statements into local field equations.

1.6 Orthogonal Curvilinear Coordinates

Every identity so far has been written in a *fixed* Cartesian basis, whose vectors are the same at every point,

$$\partial_i\boldsymbol{e}_j=\boldsymbol{0}.$$

That is precisely what let us slide ∂_i past the basis vectors and work with components alone.

Cartesian coordinates are rarely the natural ones for a transport problem. Pipes are round, drops and catalyst pellets are spherical, and a boundary condition is a statement made on a surface: no slip at the wall of a tube is $\boldsymbol{v}=\boldsymbol{0}$ at $r=a$, one clean line in cylindrical coordinates and an unusable mess in Cartesian ones. So we work in coordinates fitted to the geometry. In an *orthogonal curvilinear* system the three basis vectors remain mutually perpendicular unit vectors at every point, and all the algebra of dot and cross products

carries over untouched. One thing alone is different, and every complication below descends from it: **the basis vectors change with location**.

Take cylindrical coordinates (r, θ, z), with $x = r\cos\theta$, $y = r\sin\theta$, $z = z$, for which

$$\boldsymbol{e}_r = (\cos\theta,\ \sin\theta,\ 0), \qquad \boldsymbol{e}_\theta = (-\sin\theta,\ \cos\theta,\ 0), \qquad \boldsymbol{e}_z = (0,\ 0,\ 1).$$

Differentiating these three directly gives the only non-vanishing basis derivatives in the system,

$$\frac{\partial \boldsymbol{e}_r}{\partial\theta} = \boldsymbol{e}_\theta, \qquad \frac{\partial \boldsymbol{e}_\theta}{\partial\theta} = -\,\boldsymbol{e}_r, \tag{1.5}$$

every other derivative of $\boldsymbol{e}_r$, $\boldsymbol{e}_\theta$, $\boldsymbol{e}_z$ vanishing. These two relations generate every extra term in the formulas that follow. Nothing else is needed.

Gradient. An infinitesimal displacement is

$$\mathrm{d}\boldsymbol{x} = \mathrm{d}r\,\boldsymbol{e}_r + r\,\mathrm{d}\theta\,\boldsymbol{e}_\theta + \mathrm{d}z\,\boldsymbol{e}_z,$$

the factor r appearing because the arc length subtended by $\mathrm{d}\theta$ is $r\,\mathrm{d}\theta$ and not $\mathrm{d}\theta$. Setting the chain rule against $\mathrm{d}\phi = \boldsymbol{\nabla}\phi\cdot\mathrm{d}\boldsymbol{x}$,

$$\frac{\partial\phi}{\partial r}\mathrm{d}r + \frac{\partial\phi}{\partial\theta}\mathrm{d}\theta + \frac{\partial\phi}{\partial z}\mathrm{d}z = \boldsymbol{\nabla}\phi\cdot\mathrm{d}\boldsymbol{x},$$

and matching the coefficient of each independent differential,

$$\boldsymbol{\nabla} = \boldsymbol{e}_r\,\frac{\partial}{\partial r} + \boldsymbol{e}_\theta\,\frac{1}{r}\frac{\partial}{\partial\theta} + \boldsymbol{e}_z\,\frac{\partial}{\partial z}.$$

The $1/r$ is not a correction bolted on afterwards; it is the reciprocal of the scale factor that turns $\mathrm{d}\theta$ into a length.

Divergence. Apply that operator to $\boldsymbol{v} = v_r\boldsymbol{e}_r + v_\theta\boldsymbol{e}_\theta + v_z\boldsymbol{e}_z$, differentiating the basis vectors as well as the components. Only the θ-derivative produces anything new, and it does so through Eq. (1.5):

$$\frac{\partial\boldsymbol{v}}{\partial\theta} = \left(\frac{\partial v_r}{\partial\theta} - v_\theta\right)\boldsymbol{e}_r + \left(\frac{\partial v_\theta}{\partial\theta} + v_r\right)\boldsymbol{e}_\theta + \frac{\partial v_z}{\partial\theta}\,\boldsymbol{e}_z.$$

Projecting the three derivatives onto their own basis vectors,

$$\boldsymbol{\nabla}\cdot\boldsymbol{v} = \boldsymbol{e}_r\cdot\frac{\partial\boldsymbol{v}}{\partial r} + \boldsymbol{e}_\theta\cdot\frac{1}{r}\frac{\partial\boldsymbol{v}}{\partial\theta} + \boldsymbol{e}_z\cdot\frac{\partial\boldsymbol{v}}{\partial z} = \frac{\partial v_r}{\partial r} + \frac{1}{r}\left(v_r + \frac{\partial v_\theta}{\partial\theta}\right) + \frac{\partial v_z}{\partial z},$$

and gathering the two radial terms into one,

$$\boldsymbol{\nabla}\cdot\boldsymbol{v} = \frac{1}{r}\frac{\partial(rv_r)}{\partial r} + \frac{1}{r}\frac{\partial v_\theta}{\partial\theta} + \frac{\partial v_z}{\partial z}.$$

That lone v_r/r carries the whole difference from the Cartesian formula, and it came from $\partial \boldsymbol{e}_r/\partial\theta = \boldsymbol{e}_\theta$.

Curl. The same expansion with a cross product in place of the dot, using $\boldsymbol{e}_r \times \boldsymbol{e}_\theta = \boldsymbol{e}_z$ together with its cyclic partners. Once again it is the θ-derivative, and specifically the $-v_\theta\,\boldsymbol{e}_r$ term above, that supplies the extra piece:

$$\boldsymbol{\nabla}\times\boldsymbol{v} = \left(\frac{1}{r}\frac{\partial v_z}{\partial\theta} - \frac{\partial v_\theta}{\partial z}\right)\boldsymbol{e}_r + \left(\frac{\partial v_r}{\partial z} - \frac{\partial v_z}{\partial r}\right)\boldsymbol{e}_\theta + \frac{1}{r}\left(\frac{\partial (r v_\theta)}{\partial r} - \frac{\partial v_r}{\partial\theta}\right)\boldsymbol{e}_z.$$

The axial component is the one to remember. Written as $r^{-1}\partial(rv_\theta)/\partial r$ rather than $\partial v_\theta/\partial r$, it is what makes solid-body rotation $v_\theta = \Omega r$ carry the uniform vorticity 2Ω while the point vortex $v_\theta \propto 1/r$ carries none.

Laplacian. Nothing further is required, since $\nabla^2\phi = \boldsymbol{\nabla}\cdot(\boldsymbol{\nabla}\phi)$: feed the three components of $\boldsymbol{\nabla}\phi$ into the divergence just obtained,

$$\nabla^2\phi = \frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial\phi}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2\phi}{\partial\theta^2} + \frac{\partial^2\phi}{\partial z^2}.$$

The $1/r^2$ appears because the $1/r$ carried by the gradient meets the $1/r$ carried by the divergence.

Spherical coordinates (r,θ,ϕ) follow the identical route with scale factors $(1,\, r,\, r\sin\theta)$ in place of $(1,\, r,\, 1)$; only the algebra is longer. Every operator in both systems is collected in the table of Appendix 1.A at the end of this chapter, and in practice it is that table, not these derivations, that you will consult. The derivations are here so that the $1/r$ factors read as geometry rather than as something to be memorized.

Remark 1.19 | Covariant derivatives

The systematic way to handle a moving basis is to replace the ordinary partial derivative by the *covariant derivative*, which differentiates a field while automatically accounting for the change in the basis vectors. It is the backbone of tensor calculus in general curvilinear coordinates and on curved manifolds; we shall not need its machinery, only the awareness that the curvilinear formulas of later chapters are its concrete output. If you ever take an advanced course in differential geometry, you will see that the covariant derivative is the natural generalization of the gradient to curved spaces.

Appendix 1.A: Vector Operators in Curvilinear Coordinates

For reference, the first-order operators and the scalar Laplacian in the two coordinate systems used in later chapters. The unit vectors are orthonormal but position-dependent; the scale factors (arc-length coefficients) are $(1, r, 1)$ for cylindrical and $(1, r, r\sin\theta)$ for spherical

coordinates, and every $1/r$ and $\sin\theta$ below traces back to them.

Cylindrical Coordinates (r, θ, z)

$$\boldsymbol{\nabla} f = \frac{\partial f}{\partial r}\,\boldsymbol{e}_r + \frac{1}{r}\frac{\partial f}{\partial \theta}\,\boldsymbol{e}_\theta + \frac{\partial f}{\partial z}\,\boldsymbol{e}_z,$$

$$\begin{aligned}
\boldsymbol{\nabla}\cdot\boldsymbol{v} &= \frac{1}{r}\frac{\partial (r v_r)}{\partial r} + \frac{1}{r}\frac{\partial v_\theta}{\partial \theta} + \frac{\partial v_z}{\partial z},\\
\nabla^2 f &= \frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial f}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2 f}{\partial \theta^2} + \frac{\partial^2 f}{\partial z^2},
\end{aligned}$$

$$\begin{aligned}
(\boldsymbol{\nabla}\times\boldsymbol{v})_r &= \frac{1}{r}\frac{\partial v_z}{\partial \theta} - \frac{\partial v_\theta}{\partial z},\\
(\boldsymbol{\nabla}\times\boldsymbol{v})_\theta &= \frac{\partial v_r}{\partial z} - \frac{\partial v_z}{\partial r},\\
(\boldsymbol{\nabla}\times\boldsymbol{v})_z &= \frac{1}{r}\left(\frac{\partial (r v_\theta)}{\partial r} - \frac{\partial v_r}{\partial \theta}\right).
\end{aligned}$$

Spherical Coordinates (r, θ, ϕ)

Here θ is the polar angle and ϕ the azimuthal angle.

$$\boldsymbol{\nabla} f = \frac{\partial f}{\partial r}\,\boldsymbol{e}_r + \frac{1}{r}\frac{\partial f}{\partial \theta}\,\boldsymbol{e}_\theta + \frac{1}{r\sin\theta}\frac{\partial f}{\partial \phi}\,\boldsymbol{e}_\phi,$$

$$\boldsymbol{\nabla}\cdot\boldsymbol{v} = \frac{1}{r^2}\frac{\partial (r^2 v_r)}{\partial r} + \frac{1}{r\sin\theta}\frac{\partial (\sin\theta\, v_\theta)}{\partial \theta} + \frac{1}{r\sin\theta}\frac{\partial v_\phi}{\partial \phi},$$

$$\nabla^2 f = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial f}{\partial r}\right) + \frac{1}{r^2\sin\theta}\frac{\partial}{\partial \theta}\left(\sin\theta\,\frac{\partial f}{\partial \theta}\right) + \frac{1}{r^2\sin^2\theta}\frac{\partial^2 f}{\partial \phi^2},$$

$$\begin{aligned}
(\boldsymbol{\nabla}\times\boldsymbol{v})_r &= \frac{1}{r\sin\theta}\left(\frac{\partial (\sin\theta\, v_\phi)}{\partial \theta} - \frac{\partial v_\theta}{\partial \phi}\right),\\
(\boldsymbol{\nabla}\times\boldsymbol{v})_\theta &= \frac{1}{r}\left(\frac{1}{\sin\theta}\frac{\partial v_r}{\partial \phi} - \frac{\partial (r v_\phi)}{\partial r}\right),\\
(\boldsymbol{\nabla}\times\boldsymbol{v})_\phi &= \frac{1}{r}\left(\frac{\partial (r v_\theta)}{\partial r} - \frac{\partial v_r}{\partial \theta}\right).
\end{aligned}$$

Important | Notation contract

Throughout these notes the derivatives of vector and tensor fields are fixed by the following component definitions:

$$(\boldsymbol{\nabla v})_{ij} = \partial_j v_i, \qquad (\boldsymbol{\nabla}\cdot\boldsymbol{T})_i = \partial_j T_{ij}, \qquad (\boldsymbol{T}\cdot\boldsymbol{v})_i = T_{ij}v_j,$$

$$\boldsymbol{T} : \boldsymbol{\nabla v} = T_{ij}\,\partial_j v_i, \qquad (\boldsymbol{v}\otimes\boldsymbol{w})_{ij} = v_i w_j.$$

These conventions are *not* universal: other texts place the derivative index of $\boldsymbol{\nabla v}$ first, or contract $\boldsymbol{\nabla}\cdot\boldsymbol{T}$ on the other index. It is the component definition, not the visual placement of the $\boldsymbol{\nabla}$ symbol, that fixes the meaning, so when comparing with another source, check the indices rather than the typography.

Exercises

Unless stated otherwise, all indices range over $1, 2, 3$, repeated indices are summed, and fields are as smooth as needed. Problems and parts marked [Practice] are for self-study.

1. **(Index gymnastics)** Evaluate each contraction, stating the rank of the result:

   (a) δ_{ii}

   (b) $\delta_{ij}\delta_{ij}$

   (c) $\delta_{ij}\epsilon_{ijk}$

   (d) $\epsilon_{ijk}\epsilon_{ijk}$

2. **(The two special symbols)** Prove the following, using the epsilon–delta identity Eq. (1.3) where helpful.

   (a) $\epsilon_{ijk}\epsilon_{ijm} = 2\,\delta_{km}$.

   (b) The determinant representation

   $$\epsilon_{ijk} = \det\begin{pmatrix} \delta_{i1} & \delta_{i2} & \delta_{i3} \\ \delta_{j1} & \delta_{j2} & \delta_{j3} \\ \delta_{k1} & \delta_{k2} & \delta_{k3} \end{pmatrix}.$$

   (c) [Practice] The general product

   $$\epsilon_{ijk}\epsilon_{lmn} = \det\big(\delta_{(\cdot)(\cdot)}\big),$$

where the right-hand side is the 3×3 determinant whose entries are $\delta_{il}, \delta_{im}, \dots, \delta_{kn}$. Recover Eq. (1.3) by setting $l = i$.

3. **(Product rules for the balance laws)** Let ϕ be a scalar field and $\boldsymbol{v}, \boldsymbol{w}$ vector fields. Using only the Leibniz rule for ∂_i and the symbols $\delta_{ij}, \epsilon_{ijk}$, verify:

   (a) $\boldsymbol{\nabla}\cdot(\phi\boldsymbol{v}) = \phi\,\boldsymbol{\nabla}\cdot\boldsymbol{v} + \boldsymbol{v}\cdot\boldsymbol{\nabla}\phi$.

   (b) $\boldsymbol{\nabla}\times(\phi\boldsymbol{v}) = \phi\,\boldsymbol{\nabla}\times\boldsymbol{v} + \boldsymbol{\nabla}\phi\times\boldsymbol{v}$.

   (c) $\boldsymbol{\nabla}\cdot(\boldsymbol{v}\times\boldsymbol{w}) = \boldsymbol{w}\cdot\boldsymbol{\nabla}\times\boldsymbol{v} - \boldsymbol{v}\cdot\boldsymbol{\nabla}\times\boldsymbol{w}$.

   (d) [PRACTICE] $\boldsymbol{\nabla}(\boldsymbol{v}\cdot\boldsymbol{w}) = (\boldsymbol{v}\cdot\boldsymbol{\nabla})\boldsymbol{w} + (\boldsymbol{w}\cdot\boldsymbol{\nabla})\boldsymbol{v} + \boldsymbol{v}\times\boldsymbol{\nabla}\times\boldsymbol{w} + \boldsymbol{w}\times\boldsymbol{\nabla}\times\boldsymbol{v}$.

   (e) The tensor rule $\boldsymbol{\nabla}\cdot(\phi\boldsymbol{T}) = \phi\,\boldsymbol{\nabla}\cdot\boldsymbol{T} + \boldsymbol{T}\cdot\boldsymbol{\nabla}\phi$, under the convention $(\boldsymbol{\nabla}\cdot\boldsymbol{T})_i = \partial_j T_{ij}$. Taking $\boldsymbol{T} = \boldsymbol{I}$, deduce $\boldsymbol{\nabla}\cdot(p\,\boldsymbol{I}) = \boldsymbol{\nabla}p$, the identity that peels the pressure gradient off the stress divergence in the equation of motion.

   Exercise 1.3(a) is exactly what turns $\frac{\partial\rho}{\partial t} + \boldsymbol{\nabla}\cdot(\rho\boldsymbol{v}) = 0$ into the material form of the continuity equation (Thm. 3.6).

4. **(Second-order operators)**

   (a) Show that $\boldsymbol{\nabla}\cdot(\boldsymbol{\nabla}\times\boldsymbol{v}) = 0$ for any twice-differentiable $\boldsymbol{v}$, and identify precisely where the argument uses the symmetry of mixed partial derivatives.

   (b) Consider the scalar field $\phi(\boldsymbol{x}) = \alpha\,x_1x_2^2x_3 + \beta\sin(\gamma x_2)$, which you may regard as a temperature or pressure distribution. Compute $\boldsymbol{\nabla}\phi$, $\boldsymbol{\nabla}\cdot(\boldsymbol{\nabla}\phi) = \nabla^2\phi$, and confirm $\boldsymbol{\nabla}\times(\boldsymbol{\nabla}\phi) = \boldsymbol{0}$ by direct differentiation.

5. **(Triple products)** For arbitrary vectors $\boldsymbol{v}, \boldsymbol{w}, \boldsymbol{c}, \boldsymbol{d}$:

   (a) Show $\boldsymbol{v}\cdot(\boldsymbol{w}\times\boldsymbol{c}) = \epsilon_{ijk}v_iw_jc_k$, hence that this "box product" equals the determinant with rows $\boldsymbol{v}, \boldsymbol{w}, \boldsymbol{c}$ and gives the signed volume of the parallelepiped they span. Deduce its invariance under cyclic permutation and its sign change under any swap.

   (b) Prove Lagrange's identity $(\boldsymbol{v}\times\boldsymbol{w})\cdot(\boldsymbol{c}\times\boldsymbol{d}) = (\boldsymbol{v}\cdot\boldsymbol{c})(\boldsymbol{w}\cdot\boldsymbol{d}) - (\boldsymbol{v}\cdot\boldsymbol{d})(\boldsymbol{w}\cdot\boldsymbol{c})$, and recover the cross-product magnitude of the worked example as the case $\boldsymbol{c} = \boldsymbol{v},\ \boldsymbol{d} = \boldsymbol{w}$.

6. **(Tensor algebra)** Work on the orthonormal basis $\{\boldsymbol{e}_i\}$ of E^3.

   (a) Let $\boldsymbol{A}$ be symmetric ($A_{ij} = A_{ji}$) and $\boldsymbol{B}$ skew-symmetric ($B_{ij} = -B_{ji}$). Show that $A_{ij}B_{ij} = 0$. (This is why, when the velocity gradient is split into its symmetric rate-of-strain and antisymmetric rotation parts, the two do not cross-couple in the rate of viscous dissipation.)

(b) Let $\boldsymbol{Q}$ be an orthogonal tensor ($Q_{ik}Q_{jk} = \delta_{ij}$) and $\boldsymbol{u}, \boldsymbol{v} \in E^3$. Show that $(\boldsymbol{Q}\cdot\boldsymbol{u})\cdot(\boldsymbol{Q}\cdot\boldsymbol{v}) = \boldsymbol{u}\cdot\boldsymbol{v}$, i.e. rotations preserve lengths and angles, the reason dot products of vectors are scalar invariants.

7. (Momentum: convective acceleration and Bernoulli)

(a) Using the BAC–CAB rule (Thm. 1.11), prove the identity

$$(\boldsymbol{\nabla v})\,\boldsymbol{v} = \boldsymbol{\nabla}\left(\tfrac{1}{2}\,|\boldsymbol{v}|^2\right) - \boldsymbol{v}\times\boldsymbol{\omega}, \qquad \boldsymbol{\omega} = \boldsymbol{\nabla}\times\boldsymbol{v}.$$

(b) For a steady, inviscid, incompressible flow under gravity, the momentum balance reads $(\boldsymbol{\nabla v})\,\boldsymbol{v} = -\boldsymbol{\nabla}(p/\rho) - g\,\boldsymbol{e}_z$. Dot this with $\boldsymbol{v}$ and use Exercise 1.7(a) to show that the Bernoulli quantity $H = \tfrac{1}{2}\,|\boldsymbol{v}|^2 + p/\rho + gz$ is constant along streamlines.

8. (Kinematics: solid-body rotation) A fluid rotates rigidly with constant angular velocity $\boldsymbol{\Omega}$, so that $\boldsymbol{v}(\boldsymbol{x}) = \boldsymbol{\Omega}\times\boldsymbol{x}$, i.e. $v_i = \epsilon_{ijk}\Omega_j x_k$.

(a) Show $\boldsymbol{\nabla}\cdot\boldsymbol{v} = 0$ (the motion is incompressible).

(b) Show $\boldsymbol{\nabla}\times\boldsymbol{v} = 2\,\boldsymbol{\Omega}$. Conclude that vorticity measures *twice* the local angular velocity of a fluid element.

9. (Heat and mass: the Laplace equation) In a medium at rest with no internal sources, the steady energy balance reduces to $\boldsymbol{\nabla}\cdot\boldsymbol{q} = 0$, where Fourier's law gives the heat flux $\boldsymbol{q} = -k\,\boldsymbol{\nabla}T$ with constant conductivity k.

(a) Show that T satisfies Laplace's equation, $\nabla^2 T = 0$.

(b) State the exact analog for steady dilute diffusion, where Fick's law gives the molar flux $\boldsymbol{j} = -D\,\boldsymbol{\nabla}c$ and $\boldsymbol{\nabla}\cdot\boldsymbol{j} = 0$. Comment on the heat–mass transfer analogy this exposes.

10. (Momentum: the viscous term) For an incompressible Newtonian fluid the deviatoric stress is $2\mu\boldsymbol{D}$ with $\boldsymbol{D} = \tfrac{1}{2}(\boldsymbol{\nabla v} + \boldsymbol{\nabla v}^{\top})$ and constant viscosity μ. Working in indices and using $\boldsymbol{\nabla}\cdot\boldsymbol{v} = 0$, show that

$$\boldsymbol{\nabla}\cdot(2\mu\boldsymbol{D}) = \mu\,\nabla^2\boldsymbol{v},$$

so the divergence of the viscous stress supplies the familiar Laplacian term of the Navier–Stokes equations.

11. (Cylindrical divergence from a control volume) Take a small control volume bounded by the coordinate surfaces $[r, r+\mathrm{d}r]$, $[\theta, \theta+\mathrm{d}\theta]$, and $[z, z+\mathrm{d}z]$, whose θ-faces are separated by the arc length $r\,\mathrm{d}\theta$. By balancing the outward flux of $\boldsymbol{v} = v_r\boldsymbol{e}_r + v_\theta\boldsymbol{e}_\theta + v_z\boldsymbol{e}_z$ through the six faces and dividing by the volume $r\,\mathrm{d}r\,\mathrm{d}\theta\,\mathrm{d}z$, derive

$$\boldsymbol{\nabla}\cdot\boldsymbol{v} = \frac{1}{r}\frac{\partial (r v_r)}{\partial r} + \frac{1}{r}\frac{\partial v_\theta}{\partial \theta} + \frac{\partial v_z}{\partial z},$$

and identify which term owes its $1/r$ to the r-dependence of the face area rather than to a derivative of a basis vector (Appendix 1.A).

2
Kinematics of Fluid Motion

Kinematics is the description of motion *without reference to the forces that cause it.* Before we can write balance laws for momentum and energy (Ch. 3), we need a precise language for how a fluid moves, stretches, and rotates. This chapter develops that language. Its centerpiece is the decomposition of the local motion into three elementary pieces, a translation, a straining, and a rotation, encoded respectively by the velocity, the *rate-of-strain tensor*, and the *rotation tensor*. Along the way we give the incompressibility condition $\boldsymbol{\nabla}\cdot\boldsymbol{v}=0$ its precise kinematic meaning: simply put, it is the statement that the volume of any material fluid element is preserved as it moves. In technical terms, incompressibility is a kinematic constraint, not a constitutive assumption: it holds for any fluid, Newtonian or not, if the flow is slow enough that density variations are negligible. The incompressibility condition is therefore a statement about the motion of the fluid, not about its material properties; in particular, it does not require that the fluid be a liquid.

Throughout, we use the indicial notation of Ch. 1 wherever it makes a derivation cleaner, but we quote the final result in both symbolic and indicial form. We adopt the convention

$$(\boldsymbol{\nabla}\boldsymbol{v})_{ij} \;=\; \frac{\partial v_i}{\partial x_j} \;=\; \partial_j v_i, \tag{2.1}$$

so that the first index of $\boldsymbol{\nabla}\boldsymbol{v}$ labels the velocity component and the second the direction of differentiation.[1] This convention is not universal and differs from field to field and textbook to textbook. But we will stay true to this convention, so remember it!!

[1] In continuum mechanics this component notation is often written with a comma, $v_{i,j}\equiv\partial v_i/\partial x_j$, in which the first index (i) denotes the component of the velocity and the second (j) the coordinate with respect to which it is differentiated.

2.1 Motion and the Two Descriptions

Imagine labeling every fluid particle by its position $\boldsymbol{x}_0$ in a *reference position*, taken at time $t = 0$. The subsequent motion can be formally represented by the map

$$\boldsymbol{x}(t) = \boldsymbol{\chi}(\boldsymbol{x}_0, t), \qquad \boldsymbol{\chi}(\boldsymbol{x}_0, 0) = \boldsymbol{x}_0,$$

so that each particle starts at $\boldsymbol{x}(0) = \boldsymbol{x}_0$. It gives the *current position* $\boldsymbol{x}$ of the particle labeled $\boldsymbol{x}_0$. At each instant the map $\boldsymbol{\chi}(\cdot, t)$ is smooth and *invertible*: distinct particles occupy distinct points, and none is created or destroyed, so it can be inverted, $\boldsymbol{x}_0 = \boldsymbol{\chi}^{-1}(\boldsymbol{x}, t)$, to recover the reference position of the particle now at $\boldsymbol{x}$. Invertibility is guaranteed by a positive Jacobian, $\det(\partial\boldsymbol{\chi}/\partial\boldsymbol{x}_0) > 0$. A positive Jacobian is a mathematical way of saying that we don't allow silly unphysical things like crushing a blob of material onto zero volume or turning it inside out. Holding $\boldsymbol{x}_0$ fixed and varying t traces the *pathline* of a single particle (Fig. 2.1(a)); this is also called the *Lagrangian* (or material) description. The velocity and acceleration of a particle are

$$\boldsymbol{v} = \left.\frac{\partial \boldsymbol{\chi}}{\partial t}\right|_{\boldsymbol{x}(0)=\boldsymbol{x}_0}, \qquad \boldsymbol{a} = \left.\frac{\partial^2 \boldsymbol{\chi}}{\partial t^2}\right|_{\boldsymbol{x}(0)=\boldsymbol{x}_0}.$$

In practice we rarely track individual particles. Instead we observe fields at fixed points in space, the velocity $\boldsymbol{v}(\boldsymbol{x}, t)$, pressure $p(\boldsymbol{x}, t)$, and so on, where $\boldsymbol{x}$ now denotes a fixed point in space rather than the position of any particular particle. This is the *Eulerian* (or spatial) description, and it is the one we use almost exclusively.[2] In this description the velocity is not computed from a particle label but read off directly as the field that takes, at each point $\boldsymbol{x}$ and time t, the velocity of whichever particle instantaneously occupies that point,

$$\boldsymbol{v}(\boldsymbol{x}, t) = \left.\frac{\partial \boldsymbol{\chi}}{\partial t}\right|_{\boldsymbol{x}_0=\boldsymbol{\chi}^{-1}(\boldsymbol{x},t)}.$$

The price is that the acceleration, i.e. the rate of change *following a particle*, is no longer a simple time derivative and must be reconstructed (Sec. 2.2).

Two families of curves help to visualize a flow. A *pathline* is the trajectory actually followed by a single material particle, i.e., the curve traced by $\boldsymbol{\chi}(\boldsymbol{x}_0, t)$ as t varies (Fig. 2.1(a)); it is the natural object of the Lagrangian description. A *streamline*, by contrast, is a curve everywhere tangent to the velocity field at one *fixed* instant, obtained by holding t fixed and integrating $\mathrm{d}\boldsymbol{x}/\mathrm{d}s = \boldsymbol{v}(\boldsymbol{x}, t)$; it belongs to the Eulerian description. The two answer

[2] The spatial description is conventionally associated with L. Euler, particularly his "Principes généraux du mouvement des fluides," *Mémoires de l'Académie des sciences de Berlin* **11** (1757), 274–315; the material description is commonly associated with J.-L. Lagrange, *Méchanique analitique* (Chez la veuve Desaint, Paris, 1788).



(a) Lagrangian: one particle, followed. **(b)** Eulerian: fixed points, watched.

Figure 2.1: The two descriptions of motion. **(a)** The *Lagrangian* view follows a single fluid particle along its pathline, labeling it by its reference position $\boldsymbol{x}_0$ and recording its properties at successive times. **(b)** The *Eulerian* view instead records the fields (here the velocity) at fixed points $\boldsymbol{x}$ in space. Fluid mechanics uses the Eulerian view almost exclusively; solid mechanics favors the Lagrangian view.

different questions, where one particle goes over time, versus the instantaneous direction of the flow. And in general they differ. In a *steady* flow ($\partial \boldsymbol{v}/\partial t = \boldsymbol{0}$) the velocity pattern never changes, and pathlines and streamlines coincide.[3]

The choice between the two descriptions is partly a matter of discipline (Fig. 2.1). Solid mechanics, and with it much of mechanical and civil engineering, naturally adopts the *Lagrangian* viewpoint: a solid has a definite reference configuration to which its stresses and strains are referred, and its particles make only small excursions from it. In a fluid the particles wander arbitrarily far, no such reference configuration remains useful, and the *Eulerian* viewpoint prevails.[4]

2.2 The Material Derivative

Consider any field, scalar, vector, or tensor, and follow a particle whose position is $\boldsymbol{x}(t)$ with $\dot{\boldsymbol{x}} = \boldsymbol{v}$. Taking a scalar field $\phi(\boldsymbol{x}, t)$ for simplicity, the rate of change experienced by that particle is, by the chain rule,

$$\frac{\mathrm{d}}{\mathrm{d}t}\,\phi(\boldsymbol{x}(t), t) = \frac{\partial \phi}{\partial t} + \frac{\mathrm{d}x_i}{\mathrm{d}t}\,\partial_i \phi = \frac{\partial \phi}{\partial t} + v_i\,\partial_i \phi.$$

The second term arises precisely because the particle's own location changes with time: as it is carried through a spatially varying field it samples different values of ϕ, and the chain rule turns $\mathrm{d}x_i/\mathrm{d}t = v_i$ into the spatial-gradient contribution $v_i\,\partial_i\phi$. This total derivative, taken along the path of a fluid particle, is the *material derivative*. We write the operator with a capital D, as $\mathrm{D}/\mathrm{D}t$, to distinguish it from the ordinary derivative $\mathrm{d}/\mathrm{d}t$ that appears

[3] A third curve, the *streakline*, supposedly joins at one instant all particles that have previously passed through a fixed point (the pattern a dye filament makes); it too coincides with the others in steady flow. I say *supposedly* because I had happily forgotten about it until I re-read a textbook to prepare these notes.

[4] The same dichotomy appears elsewhere in physics: the Lagrangian and Eulerian descriptions are loosely analogous to the Heisenberg and Schrödinger pictures of quantum (and statistical) mechanics, respectively. In the Heisenberg picture the state is held fixed while the operators carry the time dependence, as the Lagrangian view fixes the particle label $\boldsymbol{x}_0$ and lets its observed properties evolve, whereas in the Schrödinger picture the operators are fixed and the state evolves, as the Eulerian view fixes points in space and watches the fields change there.

just above and from the partial derivative $\partial/\partial t$: the capital D signals a rate of change taken *following the fluid particle* (along $\mathrm{d}\boldsymbol{x}/\mathrm{d}t = \boldsymbol{v}$), rather than one at a fixed point ($\partial/\partial t$) or along an arbitrary prescribed path ($\mathrm{d}/\mathrm{d}t$).

Definition 2.1 | Material derivative

For any field $\phi(\boldsymbol{x}, t)$,

$$\frac{\mathrm{D}\phi}{\mathrm{D}t} = \frac{\partial \phi}{\partial t} + (\boldsymbol{v}\cdot\boldsymbol{\nabla})\phi = \frac{\partial \phi}{\partial t} + v_i\,\partial_i\phi.$$

The first term is the *local* (unsteady) rate of change at a fixed point; the second is the *convective* rate of change due to the particle being carried through a spatially varying field.

The material derivative operator $\mathrm{D}/\mathrm{D}t$ can be applied to any field, not just scalars. When applied to the velocity itself, the material derivative gives the acceleration that enters Newton's second law:

$$\boldsymbol{a} = \frac{\mathrm{D}\boldsymbol{v}}{\mathrm{D}t} = \frac{\partial \boldsymbol{v}}{\partial t} + (\boldsymbol{\nabla}\boldsymbol{v})\,\boldsymbol{v}, \qquad a_i = \frac{\partial v_i}{\partial t} + (\partial_j v_i)\,v_j. \tag{2.2}$$

We have written the convective acceleration as the matrix product of the velocity gradient with the velocity, that is, it is to be read as the 3×3 matrix $(\boldsymbol{\nabla}\boldsymbol{v})$ multiplying the column vector $\boldsymbol{v}$, which yields another vector. With the convention Eq. (2.1), $(\boldsymbol{\nabla}\boldsymbol{v})\,\boldsymbol{v}$ has components $(\boldsymbol{\nabla}\boldsymbol{v})_{ij}\,v_j = (\partial_j v_i)\,v_j = v_j\,\partial_j v_i$, so this is the familiar term $(\boldsymbol{v}\cdot\boldsymbol{\nabla})\boldsymbol{v}$ written in a form that makes its tensor structure explicit; we use the matrix-product form $(\boldsymbol{\nabla}\boldsymbol{v})\,\boldsymbol{v}$ throughout these notes. The convective term is nonlinear in $\boldsymbol{v}$; it is the ultimate source of all of the difficulty, and all of the richness, of fluid dynamics.

Example 2.2 | A steady field can still accelerate a particle

This is a simple example to illustrate the importance of the second term in the material derivative. Take the one-dimensional steady field $v(x) = \alpha x$ with constant α. Although it is steady,

$$\frac{\partial v}{\partial t} = 0,$$

it **does not** mean a particle is unaccelerated. Following one particle, $\mathrm{d}x/\mathrm{d}t = v = \alpha x$, so

$$x(t) = x_0\,e^{\alpha t},$$

and its acceleration is

$$\frac{\mathrm{D}v}{\mathrm{D}t} = \frac{\partial v}{\partial t} + v\,\frac{\partial v}{\partial x} = 0 + (\alpha x)(\alpha) = \alpha^2 x.$$

The particle accelerates because it is convected into regions of larger velocity: a steady Eulerian field can still produce a nonzero particle acceleration.

Remark 2.3 | Following the particle

The material derivative measures the rate of change of a property *as experienced by the moving fluid particle*, not the rate seen at a fixed point. A quantity is called *objective* (or *frame-indifferent*) when it is the same for every observer, whatever their state of motion; the change a particle actually undergoes, for a scalar such as its own temperature, is objective in this sense, while the local rate $\partial/\partial t$ is tied to the particular frame in which one chooses to stand still. This objectivity is special to scalars: it does *not* extend automatically to the material derivative of every vector or tensor. Among the kinematic tensors introduced later in this chapter, the rate-of-strain tensor $\boldsymbol{D}$ *is* objective, two observers in relative rigid motion measure the same stretching rates, whereas the rotation tensor $\boldsymbol{W}$ and the vorticity $\boldsymbol{\omega}$ are *not*: they carry the spin of the observer's own frame, so their measured values depend on the observer's rotation. The transformation rules that make this precise are derived in Exercise 2.10. Note that $\mathrm{D}/\mathrm{D}t$ reduces to $\partial/\partial t$ for a field uniform in space, and to $\boldsymbol{v}\cdot\boldsymbol{\nabla}$ for a steady field. A steady Eulerian field can still be experienced as time-varying by a moving particle, as discussed in Ex. 2.2.

2.3 The Velocity Gradient and Its Decomposition

Deformation is a matter of *relative* motion, so we follow the treatment of W. M. Deen[5] and begin with two neighboring material points: point 1 at position $\boldsymbol{r}_1$ and point 2 a short distance away at $\boldsymbol{r}_2 = \boldsymbol{r}_1 + \Delta\boldsymbol{r}$ (Fig. 2.2(a)). At the instant considered they move with the local fluid velocities $\boldsymbol{v}_1 = \boldsymbol{v}(\boldsymbol{r}_1, t)$ and $\boldsymbol{v}_2 = \boldsymbol{v}(\boldsymbol{r}_2, t)$, and whether the material between them stretches, shears, or merely turns is decided by their *relative velocity* $\Delta\boldsymbol{v} = \boldsymbol{v}_2 - \boldsymbol{v}_1$. Taylor-expanding for small $\Delta\boldsymbol{r}$,

$$\Delta v_i = (\partial_j v_i)\,\Delta r_j + \mathcal{O}\Big(|\Delta\boldsymbol{r}|^2\Big), \qquad \Delta\boldsymbol{v} = (\boldsymbol{\nabla}\boldsymbol{v})\cdot\Delta\boldsymbol{r} + \mathcal{O}\Big(|\Delta\boldsymbol{r}|^2\Big). \tag{2.3}$$

[5] W. M. Deen, *Analysis of Transport Phenomena* (Oxford University Press, New York, 1998).

To leading order the relative motion of nearby material points is therefore controlled by a single object, the *velocity gradient tensor* introduced earlier

$$\boldsymbol{L} = \boldsymbol{\nabla} \boldsymbol{v}, \qquad L_{ij} = \partial_j v_i, \tag{2.4}$$

with the convention Eq. (2.1). Everything local about the flow, stretching, shearing, spinning, is contained in $\boldsymbol{L}$.

Not all of this relative motion, however, is deformation. A body in rigid rotation, which does not deform at all, still shows a relative velocity between any two of its points; to extract the genuine deformation we must first subtract the rotation. The tool is the unique decomposition of a second-rank tensor into a symmetric and an antisymmetric part:

$$\boldsymbol{L} = \boldsymbol{D} + \boldsymbol{W}, \qquad \boldsymbol{D} = \tfrac{1}{2}(\boldsymbol{L} + \boldsymbol{L}^{\top}), \qquad \boldsymbol{W} = \tfrac{1}{2}(\boldsymbol{L} - \boldsymbol{L}^{\top}), \tag{2.5}$$

or in components,

$$D_{ij} = \tfrac{1}{2}(\partial_j v_i + \partial_i v_j), \qquad W_{ij} = \tfrac{1}{2}(\partial_j v_i - \partial_i v_j). \tag{2.6}$$

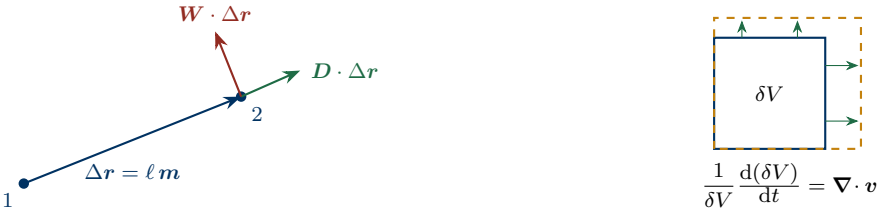
Note that $\boldsymbol{D} = \boldsymbol{D}^{\top}$ (symmetric) and $\boldsymbol{W} = -\boldsymbol{W}^{\top}$ (antisymmetric). The relative velocity splits accordingly into a deformation part and a rotation part,

$$\Delta \boldsymbol{v} = \underbrace{\boldsymbol{D} \cdot \Delta \boldsymbol{r}}_{\text{deformation}} + \underbrace{\boldsymbol{W} \cdot \Delta \boldsymbol{r}}_{\text{rotation}}. \tag{2.7}$$

It is easy to see that $\boldsymbol{W}$ is genuinely a rotation. A rigid body spinning with angular velocity $\boldsymbol{w}$ moves so that $\boldsymbol{v} = \boldsymbol{w} \times \boldsymbol{R}$, giving a relative velocity $\Delta \boldsymbol{v} = \boldsymbol{w} \times \Delta \boldsymbol{r}$ between two of its points. We show in Sec. 2.5 that the antisymmetric part acts in exactly this way,

$$\boldsymbol{W} \cdot \Delta \boldsymbol{r} = \boldsymbol{w} \times \Delta \boldsymbol{r}, \qquad \boldsymbol{w} = \tfrac{1}{2} \boldsymbol{\nabla} \times \boldsymbol{v},$$

so $\boldsymbol{W}$ turns the line element $\Delta \boldsymbol{r}$ rigidly, without changing its length (Fig. 2.2(a)). All the stretching and shearing is thus carried by the symmetric part $\boldsymbol{D}$, the **rate-of-strain** (or rate-of-deformation) tensor; $\boldsymbol{W}$ is the **rotation** (or vorticity) tensor. The next section makes the deformation content of $\boldsymbol{D}$ precise.



(a) Line element: $\boldsymbol{D}$ deforms, $\boldsymbol{W}$ rotates. **(b)** Volume element: dilation rate $= \boldsymbol{\nabla}\cdot\boldsymbol{v}$.

Figure 2.2: Deformation of material elements. **(a)** A line element $\Delta\boldsymbol{r} = \ell\boldsymbol{m}$ joining two material points. Its relative velocity splits into a deformation part $\boldsymbol{D}\cdot\Delta\boldsymbol{r}$, which stretches and shears the element (changing ℓ), and a rotation part $\boldsymbol{W}\cdot\Delta\boldsymbol{r} = \boldsymbol{w}\times\Delta\boldsymbol{r}$, which turns it rigidly without changing ℓ. **(b)** A volume element δV; the net outflow of its faces makes it dilate at the fractional rate $\operatorname{tr}\boldsymbol{D} = \boldsymbol{\nabla}\cdot\boldsymbol{v}$.

2.4 The Rate-of-Strain Tensor

Return to the line element $\Delta\boldsymbol{r} = \ell\,\boldsymbol{m}$ joining our two material points, with length $\ell = |\Delta\boldsymbol{r}|$ and unit direction $\boldsymbol{m}$. Because both endpoints are convected with the flow, the connecting vector evolves at the rate given by the relative velocity Eq. (2.3). In the infinitesimal-line-element limit, the separation vector evolves as

$$\frac{\mathrm{d}}{\mathrm{d}t}(\Delta\boldsymbol{r}) = \Delta\boldsymbol{v} = \boldsymbol{L}\cdot\Delta\boldsymbol{r}. \tag{2.8}$$

What we want is the rate of change of its *length*. Differentiating $\ell^2 = \Delta\boldsymbol{r}\cdot\Delta\boldsymbol{r}$ following the motion and using Eq. (2.8),

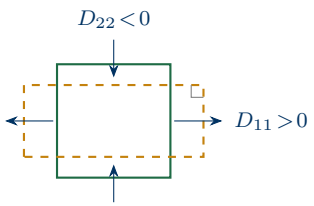
$$\ell\,\frac{\mathrm{d}\ell}{\mathrm{d}t} = \Delta\boldsymbol{r}\cdot\frac{\mathrm{d}}{\mathrm{d}t}(\Delta\boldsymbol{r}) = \Delta\boldsymbol{r}\cdot(\boldsymbol{L}\cdot\Delta\boldsymbol{r}) = \Delta\boldsymbol{r}\cdot(\boldsymbol{D}\cdot\Delta\boldsymbol{r}),$$

where the rotation part drops out because $\Delta\boldsymbol{r}\cdot(\boldsymbol{W}\cdot\Delta\boldsymbol{r}) = W_{ij}\,\Delta r_i\,\Delta r_j = 0$ for the antisymmetric $\boldsymbol{W}$, a rigid rotation cannot change a length. Dividing by ℓ^2 gives the *rate of strain* of the fiber pointing along $\boldsymbol{m}$:

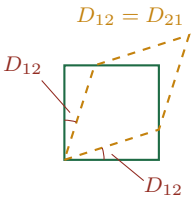
$$\frac{1}{\ell}\,\frac{\mathrm{d}\ell}{\mathrm{d}t} = m_i\,D_{ij}\,m_j = \boldsymbol{m}\cdot\boldsymbol{D}\cdot\boldsymbol{m}. \tag{2.9}$$

This is the central meaning of $\boldsymbol{D}$: contracted twice with a unit vector, it returns the fractional rate at which a material fiber in that direction stretches. In particular,

- the **diagonal** components D_{11}, D_{22}, D_{33} are the normal strain rates, taking $\boldsymbol{m} = \boldsymbol{e}_1$ gives $\frac{1}{\ell}\frac{\mathrm{d}\ell}{\mathrm{d}t} = D_{11}$, the stretching rate of a fiber along the x-axis;
- the **off-diagonal** components D_{ij} $(i \neq j)$ are the *tensorial shear-strain rates*, each equal to half the rate at which the angle between two initially perpendicular fibers decreases (Exercise 2.3); the corresponding engineering shear rate is $\dot\gamma_{ij} = 2D_{ij}$.



(a) Extension: lengths change, angles preserved.



(b) Shear: angles change, lengths preserved.

Figure 2.3: The two elementary actions of the rate-of-strain tensor $\boldsymbol{D}$ on a material element (original in green, deformed dashed). **(a)** Its *diagonal* components produce *normal* (extensional) strain: fibers along the axes stretch or compress, here $D_{11} > 0$ elongates the element and $D_{22} < 0$ thins it, while the corners remain right angles. **(b)** Its *off-diagonal* components produce *shear* strain: two initially perpendicular fibers each rotate toward the other by D_{12}, closing the angle between them while their lengths scarcely change. Because $\boldsymbol{D}$ is symmetric the two fibers tilt *equally*, so pure strain carries no net rotation, that is the separate work of $\boldsymbol{W}$.

Because $\boldsymbol{D}$ is symmetric it has three real eigenvalues (the principal strain rates) and three mutually orthogonal eigenvectors (the principal axes of strain); along those axes the deformation is a pure stretch with no shear. The two elementary modes that $\boldsymbol{D}$ superposes, *extension* along the axes and *shear* between them, are shown in Fig. 2.3.

Theorem 2.4 | Rate of change of a material volume

An infinitesimal material volume $\mathrm{d}V$ convected with the flow changes at the rate

$$\frac{1}{\mathrm{d}V}\frac{\mathrm{d}}{\mathrm{d}t}(\mathrm{d}V) = \operatorname{tr}\boldsymbol{D} = D_{ii} = \partial_i v_i = \boldsymbol{\nabla}\cdot\boldsymbol{v}.$$

Proof. Let three material line elements $\boldsymbol{a}, \boldsymbol{b}, \boldsymbol{c}$ span the infinitesimal parallelepiped, so that its volume is the scalar triple product

$$\mathrm{d}V = \boldsymbol{a}\cdot(\boldsymbol{b}\times\boldsymbol{c}).$$

Each edge is convected with the flow and so evolves by Eq. (2.8), $\frac{\mathrm{d}\boldsymbol{a}}{\mathrm{d}t} = \boldsymbol{L}\cdot\boldsymbol{a}$, and likewise for $\boldsymbol{b}$ and $\boldsymbol{c}$. Differentiating the triple product with the product rule,

$$\frac{\mathrm{d}}{\mathrm{d}t}(\mathrm{d}V) = (\boldsymbol{L}\boldsymbol{a})\cdot(\boldsymbol{b}\times\boldsymbol{c}) + \boldsymbol{a}\cdot[(\boldsymbol{L}\boldsymbol{b})\times\boldsymbol{c}] + \boldsymbol{a}\cdot[\boldsymbol{b}\times(\boldsymbol{L}\boldsymbol{c})] = (\operatorname{tr}\boldsymbol{L})\,\mathrm{d}V,$$

the last equality being the standard triple-product identity, proved in Appendix 2.B. It is worth trying in components before reading the proof: summing L_{ii} is exactly what the three terms produce. As $\operatorname{tr}\boldsymbol{L} = \partial_i v_i = \boldsymbol{\nabla}\cdot\boldsymbol{v}$,

$$\frac{1}{\mathrm{d}V}\frac{\mathrm{d}}{\mathrm{d}t}(\mathrm{d}V) = \operatorname{tr}\boldsymbol{L} = \boldsymbol{\nabla}\cdot\boldsymbol{v}.$$

■



(a) The element shears into a parallelogram. **(b)** Strain along $\pm 45°$, plus rotation.

Figure 2.4: Simple shear $\boldsymbol{v} = (\dot\gamma\, x_2, 0, 0)$ as an equal blend of straining and rotation. **(a)** The shear flow carries the top of a square element faster than its base, deforming it into a parallelogram. **(b)** Equivalently, an inscribed circle becomes an ellipse stretched along the $+45°$ principal axis and compressed along the $-45°$ axis (the eigenvalues $\pm\dot\gamma/2$ of $\boldsymbol{D}$), while the rotation tensor $\boldsymbol{W}$ rotates the element.

This identity (Thm. 2.4) is the kinematic heart of the incompressibility condition (Sec. 2.6): the divergence of the velocity is precisely the fractional rate of change of a material volume, the local rate of volumetric expansion (Fig. 2.2(b)). An alternative derivation, which follows the motion map directly, is developed in Exercise 2.9.

Example 2.5 | Simple shear

Consider the shear flow $\boldsymbol{v} = (\dot\gamma\, x_2,\, 0,\, 0)$ with constant shear rate $\dot\gamma$. The only nonzero velocity derivative is $\partial_2 v_1 = \dot\gamma$, so

$$\boldsymbol{L} = \begin{pmatrix} 0 & \dot\gamma & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \qquad \boldsymbol{D} = \frac{\dot\gamma}{2}\begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \qquad \boldsymbol{W} = \frac{\dot\gamma}{2}\begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

Since $\operatorname{tr}\boldsymbol{D} = 0$, the flow is volume-preserving. The eigenvalues of $\boldsymbol{D}$ are $\pm\dot\gamma/2$, with principal axes at $45°$ to the flow: a small fluid disk is stretched along one diagonal and compressed along the other while $\boldsymbol{W}$ spins it (Fig. 2.4). Simple shear is thus an equal blend of straining and rotation, a fact of central importance for the behavior of suspended particles and polymers.

A word of caution on what this decomposition does and does not say. The splitting $\boldsymbol{L} = \boldsymbol{D} + \boldsymbol{W}$ describes only the *instantaneous* local motion: at each instant $\boldsymbol{D}$ gives the stretching rates and $\boldsymbol{W}$ the local spin *relative to the observer*. The *finite* deformation accumulated over a time interval must be found by following the material particles through the flow; it is *not* recovered by integrating $\boldsymbol{D}$ and $\boldsymbol{W}$ separately and recombining them, because each acts on a material element whose own orientation is continually changing (Exercise 2.11).

2.5 The Rotation Tensor and Vorticity

The antisymmetric tensor $\boldsymbol{W}$ has only three independent components, so it can be represented by a single vector. For any antisymmetric $\boldsymbol{W}$ there is a unique *axial vector* $\boldsymbol{w}$ such that

$$\boldsymbol{W}\cdot\boldsymbol{z}=\boldsymbol{w}\times\boldsymbol{z}\quad\text{for all }\boldsymbol{z},\qquad\Longleftrightarrow\qquad W_{ij}=-\epsilon_{ijk}\,w_k. \tag{2.10}$$

This is a statement about *every* antisymmetric tensor, not a definition, and it is proved in Appendix 2.B: the vector exists, it is unique, and the correspondence is peculiar to three dimensions. Contracting Eq. (2.10) with $-\epsilon_{ijm}$ and using $\epsilon_{ijm}\epsilon_{ijk}=2\delta_{mk}$ inverts the relation to $w_m=-\tfrac{1}{2}\epsilon_{mij}W_{ij}$. Substituting W_{ij} from Eq. (2.6) yields (Exercise 2.4)

$$\boldsymbol{w}=\tfrac{1}{2}\,\boldsymbol{\nabla}\times\boldsymbol{v}. \tag{2.11}$$

Thus $\boldsymbol{W}$ generates a local rigid-body rotation with angular velocity $\boldsymbol{w}=\frac{1}{2}\boldsymbol{\nabla}\times\boldsymbol{v}$. It is conventional in fluid mechanics to work not with $\boldsymbol{w}$ but with twice it, the **vorticity**

$$\boldsymbol{\omega}=\boldsymbol{\nabla}\times\boldsymbol{v},\qquad\omega_i=\epsilon_{ijk}\,\partial_j v_k, \tag{2.12}$$

so that a fluid element rotates with angular velocity $\frac{1}{2}\boldsymbol{\omega}$.[6] A flow with $\boldsymbol{\omega}=\boldsymbol{0}$ everywhere is called *irrotational*; by Ch. 1 it then admits a velocity potential $\boldsymbol{v}=\boldsymbol{\nabla}\varphi$.

Example 2.6 | Vorticity is not "obvious" from the streamlines

Two circular flows illustrate that swirl and vorticity are distinct concepts (Fig. 2.5). *Solid-body rotation* $\boldsymbol{v}=\Omega\,(-x_2,\,x_1,\,0)$ has straight-line strain $\boldsymbol{D}=\boldsymbol{0}$ and uniform vorticity $\boldsymbol{\omega}=(0,0,2\Omega)$: every element spins. The *point vortex* $v_\theta=\Gamma/(2\pi r)$ (with $v_r=0$) has *circular* streamlines identical in shape, yet $\boldsymbol{\omega}=\boldsymbol{0}$ everywhere except at the origin: a small paddle wheel carried around the vortex translates on its circular orbit without spinning about its own axis. The circulation $\oint\boldsymbol{v}\cdot\mathrm{d}\boldsymbol{\ell}=\Gamma$ is nonetheless nonzero, all of it concentrated at the singular core. The velocity is irrotational for $r>0$, but the punctured plane $r>0$ is not simply connected, so the flow need not possess a globally single-valued velocity potential. Locally one may take $\varphi=(\Gamma/2\pi)\theta$, proportional to the polar angle; carried once around the core, φ increases by Γ, so no single-valued potential exists on the whole domain (Exercise 2.5).

[6] Terminology varies: a minority of texts call $\frac{1}{2}\boldsymbol{\nabla}\times\boldsymbol{v}$ the vorticity. We follow the majority (and Exercise 1.8) in defining $\boldsymbol{\omega}=\boldsymbol{\nabla}\times\boldsymbol{v}$, so that solid-body rotation at angular velocity $\boldsymbol{\Omega}$ has vorticity $\boldsymbol{\omega}=2\boldsymbol{\Omega}$.



(a) Solid-body rotation: element spins. **(b)** Point vortex: element keeps orientation.

Figure 2.5: Swirl versus vorticity, for two flows with *circular* streamlines. A small material cross (red) is carried once around the center. **(a)** In solid-body rotation the cross co-rotates with its orbit, the element spins, and the vorticity $\boldsymbol{\omega} = 2\boldsymbol{\Omega}$ is nonzero everywhere. **(b)** In the point vortex the cross keeps a fixed absolute orientation as it orbits, the element does not spin, and $\boldsymbol{\omega} = \mathbf{0}$ for all $r > 0$, even though the fluid plainly circulates. The vorticity is concentrated entirely at the core ($\times$).

2.6 The Meaning of Incompressibility

We can now state precisely what incompressibility means. By the volume theorem of Sec. 2.4, the local rate of volumetric dilation of a fluid element equals $\boldsymbol{\nabla}\cdot\boldsymbol{v}$. A motion that preserves the volume of every material element, an *isochoric* motion, therefore satisfies

Important | Incompressibility condition

$$\boldsymbol{\nabla}\cdot\boldsymbol{v} = 0, \qquad \text{equivalently} \qquad \partial_i v_i = 0.$$

A flow satisfying this is called *incompressible*; every material volume is conserved as it is carried along by the flow.

It is worth separating three ideas that are often conflated.

Kinematic meaning. As established above, this is purely kinematic, $\boldsymbol{\nabla}\cdot\boldsymbol{v} = 0$ preserves the volume of every material element, with no reference yet to mass or density.

Relation to density. The connection to density comes from the continuity equation, derived in Ch. 3 (Thm. 3.6) and quoted here only as a preview,

$$\frac{\mathrm{D}\rho}{\mathrm{D}t} + \rho\,\boldsymbol{\nabla}\cdot\boldsymbol{v} = 0.$$

If $\boldsymbol{\nabla}\cdot\boldsymbol{v} = 0$ then $\frac{\mathrm{D}\rho}{\mathrm{D}t} = 0$: each material particle keeps its own density as it moves, although different particles may carry different densities. If the fluid is in addition homogeneous, ρ is a single global constant. Incompressibility is thus a statement about the *flow*, not only about the fluid.

When incompressibility is a good approximation. Real fluids are, of course, compressible. In a low-Mach flow with only weak temperature, composition, reaction, and stratification effects, the *pressure*-induced fractional density changes are typically of order

$$\frac{\Delta\rho}{\rho} \sim \mathrm{Ma}^2, \qquad \mathrm{Ma} = \frac{U}{c},$$

with c the speed of sound; for water ($c \approx 1500\,\mathrm{m/s}$) and for air well below the sound speed ($\mathrm{Ma} \lesssim 0.3$) they are negligible and $\boldsymbol{\nabla}\cdot\boldsymbol{v} = 0$ holds to excellent accuracy. A small Mach number alone is not sufficient, however: substantial density variations can still be driven by temperature gradients, composition changes, chemical reaction, or stratification, and in such flows the incompressible model must be applied with care.

Kinematically, incompressibility has an immediate consequence: a divergence-free field is *solenoidal*, so it can be written as the curl of a vector potential, $\boldsymbol{v} = \boldsymbol{\nabla}\times\boldsymbol{A}$, automatically satisfying $\boldsymbol{\nabla}\cdot\boldsymbol{v} = 0$. In two dimensions this reduces to a single scalar *stream function* ψ with $\boldsymbol{v} = (\partial_2\psi,\, -\partial_1\psi,\, 0)$ (Exercise 2.7), a device we exploit repeatedly in Ch. 4.

Remark 2.7 | Incompressible $\neq$ constant density

It is worth stressing, following D. Tong,[a] that incompressibility is the statement $\frac{\mathrm{D}\rho}{\mathrm{D}t} = 0$, each fluid particle carries its own density unchanged, and *not* the stronger condition $\rho = \text{const}$ everywhere. The standard illustration is a stably *stratified* fluid, such as the ocean or the atmosphere, in which salinity or temperature makes the density vary with height ($\boldsymbol{\nabla}\rho \neq \boldsymbol{0}$); the flow may nonetheless be incompressible, $\boldsymbol{\nabla}\cdot\boldsymbol{v} = 0$, with each parcel preserving its own density as it moves. Uniform density is only the special case of a *homogeneous* incompressible fluid.

[a] D. Tong, *Lectures on Fluid Mechanics* (University of Cambridge).

Appendix 2.A The Velocity Gradient in Cartesian, Cylindrical, and Spherical Coordinates

Everything local about a flow is carried by the velocity gradient $\boldsymbol{L} = \boldsymbol{\nabla}\boldsymbol{v}$ of Eq. (2.4). Its components are collected here in the three coordinate systems used in these notes, for use in Ch. 4 and after. The convention is the one fixed in Eq. (2.1),

$$L_{ij} = \partial_j v_i,$$

so in each matrix below the *row* index names the component of $\boldsymbol{v}$ and the *column* index the direction along which it is differentiated. Several texts tabulate the transpose.[7] When comparing with another source, check the indices rather than the placement of $\boldsymbol{\nabla}$.

Cartesian Coordinates (x, y, z)

The basis vectors are the same at every point, so the components are plain partial derivatives,

$$\boldsymbol{L} = \begin{pmatrix} \frac{\partial v_x}{\partial x} & \frac{\partial v_x}{\partial y} & \frac{\partial v_x}{\partial z} \\ \frac{\partial v_y}{\partial x} & \frac{\partial v_y}{\partial y} & \frac{\partial v_y}{\partial z} \\ \frac{\partial v_z}{\partial x} & \frac{\partial v_z}{\partial y} & \frac{\partial v_z}{\partial z} \end{pmatrix}. \tag{2.13}$$

In the two curvilinear systems the basis rotates with position (Sec. 1.6), and differentiating $\boldsymbol{v}$ means differentiating its basis vectors as well as its components. Every algebraic term below, the ones without a derivative on them, comes from that and from nothing else.

Cylindrical Coordinates (r, θ, z)

$$\boldsymbol{L} = \begin{pmatrix} \frac{\partial v_r}{\partial r} & \frac{1}{r}\frac{\partial v_r}{\partial \theta} - \frac{v_\theta}{r} & \frac{\partial v_r}{\partial z} \\ \frac{\partial v_\theta}{\partial r} & \frac{1}{r}\frac{\partial v_\theta}{\partial \theta} + \frac{v_r}{r} & \frac{\partial v_\theta}{\partial z} \\ \frac{\partial v_z}{\partial r} & \frac{1}{r}\frac{\partial v_z}{\partial \theta} & \frac{\partial v_z}{\partial z} \end{pmatrix}. \tag{2.14}$$

The two extra terms are the ones people forget. The $-v_\theta/r$ in $L_{r\theta}$ and the $+v_r/r$ in $L_{\theta r}$ record that $\boldsymbol{e}_r$ and $\boldsymbol{e}_\theta$ turn into one another as θ advances, $\frac{\partial \boldsymbol{e}_r}{\partial \theta} = \boldsymbol{e}_\theta$ and $\frac{\partial \boldsymbol{e}_\theta}{\partial \theta} = -\boldsymbol{e}_r$. A rigid rotation $\boldsymbol{v} = \Omega r\, \boldsymbol{e}_\theta$ has no derivative of any component, yet it must have a nonzero velocity gradient; those two terms supply it.

Spherical Coordinates (r, θ, ϕ)

Here θ is the polar angle and ϕ the azimuthal angle.

$$\boldsymbol{L} = \begin{pmatrix} \frac{\partial v_r}{\partial r} & \frac{1}{r}\frac{\partial v_r}{\partial \theta} - \frac{v_\theta}{r} & \frac{1}{r\sin\theta}\frac{\partial v_r}{\partial \phi} - \frac{v_\phi}{r} \\ \frac{\partial v_\theta}{\partial r} & \frac{1}{r}\frac{\partial v_\theta}{\partial \theta} + \frac{v_r}{r} & \frac{1}{r\sin\theta}\frac{\partial v_\theta}{\partial \phi} - \frac{v_\phi \cot\theta}{r} \\ \frac{\partial v_\phi}{\partial r} & \frac{1}{r}\frac{\partial v_\phi}{\partial \theta} & \frac{1}{r\sin\theta}\frac{\partial v_\phi}{\partial \phi} + \frac{v_r}{r} + \frac{v_\theta \cot\theta}{r} \end{pmatrix}. \tag{2.15}$$

[7] R. B. Bird, W. E. Stewart, and E. N. Lightfoot, *Transport Phenomena*, 2nd ed. (Wiley, New York, 2002), list the same nine quantities as $(\boldsymbol{\nabla}\boldsymbol{v})_{ij} = \partial_i v_j$. The entries are identical; only the labelling of rows and columns is interchanged.

What Follows from L

In any of the three systems the rest of the chapter is read off from these matrices without further geometry. The rate of strain and the rotation tensor are the symmetric and antisymmetric parts Eq. (2.5),

$$\boldsymbol{D} = \tfrac{1}{2}(\boldsymbol{L} + \boldsymbol{L}^{\top}), \qquad \boldsymbol{W} = \tfrac{1}{2}(\boldsymbol{L} - \boldsymbol{L}^{\top}),$$

so, for instance, the cylindrical shear component used repeatedly in Ch. 4 is

$$D_{r\theta} = \tfrac{1}{2}(L_{r\theta} + L_{\theta r}) = \frac{r}{2}\frac{\partial}{\partial r}\left(\frac{v_\theta}{r}\right) + \frac{1}{2r}\frac{\partial v_r}{\partial \theta},$$

whose first term vanishes for rigid rotation $v_\theta \propto r$, as it must. The trace is the dilation rate, $\operatorname{tr}\boldsymbol{L} = \boldsymbol{\nabla}\cdot\boldsymbol{v}$, which reproduces the divergences of Appendix 1.A, and the vorticity is recovered from the antisymmetric part through $\boldsymbol{W}\cdot\Delta\boldsymbol{r} = \boldsymbol{w}\times\Delta\boldsymbol{r}$ with $\boldsymbol{w} = \tfrac{1}{2}\boldsymbol{\nabla}\times\boldsymbol{v}$.

Appendix 2.B Two Identities Used in the Text

Two results were quoted in this chapter without proof. Neither is difficult, but both are quoted so often elsewhere that it is easy to use them for years without ever seeing why they are true, and both say something about three-dimensional space that is worth knowing.

2.B.1 The Trace Identity for a Triple Product

The volume theorem Thm. 2.4 turned on the following claim.

Lemma 2.8 | Trace identity

For any second-rank tensor $\boldsymbol{L}$ and any vectors $\boldsymbol{a}, \boldsymbol{b}, \boldsymbol{c}$,

$$(\boldsymbol{L}\boldsymbol{a})\cdot(\boldsymbol{b}\times\boldsymbol{c}) + \boldsymbol{a}\cdot\big[(\boldsymbol{L}\boldsymbol{b})\times\boldsymbol{c}\big] + \boldsymbol{a}\cdot\big[\boldsymbol{b}\times(\boldsymbol{L}\boldsymbol{c})\big] = (\operatorname{tr}\boldsymbol{L})\,\boldsymbol{a}\cdot(\boldsymbol{b}\times\boldsymbol{c}). \tag{2.16}$$

Proof. Call the left-hand side $S(\boldsymbol{a},\boldsymbol{b},\boldsymbol{c})$. It is plainly linear in each of its three arguments, since $\boldsymbol{L}$ is linear and both products are. It is also *alternating*: if two of the three vectors coincide, S vanishes. Take $\boldsymbol{b} = \boldsymbol{c}$, for instance. The first term contains $\boldsymbol{b}\times\boldsymbol{b} = \boldsymbol{0}$; the remaining two are

$$\boldsymbol{a}\cdot\big[(\boldsymbol{L}\boldsymbol{b})\times\boldsymbol{b}\big] + \boldsymbol{a}\cdot\big[\boldsymbol{b}\times(\boldsymbol{L}\boldsymbol{b})\big] = 0,$$

because the two cross products are negatives of one another. The cases $\boldsymbol{a} = \boldsymbol{b}$ and $\boldsymbol{a} = \boldsymbol{c}$ follow in the same way.

Now, every alternating trilinear form on $\mathbb{R}^3$ is a scalar multiple of the determinant, that is, of the box product $\boldsymbol{a}\cdot(\boldsymbol{b}\times\boldsymbol{c})$ (Exercise 1.5(a)). There is nothing deep in this: an alternating form is fixed by its value on one basis, because expanding each argument in that basis and discarding every repeated-index term leaves exactly the six signed permutations that build the determinant. So $S=\lambda\,\boldsymbol{a}\cdot(\boldsymbol{b}\times\boldsymbol{c})$ for some scalar λ depending on $\boldsymbol{L}$ alone, and λ can be read off by evaluating S on the standard basis, where $\boldsymbol{e}_1\cdot(\boldsymbol{e}_2\times\boldsymbol{e}_3)=1$:

$$\begin{aligned}\lambda &= (\boldsymbol{L}\boldsymbol{e}_1)\cdot(\boldsymbol{e}_2\times\boldsymbol{e}_3)+\boldsymbol{e}_1\cdot\big[(\boldsymbol{L}\boldsymbol{e}_2)\times\boldsymbol{e}_3\big]+\boldsymbol{e}_1\cdot\big[\boldsymbol{e}_2\times(\boldsymbol{L}\boldsymbol{e}_3)\big]\\ &= (\boldsymbol{L}\boldsymbol{e}_1)\cdot\boldsymbol{e}_1+L_{22}+L_{33}=L_{11}+L_{22}+L_{33}=\operatorname{tr}\boldsymbol{L},\end{aligned}$$

the middle two terms following from $(\boldsymbol{L}\boldsymbol{e}_2)\times\boldsymbol{e}_3=L_{22}\,\boldsymbol{e}_1-L_{12}\,\boldsymbol{e}_2$ and $\boldsymbol{e}_2\times(\boldsymbol{L}\boldsymbol{e}_3)=L_{33}\,\boldsymbol{e}_1-L_{13}\,\boldsymbol{e}_3$, of which only the $\boldsymbol{e}_1$ components survive the dot product with $\boldsymbol{e}_1$. ∎

The trace appears here for a reason worth naming. The determinant of $\boldsymbol{I}+\varepsilon\boldsymbol{L}$ expands as

$$\det(\boldsymbol{I}+\varepsilon\boldsymbol{L})=1+\varepsilon\,\operatorname{tr}\boldsymbol{L}+\mathcal{O}\big(\varepsilon^2\big),$$

and Eq. (2.16) is precisely the $\mathcal{O}(\varepsilon)$ term of that expansion, written out on a triple of vectors. A determinant measures volume, so the trace is the first-order rate at which a linear map changes volume. That is the whole content of $\boldsymbol{\nabla}\cdot\boldsymbol{v}=\operatorname{tr}\boldsymbol{L}$: the theorem in the main text is this algebraic fact carried along by the flow, and the reader who knows Jacobi's formula for the derivative of a determinant will recognize it as the same statement.

2.B.2 The Axial Vector of an Antisymmetric Tensor

The decomposition $\boldsymbol{L}=\boldsymbol{D}+\boldsymbol{W}$ produced an antisymmetric $\boldsymbol{W}$, which Eq. (2.10) replaced by a vector. That replacement is always possible, and in exactly one way.

Lemma 2.9 | Axial vector

Let $\boldsymbol{W}$ be antisymmetric, $W_{ij}=-W_{ji}$. Then there is exactly one vector $\boldsymbol{w}$ with

$$\boldsymbol{W}\cdot\boldsymbol{z}=\boldsymbol{w}\times\boldsymbol{z}\qquad\text{for every }\boldsymbol{z},$$

namely $w_k=-\tfrac{1}{2}\epsilon_{kij}W_{ij}$, and equivalently $W_{ij}=-\epsilon_{ijk}w_k$.

Proof. *Existence.* Define $w_k=-\tfrac{1}{2}\epsilon_{kij}W_{ij}$ and verify that it does the job. Substituting and using the epsilon–delta identity Eq. (1.3),

$$-\epsilon_{ijk}w_k=\tfrac{1}{2}\,\epsilon_{ijk}\epsilon_{kpq}W_{pq}=\tfrac{1}{2}(\delta_{ip}\delta_{jq}-\delta_{iq}\delta_{jp})W_{pq}=\tfrac{1}{2}(W_{ij}-W_{ji})=W_{ij},$$

the last step being the only place where antisymmetry is used. Contracting this with $\boldsymbol{z}$,

$$(\boldsymbol{W}\cdot\boldsymbol{z})_i = W_{ij}z_j = -\epsilon_{ijk}w_k z_j = \epsilon_{ikj}w_k z_j = (\boldsymbol{w}\times\boldsymbol{z})_i.$$

Uniqueness. Suppose $\boldsymbol{w}$ and $\boldsymbol{w}'$ both satisfy the requirement. Subtracting, $(\boldsymbol{w}-\boldsymbol{w}')\times\boldsymbol{z} = \boldsymbol{0}$ for every $\boldsymbol{z}$. Choose $\boldsymbol{z}$ not parallel to $\boldsymbol{w}-\boldsymbol{w}'$; the cross product then has magnitude $|\boldsymbol{w}-\boldsymbol{w}'||\boldsymbol{z}|\sin\theta \neq 0$ unless $\boldsymbol{w}-\boldsymbol{w}' = \boldsymbol{0}$. ∎

Two remarks are in order. First, the counting is not an accident: an antisymmetric 3×3 tensor has 3 independent components and a vector in $\mathbb{R}^3$ has 3, so the correspondence is a bijection between spaces of equal dimension. In n dimensions an antisymmetric tensor has $n(n-1)/2$ components against the n of a vector, and the two agree only for $n=3$. This is why the vorticity is a vector in three dimensions, a single scalar in two, and neither in higher dimensions, and it is why the cross product is a three-dimensional accident. Second, $\boldsymbol{w}$ inherits the transformation behavior discussed in Rem. 1.6: because ϵ_{ijk} carries a sign under reflections, $\boldsymbol{w}$ is an *axial* vector, and so is the vorticity built from it.

Substituting $W_{ij} = \frac{1}{2}(\partial_j v_i - \partial_i v_j)$ from Eq. (2.6) into $w_k = -\frac{1}{2}\epsilon_{kij}W_{ij}$ and using the antisymmetry of ϵ twice gives $\boldsymbol{w} = \frac{1}{2}\boldsymbol{\nabla}\times\boldsymbol{v}$, which is Eq. (2.11); that short calculation is left as Exercise 2.4.

Exercises

Repeated indices are summed and fields are as smooth as needed. Problems and parts marked [Practice] are for self-study. Problems and parts marked [Advanced] are more challenging.

1. **(Anatomy of a flow)** For each velocity field below, compute the velocity gradient $\boldsymbol{L}$, the rate-of-strain tensor $\boldsymbol{D}$, the rotation tensor $\boldsymbol{W}$, the vorticity $\boldsymbol{\omega}$, and verify whether the flow is incompressible:

   (a) Pure straining, $\boldsymbol{v} = (\dot{\varepsilon}x_1,\ -\dot{\varepsilon}x_2,\ 0)$.

   (b) Solid-body rotation, $\boldsymbol{v} = \Omega(-x_2,\ x_1,\ 0)$.

   (c) Uniaxial extension, $\boldsymbol{v} = (\dot{\varepsilon}x_1,\ -\frac{1}{2}\dot{\varepsilon}x_2,\ -\frac{1}{2}\dot{\varepsilon}x_3)$.

2. **(Expansion of a material volume)** Show directly from the Leibniz rule that for a material volume $V(t)$,

$$\frac{\mathrm{d}}{\mathrm{d}t}\int_{V(t)}\mathrm{d}V = \int_{V(t)}\boldsymbol{\nabla}\cdot\boldsymbol{v}\,\mathrm{d}V,$$

and interpret the result. (This is the geometric content of the Reynolds transport theorem, Thm. 3.2 in Ch. 3.)

3. **(Shear strain rate as an angle change)** Consider two material fibers through a point, initially along $\boldsymbol{e}_1$ and $\boldsymbol{e}_2$. Using $\frac{\mathrm{d}}{\mathrm{d}t}(\Delta\boldsymbol{r}) = \boldsymbol{L}\cdot\Delta\boldsymbol{r}$ from Eq. (2.8), show that the rate at which the right angle between them decreases equals $2D_{12}$, so the off-diagonal component D_{12} is the tensorial shear-strain rate. Deduce more generally that the extreme values of the normal strain rate $\boldsymbol{m}\cdot\boldsymbol{D}\cdot\boldsymbol{m}$ over all unit directions $\boldsymbol{m}$ are the largest and smallest eigenvalues of $\boldsymbol{D}$.

4. **(Axial vector of the rotation tensor)** Verify $\epsilon_{ijm}\epsilon_{ijk} = 2\delta_{mk}$ and use it to invert $W_{ij} = -\epsilon_{ijk}w_k$ to $w_m = -\frac{1}{2}\epsilon_{mij}W_{ij}$. Substituting $W_{ij} = \frac{1}{2}(\partial_j v_i - \partial_i v_j)$, prove $\boldsymbol{w} = \frac{1}{2}\boldsymbol{\nabla}\times\boldsymbol{v}$, and hence $\boldsymbol{W}\cdot\boldsymbol{z} = \frac{1}{2}\boldsymbol{\omega}\times\boldsymbol{z}$.

5. **(Point vortex)** In cylindrical coordinates the point vortex has $v_\theta = \Gamma/(2\pi r)$, $v_r = v_z = 0$, for $r > 0$.

   (a) Show $\boldsymbol{\nabla}\cdot\boldsymbol{v} = 0$ and $\boldsymbol{\omega} = \boldsymbol{0}$ for all $r > 0$.

   (b) Compute the circulation $\oint_C \boldsymbol{v}\cdot\mathrm{d}\boldsymbol{\ell}$ on a circle of radius r centered on the axis, and reconcile the nonzero circulation with the vanishing vorticity using Stokes' theorem (Rem. 1.18).

   (c) The region $r > 0$ is not simply connected. Show that a velocity potential with $\boldsymbol{v} = \boldsymbol{\nabla}\varphi$ is given *locally* by $\varphi = (\Gamma/2\pi)\theta$, and explain why φ cannot be globally single-valued, its value increases by Γ on each circuit of the core. Relate this to the circulation found in Exercise 2.5(b).

6. **(Principal strain rates)** For a two-dimensional incompressible flow, show that $\operatorname{tr}\boldsymbol{D} = 0$ forces the two principal strain rates to be equal and opposite, $\pm\lambda$. Interpret geometrically.

7. **(Stream function)** For a two-dimensional incompressible flow, show that $\boldsymbol{\nabla}\cdot\boldsymbol{v} = 0$ guarantees the existence of a scalar $\psi(x_1, x_2)$ with $v_1 = \partial_2\psi$ and $v_2 = -\partial_1\psi$. Verify that lines of constant ψ are streamlines, and that ψ differs between two points by the volumetric flow rate between them.

8. [Practice] **(When is a flow incompressible?)** Assuming weak temperature, composition, reaction, and stratification effects, estimate the *pressure*-induced fractional density change $\Delta\rho/\rho \sim \mathrm{Ma}^2$ in the following cases:

(a) Water flowing at $2\,\mathrm{m/s}$ in a pipe, with $c \approx 1500\,\mathrm{m/s}$.

(b) Air in a room ventilation duct at $5\,\mathrm{m/s}$, with $c \approx 340\,\mathrm{m/s}$.

Comment on the validity of $\boldsymbol{\nabla}\cdot\boldsymbol{v} = 0$ in each case, and give one example of a flow in which the density could vary substantially *despite* a small Mach number.

9. [Advanced] **(Motion map and material volume)** Let the motion be $\boldsymbol{x} = \boldsymbol{\chi}(\boldsymbol{X}, t)$, with deformation gradient $\boldsymbol{F} = \partial\boldsymbol{x}/\partial\boldsymbol{X}$ and Jacobian $J = \det\boldsymbol{F} > 0$.

(a) Show that the velocity gradient is $\boldsymbol{L} = \dot{\boldsymbol{F}}\,\boldsymbol{F}^{-1}$, where the overdot denotes the material derivative.

(b) Using Jacobi's determinant identity $\dot{J} = J\,\mathrm{tr}(\dot{\boldsymbol{F}}\,\boldsymbol{F}^{-1})$, show that $\dot{J} = J\,\mathrm{tr}\,\boldsymbol{L} = J\,\boldsymbol{\nabla}\cdot\boldsymbol{v}$.

(c) Conclude that $\boldsymbol{\nabla}\cdot\boldsymbol{v} = 0$ if and only if $J \equiv 1$, i.e. every material volume is preserved.

(d) Explain the physical meaning of J as the local ratio of deformed to reference volume.

10. [Advanced] **(Objectivity and a rotating observer)** Two observers record positions related by $\boldsymbol{x}^{*} = \boldsymbol{Q}(t)\,\boldsymbol{x} + \boldsymbol{c}(t)$, with $\boldsymbol{Q}\boldsymbol{Q}^{\top} = \boldsymbol{I}$.

(a) Differentiate the relation to obtain the transformation of velocity.

(b) Show that the velocity gradient transforms as $\boldsymbol{L}^{*} = \boldsymbol{Q}\boldsymbol{L}\boldsymbol{Q}^{\top} + \dot{\boldsymbol{Q}}\boldsymbol{Q}^{\top}$.

(c) Show that $\dot{\boldsymbol{Q}}\boldsymbol{Q}^{\top}$ is antisymmetric.

(d) Deduce that $\boldsymbol{D}^{*} = \boldsymbol{Q}\boldsymbol{D}\boldsymbol{Q}^{\top}$ (so $\boldsymbol{D}$ is objective), while $\boldsymbol{W}^{*} = \boldsymbol{Q}\boldsymbol{W}\boldsymbol{Q}^{\top} + \dot{\boldsymbol{Q}}\boldsymbol{Q}^{\top}$.

(e) Apply the result to solid-body rotation, viewed first from the laboratory frame and then from a co-rotating frame.

(f) Explain why the deformation is observer-independent while the measured spin is not.

11. **(Finite simple shear)** For the shear flow $\boldsymbol{v} = (\dot{\gamma}\,y,\,0,\,0)$ with constant $\dot{\gamma}$:

(a) Solve for the particle trajectories.

(b) Obtain the motion map from the initial position (X, Y, Z).

(c) Compute $\boldsymbol{L}$, $\boldsymbol{D}$, $\boldsymbol{W}$, and the vorticity.

(d) Map an initially circular material curve $X^{2} + Y^{2} = a^{2}$ to its shape at time t.

(e) Show that the circle becomes an ellipse.

(f) Compare the orientation of the finite-strain ellipse with the instantaneous principal directions of $\boldsymbol{D}$ (which lie at 45° to the flow).

(g) Explain why the finite deformation is *not* obtained by treating $\boldsymbol{D}$ and $\boldsymbol{W}$ as two independently accumulated motions.

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3
Conservation Laws

Ch. 2 described the kinematics of fluid motion. We now ask what determines that motion. We now formulate the balance laws governing how mass, momentum, and energy are redistributed by transport across boundaries and sources within the fluid. Conservation of mass and momentum holds for every continuum, but these laws are not sufficient by themselves: the momentum balance contains the stress, but the balance laws do not specify the nature of the stress. That relation depends on the material, and supplying it is the role of a *constitutive relation*. Together with material properties and with initial and boundary conditions, the two make a transport phenomena problem well-posed:

$$\boxed{\begin{aligned}&\text{balance laws} \;+\; \text{constitutive relations} \;+\; \text{material properties}\\ &\qquad +\; \text{initial and boundary conditions} \;=\; \text{a formulated transport problem.}\end{aligned}}$$

We develop each of these ingredients sequentially in the sections below.[1]

This chapter develops the balances of mass, linear momentum, and angular momentum. We begin with the Reynolds transport theorem, which connects integral balances over a moving material region to local field equations. This will be our workhorse to derive the balance laws, and we begin by deriving the conservation of mass. Next, we introduce the stress tensor through Cauchy's hypothesis, and obtain the balance of linear momentum, also known as the Cauchy momentum equation or Cauchy's first law. Following this, we use the balance of angular momentum to show that the stress tensor is symmetric. Finally, the Newtonian constitutive relation closes the momentum balance and yields the Navier–Stokes equations, which are a special case of the Cauchy momentum equation. The Navier–Stokes equations express how forces govern fluid motion, providing the foundation for understanding flows from blood circulation to ocean currents and turbulence. Chapter Ch. 4 deals with solutions of the Navier-Stokes equations for a large number of elementary settings.

[1] Assembling these four ingredients specifies the model and its data. Whether the resulting problem has a solution, whether that solution is unique, and whether it depends continuously on the data are further questions, and in general hard ones; Sec. 3.10 touches on the most famous of these "open questions" and the controversy that is just a week old, as of Sep 14th 2026.

We next derive the mechanical-energy balance from the momentum equation. This balance is not an independent conservation law, but a consequence of momentum conservation that exposes an important concept known as *viscous dissipation*. The total- and internal-energy balances, which determine temperature, are developed in **??**. A more rigorous construction of the stress tensor, including Cauchy's tetrahedron argument and Noll's theorem, is given in an appendix.

Recall from Ch. 2 that the independent fields of a simple, single-component fluid are the density $\rho(\boldsymbol{x},t)$, velocity $\boldsymbol{v}(\boldsymbol{x},t)$, and temperature $T(\boldsymbol{x},t)$. Pressure is not a fourth independent field, since for a compressible fluid an equation of state[2] relates it to density and temperature:

$$p = p(\rho, T).$$

The strictly incompressible model admits no such relation, and there the pressure plays a different role altogether; we take this up in Sec. 3.7. Mass balance determines ρ, momentum balance determines $\boldsymbol{v}$, and energy balance determines T. Because the energy balance is deferred to **??**, temperature enters the mechanical problem here only as a prescribed or indirectly determined field. We use the notation best suited to each derivation and present the final results in both symbolic and indicial form.

3.1 The Reynolds Transport Theorem

Let $\phi(\boldsymbol{x},t)$ be a smooth density, the amount per unit volume of some extensive property. We shall need its integral over two kinds of moving region. A *control volume* $V_{\mathrm{CV}}(t)$ has a boundary $S_{\mathrm{CV}}(t)$ that moves with a velocity $\boldsymbol{w}(\boldsymbol{x},t)$ of our own choosing. A *material volume* $V(t)$ has a boundary $S(t)$ made of fluid particles, so that it moves with the local fluid velocity $\boldsymbol{v}$. Everything in this section follows from a single geometric observation about a moving surface.

In a time $\mathrm{d}t$ a surface element of area $\mathrm{d}A$ and outward normal $\boldsymbol{n}$, moving with velocity $\boldsymbol{w}$, sweeps out the signed volume

$$(\boldsymbol{w}\cdot\boldsymbol{n})\,\mathrm{d}A\,\mathrm{d}t, \tag{3.1}$$

positive where the surface advances outward and negative where it retreats. That is the whole geometric content of what follows.

[2]For an ideal gas, $p = \rho RT/M$, where R is the universal gas constant and M is the molar mass. A common approximation for a nearly incompressible liquid is $\rho = \rho_0\big[1 + (p-p_0)/K - \alpha(T-T_0)\big]$, where K is the isothermal bulk modulus and α is the thermal-expansion coefficient.

Theorem 3.1 | Transport identity for a moving volume

For a region $V_{\mathrm{CV}}(t)$ whose boundary moves with velocity $\boldsymbol{w}$,

$$\frac{d}{dt}\int_{V_{\mathrm{CV}}(t)} \phi \,\mathrm{d}V = \int_{V_{\mathrm{CV}}(t)} \frac{\partial \phi}{\partial t}\,\mathrm{d}V + \oint_{S_{\mathrm{CV}}(t)} \phi\, \boldsymbol{w}\cdot\boldsymbol{n}\,\mathrm{d}A.$$

Proof. Over an interval $\mathrm{d}t$ the integral changes for two reasons and no others.

The field changes at every point of the region already occupied, contributing

$$\int_{V_{\mathrm{CV}}(t)} \frac{\partial \phi}{\partial t}\,\mathrm{d}V\,\mathrm{d}t.$$

The region itself changes, because each surface element sweeps out the volume Eq. (3.1), and that new volume carries the local value of ϕ. This contributes

$$\oint_{S_{\mathrm{CV}}(t)} \phi\, \boldsymbol{w}\cdot\boldsymbol{n}\,\mathrm{d}A\,\mathrm{d}t.$$

Adding the two, dividing by $\mathrm{d}t$, and letting $\mathrm{d}t \to 0$ gives the result; the cross term, in which ϕ changes within the swept layer, is of order $\mathrm{d}t^2$ and does not survive. ∎

Applying the divergence theorem (Rem. 1.18) to the surface term puts this in local form,

$$\frac{d}{dt}\int_{V_{\mathrm{CV}}(t)} \phi \,\mathrm{d}V = \int_{V_{\mathrm{CV}}(t)} \left[\frac{\partial \phi}{\partial t} + \boldsymbol{\nabla}\cdot(\phi \boldsymbol{w})\right]\mathrm{d}V. \tag{3.2}$$

Nothing so far involves the fluid. The velocity $\boldsymbol{w}$ is ours to choose, and the identity is kinematics of a moving region.

A material volume is the special case in which the boundary is carried by the fluid, $\boldsymbol{w} = \boldsymbol{v}$.

Theorem 3.2 | Reynolds Transport Theorem

For a material volume $V(t)$ with boundary $S(t)$ and outward unit normal $\boldsymbol{n}$,

$$\frac{d}{dt}\int_{V(t)} \phi \,\mathrm{d}V = \int_{V(t)} \frac{\partial \phi}{\partial t}\,\mathrm{d}V + \oint_{S(t)} \phi\, \boldsymbol{v}\cdot\boldsymbol{n}\,\mathrm{d}S = \int_{V(t)} \left[\frac{\partial \phi}{\partial t} + \boldsymbol{\nabla}\cdot(\phi \boldsymbol{v})\right]\mathrm{d}V.$$

Proof. Set $\boldsymbol{w} = \boldsymbol{v}$ in Thm. 3.1 and Eq. (3.2). ∎

The theorem separates the rate of change into a local contribution and a boundary term.[3] Since the boundary is made of fluid particles, nothing crosses it, and the surface integral

[3] Named for O. Reynolds, who formulated such control-volume balances in *Papers on Mechanical and Physical Subjects*, Vol. 3 (Cambridge University Press, 1903).

is not a flux: it is the rate at which the moving boundary sweeps up ϕ, which is Eq. (3.1) integrated over the surface. The corollary below handles quantities carried per unit mass.

Remark 3.3 | Per-unit-mass form

If $\phi = \rho\,\psi$ with ψ a property per unit mass, then combining the transport theorem (Thm. 3.2) with conservation of mass (Sec. 3.2) gives the compact result

$$\frac{d}{dt}\int_{V(t)} \rho\,\psi \,\mathrm{d}V = \int_{V(t)} \rho\,\frac{\mathrm{D}\psi}{\mathrm{D}t}\,\mathrm{d}V.$$

All the boundary and dilation terms are absorbed by continuity. This is the form we use for momentum and energy (Exercise 1).

Remark 3.4 | Material systems versus control volumes

An engineer rarely follows a material volume; the region of interest is a pipe section or a vessel, chosen independently of the fluid. The two descriptions are related by writing Thm. 3.1 twice.

Fix an instant t and let the material volume $V(t)$ and the control volume $V_{\mathrm{CV}}(t)$ occupy the *same* region at that instant. Their geometries then agree, and the integrals $\int \frac{\partial \phi}{\partial t}\,\mathrm{d}V$ agree with them; only the boundary velocities differ, $\boldsymbol{v}$ against $\boldsymbol{w}$. Subtracting the two identities leaves only the surface terms,

$$\frac{d}{dt}\int_{V(t)} \phi\,\mathrm{d}V = \frac{d}{dt}\int_{V_{\mathrm{CV}}(t)} \phi\,\mathrm{d}V + \oint_{S_{\mathrm{CV}}(t)} \phi\,(\boldsymbol{v}-\boldsymbol{w})\cdot\boldsymbol{n}\,\mathrm{d}A,$$

or, labelled by what each term does,

$$\underbrace{\frac{d}{dt}\int_{V(t)} \phi\,\mathrm{d}V}_{\text{carried by the material}} = \underbrace{\frac{d}{dt}\int_{V_{\mathrm{CV}}(t)} \phi\,\mathrm{d}V}_{\text{accumulation inside the CV}} + \underbrace{\oint_{S_{\mathrm{CV}}(t)} \phi\,(\boldsymbol{v}-\boldsymbol{w})\cdot\boldsymbol{n}\,\mathrm{d}A}_{\text{net efflux across the CV}}.$$

The quantity $(\boldsymbol{v}-\boldsymbol{w})\cdot\boldsymbol{n}$ is the normal velocity of the material *relative to the moving control surface*, and it is the central idea of the theorem. Fluid crosses the surface at that speed and no other; a surface running alongside the flow collects nothing, however fast the fluid moves.

For an extensive property carried at ψ per unit mass, put $\phi = \rho\psi$:

$$\frac{d}{dt}\int_{V(t)} \rho\psi\,\mathrm{d}V = \frac{d}{dt}\int_{V_{\mathrm{CV}}(t)} \rho\psi\,\mathrm{d}V + \oint_{S_{\mathrm{CV}}(t)} \rho\psi\,(\boldsymbol{v}-\boldsymbol{w})\cdot\boldsymbol{n}\,\mathrm{d}A.$$

This is the form used for mass, momentum, and energy in the chapters that follow.

Two choices of $\boldsymbol{w}$ recover the familiar statements.

- $\boldsymbol{w} = \boldsymbol{v}$ (material volume). The relative flux vanishes identically and the control volume is the material volume, returning Thm. 3.2.
- $\boldsymbol{w} = \mathbf{0}$ (control volume fixed in the laboratory). The region is fixed, so the accumulation term may be written $\frac{\partial}{\partial t}\int_{V_{\mathrm{CV}}} \rho\psi \,\mathrm{d}V$, and

$$\frac{d}{dt}\int_{V(t)} \rho\psi \,\mathrm{d}V = \frac{\partial}{\partial t}\int_{V_{\mathrm{CV}}} \rho\psi \,\mathrm{d}V + \oint_{S_{\mathrm{CV}}} \rho\psi\, \boldsymbol{v}\cdot\boldsymbol{n}\,\mathrm{d}A.$$

An intermediate $\boldsymbol{w}$ describes a control volume that translates with a moving vehicle, or one that expands with an inflating balloon. These require no new theory, only the correct $\boldsymbol{w}$.

The theorem separates the change of a quantity carried by a material system into accumulation within the chosen control volume and transport through its boundary. When the control surface moves, the velocity that matters for that transport is not $\boldsymbol{v}$ but the relative velocity $\boldsymbol{v} - \boldsymbol{w}$.

A word on notation, once. We write $\mathrm{d}/\mathrm{d}t$ for the derivative of an integral over a moving region, whether that region is material or a control volume, since in both cases it is an ordinary derivative of a function of t alone. When the control volume is fixed the domain does not move, and $\frac{\partial}{\partial t}\int_{V_{\mathrm{CV}}}$ is then the natural way to write it. The material derivative

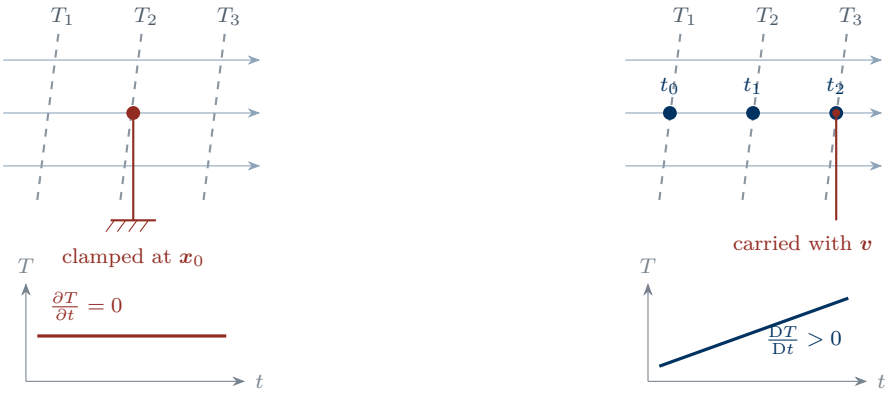
$$\frac{\mathrm{D}}{\mathrm{D}t} = \frac{\partial}{\partial t} + \boldsymbol{v}\cdot\boldsymbol{\nabla}$$

is reserved for local fields, as in Ch. 2, and is not used on the left of any of the integral statements above (Fig. 3.1).

One step recurs in every derivation that follows and is worth stating once. Each balance produces an integral statement of the form

$$\int_V f \,\mathrm{d}V = 0 \quad \text{for every admissible } V \qquad \Longrightarrow \qquad f = 0,$$

valid because a continuous integrand that did not vanish at some point would keep its sign on a small region about that point, and the integral over that region alone would then be nonzero. This is the *localization theorem*, and it is what turns a statement about bodies into a field equation. In what follows we simply write "localizing gives".



(a) Observer fixed in the laboratory.

(b) Observer carried with the fluid particle.

Figure 3.1: Two observers in the same steady temperature field, whose isotherms $T_1 < T_2 < T_3$ do not move. **(a)** A probe clamped at the fixed station $\boldsymbol{x}_0$ always samples the same isotherm, so its reading never changes and it reports $\frac{\partial T}{\partial t} = 0$. **(b)** A probe carried along with a fluid particle is swept across the isotherms, so its reading climbs and it reports $\frac{\mathrm{D}T}{\mathrm{D}t} \neq 0$. The two rates differ by exactly the transport of the particle through the spatial gradient, $\frac{\mathrm{D}T}{\mathrm{D}t} = \frac{\partial T}{\partial t} + \boldsymbol{v} \cdot \boldsymbol{\nabla} T$; here the first term vanishes and the whole signal is convective. A steady field is therefore not the same thing as an unchanging experience of that field.

Remark 3.5 | Anatomy of a local balance law

It is now useful to remark that every balance shares the same mathematical structure,

$$\underbrace{\frac{\partial(\text{density})}{\partial t}}_{\text{accumulation}} + \underbrace{\boldsymbol{\nabla}\cdot(\text{flux})}_{\text{net transport}} = \underbrace{(\text{source})}_{\text{production}},$$

differing only in what fills each slot:

balance	density	flux	source
mass	ρ	ρv_j	0
i-momentum	ρv_i	$\rho v_i v_j - T_{ij}$	ρb_i
total energy		developed in **??**	

The momentum flux carries two mechanisms: $\rho v_i v_j$ is *convective* transport (momentum carried bodily by the flow), while $-T_{ij}$ is *molecular* transport (momentum passed across surfaces by stress).

3.2 Conservation of Mass

The simplest balance is that of mass. A material volume can neither gain nor lose mass, since its boundary consists of the same particles at all times. For a material volume (Rem. 3.4)

$\frac{d}{dt}\int_{V(t)}\rho\,\mathrm{d}V = 0$. Applying the transport theorem (Thm. 3.2) with $\phi = \rho$ and localizing gives the first of our field equations.

Theorem 3.6 | Continuity equation

Conservation of mass is expressed locally as

$$\frac{\partial\rho}{\partial t} + \boldsymbol{\nabla}\cdot(\rho\boldsymbol{v}) = 0, \qquad \text{equivalently} \qquad \frac{\partial\rho}{\partial t} + \partial_i(\rho v_i) = 0.$$

In material form, using $\boldsymbol{\nabla}\cdot(\rho\boldsymbol{v}) = \boldsymbol{v}\cdot\boldsymbol{\nabla}\rho + \rho\boldsymbol{\nabla}\cdot\boldsymbol{v}$,

$$\frac{\mathrm{D}\rho}{\mathrm{D}t} + \rho\,\boldsymbol{\nabla}\cdot\boldsymbol{v} = 0.$$

For an incompressible flow, $\mathrm{D}\rho/\mathrm{D}t = 0$ and the continuity equation collapses to the purely kinematic constraint

$$\boldsymbol{\nabla}\cdot\boldsymbol{v} = 0, \qquad \partial_i v_i = 0,$$

exactly the isochoric condition of Sec. 2.6. Incompressibility and uniform density are distinct statements. Uniform density implies incompressibility, but the reverse is not true. Incompressibility is defined by $\boldsymbol{\nabla}\cdot\boldsymbol{v} = 0$, which implies $\mathrm{D}\rho/\mathrm{D}t = 0$: each material particle preserves *its own* density as it moves. Uniform density is $\boldsymbol{\nabla}\rho = \boldsymbol{0}$: all particles at one instant share the same density. A so-called stratified fluid can be modelled as incompressible while its density varies with position, so that $\mathrm{D}\rho/\mathrm{D}t = 0$ and $\boldsymbol{\nabla}\rho \neq \boldsymbol{0}$ together. A single constant density in space *and* time is an additional restriction beyond either, and we state it explicitly wherever it is used. The continuity equation is the first of our field equations; it advances ρ (or, when ρ is constant, constrains $\boldsymbol{v}$).

3.3 Linear Momentum: Forces, Traction, and Stress

We begin by choosing an arbitrary material volume $V(t)$, with boundary $S(t)$, and describe its motion in an inertial frame. Its linear momentum is

$$\boldsymbol{P}(t) = \int_{V(t)}\rho\boldsymbol{v}\,\mathrm{d}V.$$

The balance of linear momentum states that the rate of change of $\boldsymbol{P}$ equals the resultant external force,

$$\frac{\mathrm{d}\boldsymbol{P}}{\mathrm{d}t} = \frac{\mathrm{d}}{\mathrm{d}t}\int_{V(t)}\rho\boldsymbol{v}\,\mathrm{d}V = \boldsymbol{F}_{\mathrm{ext}} = \boldsymbol{F}_{\mathrm{b}} + \boldsymbol{F}_{\mathrm{s}},$$

the two subscripts distinguishing forces distributed through the volume from forces transmitted across its boundary. Nothing has yet been said about what either force is, which is our next question.

Two kinds of force act on a fluid body. *Body forces* act at a distance on the mass throughout the volume. Writing the body force per unit mass as $\boldsymbol{b}(\boldsymbol{x}, t)$, a volume element experiences $\mathrm{d}\boldsymbol{F}_{\mathrm{b}} = \rho \boldsymbol{b}\, \mathrm{d}V$, so

$$\boldsymbol{F}_{\mathrm{b}} = \int_{V(t)} \rho\, \boldsymbol{b}\, \mathrm{d}V.$$

One example of a body force is gravity, with $\boldsymbol{b} = \boldsymbol{g}$. By contrast, *surface forces* act across internal surfaces by short-range molecular interactions. They are described by the *traction* $\boldsymbol{t}_S$, the contact force per unit current area exerted on the material inside $V(t)$ by its surroundings, so that

$$\mathrm{d}\boldsymbol{F}_{\mathrm{s}} = \boldsymbol{t}_S\, \mathrm{d}S, \qquad \boldsymbol{F}_{\mathrm{s}} = \oint_{S(t)} \boldsymbol{t}_S\, \mathrm{d}S.$$

The subscript S allows, for the moment, a dependence on the surface itself and not merely on a direction. Body force per unit mass has the units of acceleration; traction has units of force per area. Assembling the pieces gives the integral balance of linear momentum,

$$\frac{\mathrm{d}}{\mathrm{d}t} \int_{V(t)} \rho \boldsymbol{v}\, \mathrm{d}V = \int_{V(t)} \rho\, \boldsymbol{b}\, \mathrm{d}V + \oint_{S(t)} \boldsymbol{t}_S\, \mathrm{d}S.$$

What qualifies as an external force? The forces on an imagined cut through a body are external to each of the two parts it separates, although internal to their union; which they are depends entirely on the sub-body one has chosen [KS: Need a Fig]. Everything that follows in this section is an attempt to describe $\boldsymbol{t}_S$ locally.

Traction can vary with the orientation of the cut, as a bar under tension already shows [KS: Fig]. This brings us to the hypothesis of Cauchy.

Definition 3.7 | Cauchy's stress hypothesis

The traction at a point depends on the surface there only through its unit normal:

$$\boldsymbol{t} = \boldsymbol{t}(\boldsymbol{x}, t; \boldsymbol{n}).$$

Thus, two surfaces with the same normal at that point therefore carry the same traction there, even if their shapes differ elsewhere.The hypothesis is actually more profound than it may seem on first glance. Infinitely many surfaces pass through a given point, and the claim is that the traction depends on none of their features except the normal direction $\boldsymbol{n}$ (Fig. 3.2). This is an assumption about how molecular interactions transmit force across a



(a) Two cuts through one point. **(b)** Components of $\boldsymbol{T}$.

Figure 3.2: The Cauchy stress hypothesis and the stress tensor. **(a)** Two different surfaces cut through the same material point $\boldsymbol{x}$. The traction depends on which cut is taken, $\boldsymbol{t}(\boldsymbol{n}) \neq \boldsymbol{t}(\boldsymbol{n}')$, but Cauchy's hypothesis (Def. 3.7) says it depends on the cut *only* through the unit normal, not through the curvature or extent of the surface. The tetrahedron argument of Appendix 3.A then forces that dependence to be linear. **(b)** Linearity means the tractions on the two coordinate faces already determine the traction on every other plane, and their components are the entries of $\boldsymbol{T}$. The first index of T_{ij} names the direction of the force, the second names the face it acts on: diagonal entries are normal stresses, off-diagonal entries are shears.

surface. What the hypothesis does is specify which geometric information is relevant: not the curvature of the cut, nor its extent away from the point, but the normal alone. It does not exclude the fact that the stress vector can vary along different points on the material.

A one-line momentum balance on a thin "pillbox" straddling a surface shows that traction obeys Newton's third law,

$$\boldsymbol{t}(\boldsymbol{x}, t; -\boldsymbol{n}) = -\boldsymbol{t}(\boldsymbol{x}, t; \boldsymbol{n}), \tag{3.3}$$

a result known as Cauchy's lemma (Exercise 6). However, Cauchy's greatest contribution to continuum mechanics is his tetrahedron argument (Appendix 3.A), is that the traction is *linear* in $\boldsymbol{n}$. This leads to:

Theorem 3.8 | Cauchy's stress theorem

There exists a second-rank tensor field $\boldsymbol{T}(\boldsymbol{x}, t)$, the **Cauchy stress tensor**, such that

$$\boldsymbol{t}(\boldsymbol{n}) = \boldsymbol{T} \cdot \boldsymbol{n}, \qquad \text{equivalently} \qquad t_i = T_{ij}\, n_j.$$

The component T_{ij} is the i-component of the traction on a surface whose normal points along $\boldsymbol{e}_j$. Its diagonal entries are normal stresses (tension/compression) and its off-diagonal entries are shear stresses. The same tensor is the *molecular flux of momentum*: $-T_{ij}$ is the outward flux of i-momentum in the j-direction, so momentum transport and stress are two descriptions of the same quantity. The minus sign is a matter of convention, in that a positive traction on the material inside a surface transfers momentum *inward*, so the *outward* flux carries the opposite sign.[4]

[4]The stress tensor and the tetrahedron construction are due to A.-L. Cauchy, "Recherches sur l'équilibre

Example 3.9 | Reading a two-dimensional stress tensor

Recall the index convention: T_{ij} is the i-component of the traction acting on a surface whose outward normal points in the j-direction. Take a two-dimensional stress tensor and a surface whose normal makes an angle θ with the x-axis,

$$\boldsymbol{T} = \begin{pmatrix} T_{11} & T_{12} \\ T_{21} & T_{22} \end{pmatrix}, \qquad \boldsymbol{n} = (\cos\theta,\, \sin\theta).$$

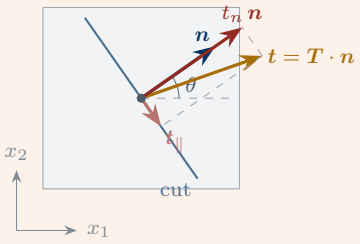
Cauchy's theorem gives the traction vector on that surface as $\boldsymbol{t} = \boldsymbol{T} \cdot \boldsymbol{n}$,

$$\boldsymbol{t} = \begin{pmatrix} T_{11}\cos\theta + T_{12}\sin\theta \\ T_{21}\cos\theta + T_{22}\sin\theta \end{pmatrix}.$$

Its projection onto $\boldsymbol{n}$ is the *normal stress*, and what remains is the *shear* (tangential) traction,

$$t_n = \boldsymbol{n} \cdot \boldsymbol{T} \cdot \boldsymbol{n}, \qquad \boldsymbol{t}_{\parallel} = \boldsymbol{T} \cdot \boldsymbol{n} - t_n\, \boldsymbol{n}.$$

The geometry is worth drawing once, because it is the picture behind every Mohr-circle construction:



The traction $\boldsymbol{t}$ generally points neither along the normal nor along the cut; resolving it into $t_n\,\boldsymbol{n}$ and $\boldsymbol{t}_{\parallel}$ separates the pull perpendicular to the surface from the drag along it. Sweeping $\boldsymbol{n}$ through all angles θ, the orientations for which $\boldsymbol{t}_{\parallel} = \boldsymbol{0}$ are the *principal directions* and the corresponding t_n the *principal stresses*, the eigenvectors and eigenvalues of $\boldsymbol{T}$ (Exercise 5).

3.4 Local and Control-Volume Forms of Momentum Balance

With the Cauchy stress theorem in hand we can now convert the linear momentum balance in Sec. 3.3 into a field equation. The left side, by the per-unit-mass transport theorem

et le mouvement intérieur des corps solides ou fluides," *Bulletin de la Société Philomathique* (1823), and *Exercices de Mathématiques* **2** (1827).

(Rem. 3.3), is $\int_V \rho\, \mathrm{D}\boldsymbol{v}/\mathrm{D}t\, \mathrm{d}V$. In the surface integral, substitute $t_i = T_{ij} n_j$ (Thm. 3.8) and apply the divergence theorem (Rem. 1.18):

$$\oint_S T_{ij} n_j \, \mathrm{d}S = \int_V \partial_j T_{ij} \, \mathrm{d}V.$$

Collecting everything under one integral,

$$\int_{V(t)} \left(\rho \frac{\mathrm{D}v_i}{\mathrm{D}t} - \rho\, b_i - \partial_j T_{ij} \right) \mathrm{d}V = 0 \qquad \text{for every material volume } V(t).$$

The integrand is continuous and the volume element is arbitrary, so localizing gives the Cauchy momentum equation.

Important | Cauchy momentum equation

$$\rho \frac{\mathrm{D}\boldsymbol{v}}{\mathrm{D}t} = \boldsymbol{\nabla}\cdot \boldsymbol{T} + \rho\, \boldsymbol{b}, \qquad \text{equivalently} \qquad \rho\left(\frac{\partial v_i}{\partial t} + (\partial_j v_i)\, v_j \right) = \partial_j T_{ij} + \rho\, b_i.$$

Here $(\boldsymbol{\nabla}\cdot \boldsymbol{T})_i = \partial_j T_{ij}$ is the divergence of the stress tensor.

This is the exact equation of motion for *any* continuous medium, fluid or solid. It is, however, not yet solvable. The balance determines what the stress must accomplish, but not how the stress is related to the motion. That relation is the content of a constitutive relation (Sec. 3.6), and it is where one material begins to differ from another.

The same balance can be written without the material derivative, in a form that displays momentum as a transported quantity. Start from the indicial equation of motion and add v_i times continuity, $\frac{\partial \rho}{\partial t} + \partial_j(\rho v_j) = 0$, which is identically zero:

$$\rho\left(\frac{\partial v_i}{\partial t} + v_j\, \partial_j v_i \right) = \partial_j T_{ij} + \rho\, b_i, \qquad v_i\left(\frac{\partial \rho}{\partial t} + \partial_j(\rho v_j) \right) = 0.$$

Adding them and collecting terms with the product rule, the conservative form follows:

$$\frac{\partial(\rho v_i)}{\partial t} + \partial_j(\rho v_i v_j) = \rho \frac{\mathrm{D}v_i}{\mathrm{D}t} + v_i \left[\frac{\partial \rho}{\partial t} + \partial_j(\rho v_j) \right] = \rho \frac{\mathrm{D}v_i}{\mathrm{D}t},$$

the bracket vanishing by continuity. Hence the *conservative* form of the momentum balance:

$$\frac{\partial(\rho v_i)}{\partial t} + \partial_j(\rho v_i v_j - T_{ij}) = \rho\, b_i, \qquad \text{equivalently} \qquad \frac{\partial(\rho \boldsymbol{v})}{\partial t} + \boldsymbol{\nabla}\cdot(\rho\, \boldsymbol{v} \otimes \boldsymbol{v} - \boldsymbol{T}) = \rho\, \boldsymbol{b}.$$

Here $\rho v_i v_j$ is the convective flux of i-momentum in the j-direction, momentum carried bodily by the flow, and $-T_{ij}$ is the contribution to the outward momentum flux due to contact between neighboring material elements. The equation now has the structure "accumulation

+ divergence of flux = source" shared by every balance law (Rem. 3.5), and it is the template for the unified scalar transport equation of **??**.

For a control volume $V_{\mathrm{CV}}(t)$ whose boundary moves with velocity $\boldsymbol{w}$, the same balance reads

$$\frac{\mathrm{d}}{\mathrm{d}t}\int_{V_{\mathrm{CV}}(t)} \rho\boldsymbol{v}\,\mathrm{d}V + \oint_{S_{\mathrm{CV}}(t)} \rho\,\boldsymbol{v}\,[(\boldsymbol{v}-\boldsymbol{w})\cdot\boldsymbol{n}]\,\mathrm{d}S = \int_{V_{\mathrm{CV}}(t)} \rho\,\boldsymbol{b}\,\mathrm{d}V + \oint_{S_{\mathrm{CV}}(t)} \boldsymbol{T}\cdot\boldsymbol{n}\,\mathrm{d}S.$$

The left side is momentum accumulation inside the control volume plus net outward transport relative to the moving control surface; the right side is the applied force. Setting $\boldsymbol{w}=\boldsymbol{0}$ gives the fixed control volume, the form used in most engineering textbooks (likely the one that you used in your undergrad).

3.5 Balance of Angular Momentum and Symmetry of the Stress

The balance of angular momentum comes from the physical law that external torque changes a system's angular momentum. In classical mechanics, this is an independent conservation principle, alongside conservation of linear momentum. Applying it to continuous medium, as we shall do here, gives us an important relation about the stress tensor T_{ij}, which may not be obvious at the outset. To carry out this balance, choose a fixed origin O in the same inertial frame and let $\boldsymbol{r}=\boldsymbol{x}-\boldsymbol{x}_O$. The angular momentum about O is

$$\boldsymbol{L}_O(t) = \int_{V(t)} \rho\,(\boldsymbol{r}\times\boldsymbol{v})\,\mathrm{d}V,$$

and its balance states that

$$\frac{\mathrm{d}\boldsymbol{L}_O}{\mathrm{d}t} = \boldsymbol{M}_O^{\mathrm{ext}} = \boldsymbol{M}_O^{\mathrm{b}} + \boldsymbol{M}_O^{\mathrm{s}},$$

where $\boldsymbol{M}_O^{\mathrm{ext}}$ is the net external torque, with $\boldsymbol{M}_O^{\mathrm{b}}$ denoting the torque due to the body forces, and $\boldsymbol{M}_O^{\mathrm{s}}$ denoting the torque due to the traction forces. We take a classical *nonpolar* continuum: no independent intrinsic angular momentum, no body couples, no couple stresses (which can be relevant in many interesting cases of contemporary research known as "active matter", but not relevant for this course).

The external torques are then nothing but the moments of the forces already introduced. A force element $\mathrm{d}\boldsymbol{F}$ has moment $\boldsymbol{r}\times\mathrm{d}\boldsymbol{F}$ about O, so

$$\boldsymbol{M}_O^{\mathrm{b}} = \int_{V(t)} \rho\,(\boldsymbol{r}\times\boldsymbol{b})\,\mathrm{d}V, \qquad \boldsymbol{M}_O^{\mathrm{s}} = \oint_{S(t)} \boldsymbol{r}\times\boldsymbol{t}(\boldsymbol{n})\,\mathrm{d}S.$$

Substituting gives the integral angular-momentum balance,

$$\frac{\mathrm{d}}{\mathrm{d}t}\int_{V(t)} \rho\,(\boldsymbol{r}\times\boldsymbol{v})\,\mathrm{d}V = \int_{V(t)} \rho\,(\boldsymbol{r}\times\boldsymbol{b})\,\mathrm{d}V + \oint_{S(t)} \boldsymbol{r}\times(\boldsymbol{T}\cdot\boldsymbol{n})\,\mathrm{d}S.$$

Taking O at the coordinate origin gives $\boldsymbol{r}=\boldsymbol{x}$, which we now do. The left side simplifies before any index work, since

$$\frac{\mathrm{D}}{\mathrm{D}t}(\boldsymbol{x}\times\boldsymbol{v}) = \boldsymbol{v}\times\boldsymbol{v} + \boldsymbol{x}\times\frac{\mathrm{D}\boldsymbol{v}}{\mathrm{D}t} = \boldsymbol{x}\times\frac{\mathrm{D}\boldsymbol{v}}{\mathrm{D}t}.$$

The manipulation is then cleanest in indices. Writing the i-component with $(\boldsymbol{x}\times\boldsymbol{t})_i = \epsilon_{ijk}x_j t_k = \epsilon_{ijk}x_j T_{k\ell}n_\ell$ (Thm. 3.8) and applying the divergence theorem (Rem. 1.18) to the surface term,

$$\oint_S \epsilon_{ijk}x_j T_{k\ell}n_\ell\,\mathrm{d}S = \int_V \epsilon_{ijk}\,\partial_\ell(x_j T_{k\ell})\,\mathrm{d}V = \int_V \epsilon_{ijk}(\delta_{j\ell}T_{k\ell} + x_j\partial_\ell T_{k\ell})\mathrm{d}V.$$

Using the per-unit-mass transport theorem (Rem. 3.3) on the left and collecting everything on one side,

$$\int_V \epsilon_{ijk}\,x_j\left(\rho\,\frac{\mathrm{D}v_k}{\mathrm{D}t} - \rho\,b_k - \partial_\ell T_{k\ell}\right)\mathrm{d}V = \int_V \epsilon_{ijk}\,T_{kj}\,\mathrm{d}V.$$

The bracket on the left vanishes identically by the balance of linear momentum, so

$$\int_V \epsilon_{ijk}\,T_{kj}\,\mathrm{d}V = 0.$$

Since V is arbitrary, $\epsilon_{ijk}T_{kj}=0$ pointwise. This contraction of the antisymmetric ϵ_{ijk} with T_{kj} vanishes precisely when T_{kj} is symmetric.

Theorem 3.10 | Symmetry of the stress tensor

For a classical Cauchy continuum, one with no body couples and no couple stresses, the balance of angular momentum requires the Cauchy stress to be symmetric:

$$\boldsymbol{T} = \boldsymbol{T}^{\top}, \qquad T_{ij} = T_{ji}.$$

Action and reaction across a cut and symmetry of the stress are distinct results, and it is worth keeping them apart. The first, Cauchy's lemma, relates the tractions on the two sides of *one* surface and follows from linear momentum balance alone. The second relates components acting on *different* coordinate faces, and follows from angular momentum balance under the nonpolar assumptions stated above. Again, aeither implies the other.

Example 3.11 | Why $\epsilon_{ijk}T_{kj} = 0$ forces symmetry

The statement $\epsilon_{ijk}T_{kj} = 0$ carries one free index, so it is really three equations, one for each value of i. Work out a single component and the pattern is plain.

Take $i = 3$. The symbol ϵ_{3jk} is nonzero only for $(j,k) = (1,2)$ and $(j,k) = (2,1)$, where it equals $+1$ and -1 respectively, so the sum collapses to two terms:

$$\epsilon_{3jk}T_{kj} = \epsilon_{312}\,T_{21} + \epsilon_{321}\,T_{12} = T_{21} - T_{12} = 0 \qquad \Longrightarrow \qquad T_{12} = T_{21}.$$

Nothing was special about the choice $i = 3$. Cycling the indices $1 \to 2 \to 3 \to 1$ repeats the computation verbatim:

$$i = 1: \quad T_{32} - T_{23} = 0, \qquad i = 2: \quad T_{13} - T_{31} = 0.$$

Those three equations exhaust the free index, and together they equate every off-diagonal pair, which is exactly $\boldsymbol{T} = \boldsymbol{T}^{\top}$. The diagonal entries are never constrained, and never need to be, since $T_{ii} = T_{ii}$ already.

The converse is immediate and worth noting: if $\boldsymbol{T}$ is symmetric, then $\epsilon_{ijk}T_{kj}$ contracts an object antisymmetric in (j,k) with one symmetric in (j,k), and any such contraction vanishes. So the vanishing of $\epsilon_{ijk}T_{kj}$ is not merely implied by symmetry. It is equivalent to it.

The stress tensor therefore has only six independent components, not nine. The origin of this result is worth noting. Symmetry follows from a balance law rather than from any assumption about the material, and it therefore holds for solids and fluids alike, so long as there is no intrinsic source of angular momentum or additional couple stresses. It also sits well with the kinematics of Ch. 2: only the symmetric part $\boldsymbol{D}$ of the velocity gradient produces deformation, so a constitutive relation built on $\boldsymbol{D}$ satisfies $\boldsymbol{T} = \boldsymbol{T}^{\top}$ automatically.

3.6 Constitutive Relations

The balance laws are universal, but they are not sufficient to fully specify a transport problem. The momentum equation contains the stress tensor $\boldsymbol{T}$, and no balance law says how that stress is related to the kinematics (i.e. motion). The missing link is a *constitutive relation*, which, unlike the balance laws, describes a particular material rather than matter in general. Stresses in different materials may obey different constitutive relationships.

We build the relation in stages, and the order matters. First we ask what the stress can possibly be in a fluid *at rest*, which turns out to be settled by the definition of a fluid alone and costs us nothing. That case delivers hydrostatics. Only then do we set the fluid

in motion, introduce the viscous stress, and ask what it may depend on. Each step adds exactly one ingredient, and it is worth keeping track of which ingredient is a consequence of a balance law, which is a requirement on any constitutive relation, and which is an assumption about the material in front of us.

The Fluid at Rest

What distinguishes a fluid from a solid is not softness but the response to shear. A solid, sheared gently, deforms by a finite amount and stops; it can hold a static shear stress indefinitely. A fluid cannot. Subject a fluid to a tangential traction, however small, and it deforms continuously for as long as that traction acts. This is the defining property, and everything in this subsection follows from it.

Definition 3.12 | Fluid

A *fluid* is a material that deforms continuously under any applied shear stress, however small. Equivalently, a fluid at rest can sustain no tangential traction.

Take the second form and see what it forces. If the fluid is at rest, the traction on every surface element must be purely normal: there is no tangential component left to balance, or the element would still be deforming and the fluid would not be at rest. So for every unit normal $\boldsymbol{n}$ there is a scalar $\alpha(\boldsymbol{n})$ with

$$\boldsymbol{t}(\boldsymbol{n}) = \boldsymbol{T}_{\mathrm{eq}}\,\boldsymbol{n} = \alpha(\boldsymbol{n})\,\boldsymbol{n}.$$

Read this as an eigenvalue problem. Every direction $\boldsymbol{n}$ whatsoever is an eigenvector of $\boldsymbol{T}_{\mathrm{eq}}$. A symmetric tensor with three independent eigenvectors sharing no common eigenvalue would fail this at once, so the eigenvalues must all coincide: if $\boldsymbol{T}_{\mathrm{eq}}\boldsymbol{e}_1 = \alpha_1\boldsymbol{e}_1$ and $\boldsymbol{T}_{\mathrm{eq}}\boldsymbol{e}_2 = \alpha_2\boldsymbol{e}_2$, then applying $\boldsymbol{T}_{\mathrm{eq}}$ to $(\boldsymbol{e}_1+\boldsymbol{e}_2)/\sqrt{2}$, itself an eigenvector by hypothesis, gives $\alpha_1\boldsymbol{e}_1 + \alpha_2\boldsymbol{e}_2 = \alpha(\boldsymbol{e}_1+\boldsymbol{e}_2)$, whence $\alpha_1 = \alpha_2 = \alpha$. The common eigenvalue is the same in every direction, so

$$\boldsymbol{T}_{\mathrm{eq}} = -p\,\boldsymbol{I}, \qquad (T_{\mathrm{eq}})_{ij} = -p\,\delta_{ij},$$

where the scalar p is the *hydrostatic pressure* and the minus sign is the usual convention: a positive p pushes inward on the material, since the traction on a surface of outward normal $\boldsymbol{n}$ is then $-p\,\boldsymbol{n}$. Notice what has been established and what has not. That the equilibrium stress is isotropic is a theorem, resting on the definition of a fluid; the value of p is not, and must come from an equation of state or, in the incompressible case, from the flow itself.

Hydrostatics and the Barometric Formula

With the equilibrium stress in hand we can already solve a class of problems. Put $\boldsymbol{v} = \boldsymbol{0}$ in Cauchy's first law. The acceleration vanishes, and with $\boldsymbol{T} = -p\boldsymbol{I}$ the divergence of the stress is $\partial_j(-p\,\delta_{ij}) = -\partial_i p$, so

$$\boldsymbol{0} = -\boldsymbol{\nabla} p + \rho\,\boldsymbol{b}, \qquad \text{that is} \qquad \boldsymbol{\nabla} p = \rho\,\boldsymbol{b}.$$

This is the equation of hydrostatics. It says that pressure varies only along the body force: surfaces of constant pressure are everywhere perpendicular to $\boldsymbol{b}$, and in a fluid at rest under gravity they are horizontal planes. It also says, on taking the curl, that $\boldsymbol{\nabla}\rho \times \boldsymbol{b} = \boldsymbol{0}$, so a fluid at rest under gravity cannot carry a horizontal density gradient.

Take gravity, $\boldsymbol{b} = -g\,\boldsymbol{e}_z$ with z measured upward and g constant. Only the vertical component survives,

$$\frac{\mathrm{d}p}{\mathrm{d}z} = -\rho g.$$

If the fluid is a liquid, ρ is very nearly constant and the integration is immediate,

$$p(z) = p_0 - \rho g z,$$

the familiar linear increase of pressure with depth that underlies manometry and buoyancy. If instead the fluid is a gas, ρ is not constant, and we need an equation of state. For an ideal gas of molar mass M at uniform temperature T, $\rho = pM/(RT)$, and the hydrostatic equation becomes separable,

$$\frac{\mathrm{d}p}{\mathrm{d}z} = -\frac{Mg}{RT}\,p \qquad \Longrightarrow \qquad p(z) = p_0\,\exp\!\left(-\frac{z}{H}\right), \qquad H = \frac{RT}{Mg},$$

which is the *barometric formula*. The pressure falls exponentially with height rather than linearly, and the decay length H is the *scale height*. For air at $288\,\mathrm{K}$, with $M = 0.029\,\mathrm{kg\,mol^{-1}}$, one finds $H \approx 8.4\,\mathrm{km}$, which is why atmospheric pressure at the summit of Everest is about a third of its sea-level value. The isothermal assumption is of course crude for the real atmosphere, whose temperature falls with altitude through the troposphere, but the exponential character of the result survives.

The Moving Fluid and the Viscous Stress

Set the fluid in motion and tangential tractions appear at once: this is the everyday observation that stirring a fluid requires sustained effort. The stress can no longer be isotropic. We therefore write the stress as the equilibrium part already found plus whatever

the motion adds,

$$\boldsymbol{T} = -p\,\boldsymbol{I} + \boldsymbol{\tau}, \qquad T_{ij} = -p\,\delta_{ij} + \tau_{ij},$$

where $\boldsymbol{\tau}$ is the *viscous stress*. The split is a definition, not yet a model, and it carries one condition: $\boldsymbol{\tau}$ must vanish when the fluid is at rest, since otherwise we would contradict the result of the previous subsection. All the physics of a flowing fluid, and all the difficulty, is now in $\boldsymbol{\tau}$.

The Simple Fluid

What may $\boldsymbol{\tau}$ depend on? The modelling assumption we adopt is the simplest one that is not trivial: that the viscous stress at a point and instant is determined by the *current, local velocity gradient* at that same point and instant,

$$\boldsymbol{\tau} = \boldsymbol{\tau}(\boldsymbol{L}), \qquad \boldsymbol{L} = \boldsymbol{\nabla v}, \qquad L_{ij} = \partial_j v_i.$$

A material obeying such a law is called a *simple* fluid, and it is worth being explicit about the three separate restrictions hidden in that one line.

First, the stress depends on $\boldsymbol{L}$ and not on $\boldsymbol{v}$ itself. This is not an approximation but a requirement. A uniform velocity is not a property of the material at all: two observers in relative uniform motion assign different values to $\boldsymbol{v}$ at the same material point, and a fluid cannot begin to exert stress merely because we choose to run past it. Only derivatives of the velocity survive a change of observer, and $\boldsymbol{L}$ is the lowest such derivative. The same argument rules out any dependence on position or on time explicitly.

Second, the dependence is *local*: $\boldsymbol{\tau}$ at $\boldsymbol{x}$ is set by $\boldsymbol{L}$ at $\boldsymbol{x}$, and not by the velocity gradient a short distance away. Third, and most restrictive, it is *instantaneous*: the fluid has no memory. The stress now depends on how the fluid is deforming now, not on how it was deformed a moment ago.

It is exactly this last assumption that fails for a large and industrially important class of materials. Polymer melts and solutions, and viscoelastic media generally, carry long molecules whose configurations take a finite time to relax. Such a fluid remembers its recent deformation history, and its stress depends on that history through a relaxation time λ_{r}. Stop shearing a polymer solution and the stress decays over λ_{r} rather than vanishing at once; shear it fast enough that $\lambda_{\mathrm{r}}\dot{\gamma} \gtrsim 1$ and it develops normal-stress differences, climbing a rotating rod instead of being flung from it. None of this is describable by $\boldsymbol{\tau} = \boldsymbol{\tau}(\boldsymbol{L})$, and the constitutive models that do describe it carry their own chapter of the subject. We set them aside and keep the simple fluid.

Objectivity: Why the Stress Cannot Feel the Spin

We have narrowed $\boldsymbol{\tau}$ to a function of $\boldsymbol{L} = \boldsymbol{D} + \boldsymbol{W}$. The next restriction removes the spin $\boldsymbol{W}$ entirely, and it comes not from any experiment but from the requirement of *frame indifference*, or *objectivity*: the response of a material cannot depend on the observer who describes it. Some care is needed over what *sameness* means here, since the two observers do not in general assign the same components. Two observers whose frames translate and rotate relative to one another assign the same event the positions

$$\boldsymbol{x}^* = \boldsymbol{Q}(t) \cdot \boldsymbol{x} + \boldsymbol{c}(t), \qquad \boldsymbol{Q}^\top \cdot \boldsymbol{Q} = \boldsymbol{I},$$

where $\boldsymbol{Q}(t)$ is a time-dependent rotation between the two frames and $\boldsymbol{c}(t)$ is the moving offset between their origins. A tensor is *objective*, or frame-indifferent, when the two observers describe the same physical tensor through the corresponding rotation of components. For the stress this requires that if observer 1 sees $\boldsymbol{T}$, observer 2 sees

$$\boldsymbol{T}^* = \boldsymbol{Q} \cdot \boldsymbol{T} \cdot \boldsymbol{Q}^\top.$$

The requirement is forced rather than chosen, because the traction on a given surface is a physical fact on which both observers must agree. If observer 1 finds a traction $\boldsymbol{t} = \boldsymbol{T} \cdot \boldsymbol{n}$ on a surface of normal $\boldsymbol{n}$, observer 2 sees the *same* force and normal re-expressed in their frame, $\boldsymbol{Q}\boldsymbol{t}$ and $\boldsymbol{Q}\boldsymbol{n}$, so we need $\boldsymbol{Q}\boldsymbol{t} = \boldsymbol{T}^* \cdot (\boldsymbol{Q}\boldsymbol{n})$ for every $\boldsymbol{n}$, which gives $\boldsymbol{T}^* = \boldsymbol{Q} \cdot \boldsymbol{T} \cdot \boldsymbol{Q}^\top$. A relation violating this condition would predict different forces on the same surface for different observers.

We now clarify a common confusion. It must be noted that frame indifference is a condition on the *material response*, and it should NOT be read as invariance of the equations of motion under arbitrary observers. A rotating or accelerating frame introduces centrifugal and Coriolis terms, which appear in the momentum balance and must be retained there. Objectivity constrains the constitutive relation, not the equation of motion.

Now apply this to $\boldsymbol{L}$. The two parts of the velocity gradient transform quite differently, as Appendix 3.B derives in full:

$$\boldsymbol{D}^* = \boldsymbol{Q}\,\boldsymbol{D}\,\boldsymbol{Q}^\top, \qquad \boldsymbol{W}^* = \boldsymbol{Q}\,\boldsymbol{W}\,\boldsymbol{Q}^\top + \dot{\boldsymbol{Q}}\,\boldsymbol{Q}^\top.$$

The rate of strain is objective: it merely co-rotates with the observer. The spin is not, because it picks up $\dot{\boldsymbol{Q}}\boldsymbol{Q}^\top$, the angular velocity of the observer's own frame. This is enough to eliminate $\boldsymbol{W}$ altogether. Fix an instant t_0 and choose the observer whose rotation satisfies $\boldsymbol{Q}(t_0) = \boldsymbol{I}$ and $\dot{\boldsymbol{Q}}(t_0) = -\boldsymbol{W}$; since $\boldsymbol{W}$ is antisymmetric such a rotation exists, for instance $\boldsymbol{Q}(t) = \exp[-(t - t_0)\boldsymbol{W}]$. At that instant $\boldsymbol{D}^* = \boldsymbol{D}$ while $\boldsymbol{W}^* = \boldsymbol{W} - \boldsymbol{W} = \boldsymbol{0}$, and objectivity

of the response demands $\boldsymbol{\tau}^{*} = \boldsymbol{Q}\boldsymbol{\tau}\boldsymbol{Q}^{\top} = \boldsymbol{\tau}$. Therefore

$$\boldsymbol{\tau}(\boldsymbol{D}, \boldsymbol{W}) = \boldsymbol{\tau}(\boldsymbol{D}, \boldsymbol{0}) \quad \text{for every } \boldsymbol{D} \text{ and } \boldsymbol{W},$$

and the viscous stress can depend on the rate of strain alone, $\boldsymbol{\tau} = \boldsymbol{\tau}(\boldsymbol{D})$.

If all of this is still confusing, worry not. Just think of the physical picture. Consider a bucket of water rotating steadily, so that the liquid turns with it as a rigid body. Then

$$\boldsymbol{D} = \boldsymbol{0}, \qquad \boldsymbol{W} \neq \boldsymbol{0},$$

and although the fluid is in motion, no element of it is being stretched or sheared. A rigid-body rotation changes orientation but not shape, and it should therefore produce no viscous stress in a simple Newtonian fluid.

Isotropy and the Viscosity Tensor

We now make the final modelling assumption: that $\boldsymbol{\tau}$ depends on $\boldsymbol{D}$ *linearly*. The most general linear map from one second-rank tensor to another is a fourth-rank tensor, which we call the *viscosity tensor* $\boldsymbol{C}$,

$$\tau_{ij} = C_{ijkl}\, D_{kl}. \tag{3.4}$$

An unrestricted fourth-rank array in three dimensions has $3^4 = 81$ components, which is hardly a model at all. What cuts it down is *isotropy*.

Isotropy is an assumption about the material, and it must be kept firmly distinct from the objectivity requirement just used. Objectivity concerns a change of *observer* and is demanded of every constitutive relation whatsoever. Isotropy concerns a rotation of the *material* relative to itself, and is a property a given material may or may not have: it says the fluid has no built-in preferred direction, so that rotating the sample before shearing it changes nothing. Water is isotropic; wood, a liquid crystal, and a fibre suspension with aligned fibres are not.

Formally, $\boldsymbol{C}$ is isotropic when its components are unchanged by every rotation of the basis. Which tensors have that property? In three dimensions the answer is remarkably restrictive: up to scalar multiples the only isotropic tensors are built from the Kronecker delta and the Levi-Civita symbol,

$$\boldsymbol{\delta} = \delta_{ij}\, \boldsymbol{e}_i \otimes \boldsymbol{e}_j = \boldsymbol{e}_i \otimes \boldsymbol{e}_i, \qquad \boldsymbol{\epsilon} = \epsilon_{ijk}\, \boldsymbol{e}_i \otimes \boldsymbol{e}_j \otimes \boldsymbol{e}_k,$$

the first a dyad and the second a triad. There is no isotropic tensor of rank one, and at rank two and three these are the only ones. At rank four the isotropic tensors are exactly the

linear combinations of the three ways of pairing four indices with two deltas,

$$C_{ijkl} = a\,\delta_{ij}\delta_{kl} + b\,\delta_{ik}\delta_{jl} + c\,\delta_{il}\delta_{jk},$$

with a, b, c scalars. The Levi-Civita symbol supplies nothing new here, because a product of two of them collapses back onto deltas through the identity $\epsilon_{ijm}\epsilon_{klm} = \delta_{ik}\delta_{jl} - \delta_{il}\delta_{jk}$, which already lies in the span above. So isotropy has taken us from 81 constants to three, and the theorems licensing this reduction are known as *representation theorems*.

Contract this $\boldsymbol{C}$ with a *symmetric* $\boldsymbol{D}$, writing $\theta := D_{kk} = \boldsymbol{\nabla}\cdot\boldsymbol{v}$ for the rate of dilation:

$$\tau_{ij} = a\,\delta_{ij}D_{kk} + b\,D_{ij} + c\,D_{ji} = a\,\theta\,\delta_{ij} + (b+c)\,D_{ij}.$$

Three constants have become two. Only the *sum* $b+c$ can ever be observed, because $\delta_{ik}\delta_{jl}$ and $\delta_{il}\delta_{jk}$ act identically on a symmetric argument; no experiment can separate b from c, and one may as well take $b = c$.

What the Two Constants Are

Take first a simple shear, $\boldsymbol{v} = \dot\gamma\, y\, \boldsymbol{e}_x$, for which the only nonzero rate-of-strain components are $D_{xy} = D_{yx} = \dot\gamma/2$ and the dilation θ vanishes. The relation above gives $\tau_{xy} = (b+c)\dot\gamma/2$. The measured law for a Newtonian fluid, the one Newton wrote down and every viscometer confirms, is $\tau_{xy} = \mu\,\mathrm{d}v_x/\mathrm{d}y = \mu\dot\gamma$, with μ the **shear viscosity**. Comparing,

$$b + c = 2\mu.$$

Take next a pure uniform dilation, $\boldsymbol{v} = \alpha\,\boldsymbol{x}$, in which every element expands without changing shape. Here $\boldsymbol{D} = \alpha\boldsymbol{I}$ and $\theta = 3\alpha$, and the isotropic term is the one that responds. Writing the coefficient of $\theta\,\boldsymbol{I}$ as η, the **bulk viscosity**,

$$a = \eta.$$

So the viscous stress of an isotropic, linear, simple fluid is fixed by exactly two material constants.

Important | Newtonian constitutive relation

For an isotropic Newtonian fluid,

$$\boldsymbol{\tau} = 2\mu\,\boldsymbol{D} + \eta\,(\boldsymbol{\nabla}\cdot\boldsymbol{v})\,\boldsymbol{I}, \qquad \tau_{ij} = 2\mu\,D_{ij} + \eta\,D_{kk}\,\delta_{ij},$$

and the total stress is

$$\boldsymbol{T} = -p\,\boldsymbol{I} + 2\mu\,\boldsymbol{D} + \eta\,(\boldsymbol{\nabla}\cdot\boldsymbol{v})\,\boldsymbol{I}.$$

Here μ is the shear viscosity and η the bulk viscosity.

The two constants μ and η measure resistance to the two things that can happen to a fluid element. The shear viscosity μ resists *change of shape*: it is the constant of proportionality between a tangential traction and the rate at which the element is sheared, and it is what makes honey harder to stir than water. Its SI unit is the pascal second, and for water at room temperature $\mu \approx 10^{-3}\,\mathrm{Pa\,s}$, for air about $1.8\times10^{-5}\,\mathrm{Pa\,s}$, for glycerol roughly $1\,\mathrm{Pa\,s}$. The bulk viscosity η resists *change of volume*: it is the extra pressure, over and above the thermodynamic p, required to compress or expand the fluid at a finite rate. Because it multiplies $\boldsymbol{\nabla}\cdot\boldsymbol{v}$, it does nothing whatever in an incompressible flow, and it is measurable only in situations involving rapid compression, such as the absorption of sound.[5] G. G. Stokes proposed setting $\eta = 0$ outright. Kinetic theory supports this only as an approximation for a dilute monatomic gas; polyatomic gases and dense fluids have internal degrees of freedom that relax on a finite time, and for them $\eta > 0$, sometimes comparable with μ. In incompressible flow the question is moot.[6]

3.7 The Navier–Stokes Equations

Everything is now in place. Substitute the Newtonian stress into Cauchy's first law, $\rho\,\mathrm{D}\boldsymbol{v}/\mathrm{D}t = \boldsymbol{\nabla}\cdot\boldsymbol{T} + \rho\boldsymbol{b}$, and take the divergence term by term, holding μ and η constant. The pressure contributes $\partial_j(-p\,\delta_{ij}) = -\partial_i p$. The shear term contributes

$$\partial_j\left(2\mu D_{ij}\right) = \mu\,\partial_j\left(\partial_j v_i + \partial_i v_j\right) = \mu\left(\partial_j\partial_j v_i + \partial_i\theta\right),$$

where the second equality used the interchange of partial derivatives to write $\partial_j\partial_i v_j = \partial_i(\partial_j v_j) = \partial_i\theta$. The bulk term contributes $\partial_j(\eta\theta\,\delta_{ij}) = \eta\,\partial_i\theta$. Collecting,

$$\partial_j T_{ij} = -\partial_i p + \mu\,\partial_j\partial_j v_i + (\mu+\eta)\,\partial_i\theta,$$

[5] With the grouping used here the mean viscous normal stress is $\tfrac{1}{3}\tau_{kk} = (\eta + \tfrac{2}{3}\mu)\theta$, so some authors reserve the name "bulk viscosity" for the combination $\eta + \tfrac{2}{3}\mu$ and call η the second coefficient of viscosity. The distinction is one of bookkeeping and evaporates whenever $\boldsymbol{\nabla}\cdot\boldsymbol{v} = 0$.

[6] The "Stokes hypothesis" $\eta = 0$ appears in G. G. Stokes, "On the theories of the internal friction of fluids in motion," *Transactions of the Cambridge Philosophical Society* **8** (1845), 287–319; reprinted in *Mathematical and Physical Papers*, Vol. 1 (Cambridge University Press, 1880), 75–129.

and therefore

$$\rho \frac{\mathrm{D}\boldsymbol{v}}{\mathrm{D}t} = -\boldsymbol{\nabla} p + \mu \nabla^2 \boldsymbol{v} + (\mu + \eta)\, \boldsymbol{\nabla}(\boldsymbol{\nabla} \cdot \boldsymbol{v}) + \rho\, \boldsymbol{b}. \tag{3.5}$$

This is the compressible Navier–Stokes equation. Note that the two viscosities do not stay separate in the final equation: the shear viscosity contributes to the dilatational term as well, through the $\partial_i \theta$ left over from differentiating $2\mu \boldsymbol{D}$. Note also that this equation does not stand alone. Once the density can vary, the temperature enters, and the flow and the heat must be found together: one needs in addition the continuity equation, an energy equation, an equation of state relating p, ρ, and T, and constitutive laws for the heat flux and the material properties.

For a liquid, or a gas in low-speed flow, each fluid particle keeps a nearly constant density and the motion is *incompressible*. By the continuity equation (Sec. 3.2) this is the kinematic constraint $\boldsymbol{\nabla} \cdot \boldsymbol{v} = 0$, and the whole dilatational term disappears, taking the bulk viscosity with it. We note two further assumptions are made in what follows. We take the density to be not merely constant along particle paths but *uniform*, and the shear viscosity to be constant as well. Incompressibility alone does not imply either (Sec. 3.2), and where a stratified or strongly heated fluid is at issue, the equations below cannot be applied as written, and require to be modified.

Important | Incompressible Navier–Stokes equations

$$\rho\left(\frac{\partial \boldsymbol{v}}{\partial t} + (\boldsymbol{\nabla}\boldsymbol{v})\, \boldsymbol{v}\right) = -\boldsymbol{\nabla} p + \mu \nabla^2 \boldsymbol{v} + \rho\, \boldsymbol{b}, \qquad \boldsymbol{\nabla} \cdot \boldsymbol{v} = 0.$$

In indicial form,

$$\rho\left(\frac{\partial v_i}{\partial t} + (\partial_j v_i)\, v_j\right) = -\partial_i p + \mu\, \partial_j \partial_j v_i + \rho\, b_i, \qquad \partial_i v_i = 0.$$

Together these are the four governing mechanical equations for the four fields (p, v_1, v_2, v_3). They presuppose a uniform density ρ and constant viscosity μ.

The pressure has changed character in passing to this limit. For a compressible fluid $p = p(\rho, T)$ is fixed by the thermodynamic state through an equation of state. Here there is no such relation: p is whatever scalar field enforces $\boldsymbol{\nabla} \cdot \boldsymbol{v} = 0$, entering the problem as a *Lagrange multiplier* whose job is to enforce that constraint. It is determined only up to an additive constant unless a pressure or a traction is prescribed somewhere on the boundary, and it carries no independent thermodynamic meaning.

With initial and boundary conditions (Sec. 3.9), these four equations will specify the nature of the flow. It is worth noticing what is missing. The density and viscosity are constant, so the velocity and pressure are settled by mechanics alone, and the temperature

has not appeared yet! This is why so much of incompressible fluid mechanics can be done without ever writing an energy equation, which we will deal with in a later chapter. The luck runs out the moment the properties vary, as they do in the compressible problem above.

Setting $\mu = 0$ recovers the inviscid *Euler equations*, $\rho\,\mathrm{D}\boldsymbol{v}/\mathrm{D}t = -\boldsymbol{\nabla} p + \rho\boldsymbol{b}$, in which the fluid can no longer transmit tangential traction at all. Dropping the highest derivative changes the character of the problem, not merely its size: the no-slip condition can no longer be imposed, and the discrepancy is resolved in the thin boundary layers of **??**.[7] The inviscid equations are due to L. Euler (1757).

3.8 Mechanical Energy and Dissipation

One more exact consequence of the momentum balance is extremely important, and it couples the mechanics to the first and second laws of thermodynamics.

3.8.1 The Mechanical Energy Balance

Ask a simple question: where does the work done on a fluid go? Some of it must show up as kinetic energy, since stirring a still fluid sets it moving. But a fluid stirred steadily gains no further kinetic energy while the stirring continues to demand power, so something else is absorbing the rest. The mechanical energy balance answers the question exactly, and it is obtained by the direct route of dotting the momentum equation with the velocity.

Take Cauchy's first law $\rho\,\mathrm{D}\boldsymbol{v}/\mathrm{D}t = \boldsymbol{\nabla}\cdot\boldsymbol{T} + \rho\boldsymbol{b}$ and form the scalar product with $\boldsymbol{v}$. The left side becomes the material derivative of the kinetic energy per unit volume,

$$\rho\,\boldsymbol{v}\cdot\frac{\mathrm{D}\boldsymbol{v}}{\mathrm{D}t} = \rho\,\frac{\mathrm{D}}{\mathrm{D}t}\left(\tfrac{1}{2}\,|\boldsymbol{v}|^2\right).$$

On the right, the tensor product rule Eq. (1.4) gives $(\boldsymbol{\nabla}\cdot\boldsymbol{T})\cdot\boldsymbol{v} = \boldsymbol{\nabla}\cdot(\boldsymbol{T}^{\top}\boldsymbol{v}) - \boldsymbol{T}:\boldsymbol{\nabla}\boldsymbol{v}$. Because $\boldsymbol{T}$ is symmetric (Thm. 3.10) the transpose drops, and because $\boldsymbol{T}$ is symmetric while $\boldsymbol{W}$ is antisymmetric, $\boldsymbol{T}:\boldsymbol{\nabla}\boldsymbol{v} = \boldsymbol{T}:\boldsymbol{D}$. Hence the local statement

$$\rho\,\frac{\mathrm{D}}{\mathrm{D}t}\left(\tfrac{1}{2}\,|\boldsymbol{v}|^2\right) = \boldsymbol{\nabla}\cdot(\boldsymbol{T}\boldsymbol{v}) - \boldsymbol{T}:\boldsymbol{D} + \rho\,\boldsymbol{v}\cdot\boldsymbol{b},$$

[7] The viscous equations were arrived at, by very different routes, by C.-L. Navier, "Mémoire sur les lois du mouvement des fluides," *Mémoires de l'Académie Royale des Sciences de l'Institut de France* **6** (1823), 389–440; S.-D. Poisson (1831), A. Barré de Saint-Venant (1843), and G. G. Stokes, "On the theories of the internal friction of fluids in motion," *Transactions of the Cambridge Philosophical Society* **8** (1845), 287–319. The two names sit together on the equation for reasons that are almost accidental. Navier, working in 1822 from a molecular picture of attracting and repelling particles, obtained the correct equation from a model of matter that is not the one we now accept, and he did not think of his coefficient as a viscosity. Stokes, two decades later and in ignorance of much of the intervening French work, derived the same equation from continuum assumptions close to those used in this chapter, and it is his derivation that the modern treatment follows. That both routes end at the same place is a reminder that the equation expresses a balance, not a molecular mechanism.

in which $\boldsymbol{T} : \boldsymbol{D}$ is the *stress power*, the rate per unit volume at which the surroundings do work on an element by deforming it.

Now integrate over a material volume $V(t)$ with boundary $S(t)$, using the transport theorem on the left and the divergence theorem on the first term of the right, and rearrange so that the work supplied from outside stands alone.

Important | Mechanical energy balance

$$\underbrace{\oint_{S(t)} (\boldsymbol{T}\boldsymbol{n}) \cdot \boldsymbol{v} \, \mathrm{d}S}_{\text{traction working}} + \underbrace{\int_{V(t)} \rho \, \boldsymbol{v} \cdot \boldsymbol{b} \, \mathrm{d}V}_{\text{body-force working}} = \underbrace{\frac{\mathrm{d}}{\mathrm{d}t} \int_{V(t)} \tfrac{1}{2} \rho \, |\boldsymbol{v}|^2 \, \mathrm{d}V}_{\text{rate of change of kinetic energy}} + \underbrace{\int_{V(t)} \boldsymbol{T} : \boldsymbol{D} \, \mathrm{d}V}_{\text{stress power}} .$$

Read from the left, this says that the total rate of working on the material volume, by tractions on its surface and by body forces throughout it, is spent in exactly two ways: accelerating the fluid, and deforming it. Nothing is lost and nothing is unaccounted for; it is an identity, not a model.

The second term is where the models enter. Split the stress with $\boldsymbol{T} = -p\boldsymbol{I} + \boldsymbol{\tau}$ and contract with $\boldsymbol{D}$, using $\boldsymbol{I} : \boldsymbol{D} = \operatorname{tr} \boldsymbol{D} = \boldsymbol{\nabla} \cdot \boldsymbol{v}$:

$$\boldsymbol{T} : \boldsymbol{D} = -p \, (\boldsymbol{\nabla} \cdot \boldsymbol{v}) + \underbrace{\boldsymbol{\tau} : \boldsymbol{D}}_{\Phi} .$$

The two pieces are physically quite different. The first is reversible: compressing an element stores energy that expansion returns, and it vanishes identically whenever $\boldsymbol{\nabla} \cdot \boldsymbol{v} = 0$. The second is the *dissipation function* Φ, and it is not reversible at all. Substituting,

$$\oint_{S(t)} (\boldsymbol{T}\boldsymbol{n}) \cdot \boldsymbol{v} \, \mathrm{d}S + \int_{V(t)} \rho \, \boldsymbol{v} \cdot \boldsymbol{b} \, \mathrm{d}V = \frac{\mathrm{d}}{\mathrm{d}t} \int_{V(t)} \tfrac{1}{2} \rho \, |\boldsymbol{v}|^2 \, \mathrm{d}V - \int_{V(t)} p \, (\boldsymbol{\nabla} \cdot \boldsymbol{v}) \, \mathrm{d}V + \int_{V(t)} \Phi \, \mathrm{d}V .$$

This is the answer to the question we began with. The work supplied goes into kinetic energy, into reversible compression, and into Φ. In a steady incompressible flow the first two vanish, and every watt supplied is accounted for by Φ, which is why a stirrer must be driven continuously to maintain a motion that is not accelerating.

3.8.2 Viscous Dissipation

Definition 3.13 | Viscous dissipation

The *dissipation function* is the viscous stress power,

$$\Phi = \boldsymbol{\tau} : \boldsymbol{D} = \tau_{ij} D_{ij}.$$

For an incompressible Newtonian fluid, where $\boldsymbol{\nabla} \cdot \boldsymbol{v} = 0$ and $\boldsymbol{\tau} = 2\mu \boldsymbol{D}$,

$$\Phi = 2\mu\, \boldsymbol{D} : \boldsymbol{D} = 2\mu\, D_{ij} D_{ij} \;\geq\; 0.$$

The sign is the whole point, and in the incompressible case it is immediate: $D_{ij}D_{ij}$ is a sum of squares of real numbers, so with $\mu > 0$ the dissipation is non-negative for every conceivable motion, and vanishes only when $\boldsymbol{D} = \boldsymbol{0}$, that is, only for a rigid motion. Viscosity can remove mechanical energy; it can never supply it.

For a compressible fluid the same calculation with the full Newtonian law gives

$$\Phi = (2\mu \boldsymbol{D} + \eta\, \theta\, \boldsymbol{I}) : \boldsymbol{D} = 2\mu\, \boldsymbol{D} : \boldsymbol{D} + \eta\, \theta^2,$$

again a sum of squares, non-negative for every motion provided $\mu \geq 0$ and $\eta \geq 0$. The two viscosities now appear as the two distinct routes by which mechanical energy is lost: μ through change of shape, η through change of volume. Setting $\theta = 0$ recovers the incompressible result.

None of this energy is destroyed. What Φ removes from the mechanical balance reappears, term for term, as a volumetric heat source in the internal-energy balance of **??**: the first law is not violated but completed. What the one-signedness of Φ expresses is the *second* law. The conversion runs one way only, from ordered motion to internal energy, never back, and this is why stirring warms a fluid, why an unforced viscous flow decays, and why the churning of a fluid cannot be run in reverse. A steadily driven flow, such as flow through a pipe held at fixed pressure drop, dissipates continuously without decaying, the loss being made good by the pumping.

Example 3.14 | Dissipation in simple shear

For the shear flow $\boldsymbol{v} = (\dot{\gamma}\, x_2, 0, 0)$ the only nonzero rate-of-strain components are $D_{12} = D_{21} = \dot{\gamma}/2$, so

$$\Phi = 2\mu\, (D_{12}^2 + D_{21}^2) = 2\mu \cdot \tfrac{\dot{\gamma}^2}{2} = \mu\, \dot{\gamma}^2.$$

The power dissipated per unit volume grows as the *square* of the shear rate. This is

why viscous heating becomes severe in high-speed lubrication films, polymer extrusion, and journal bearings, where $\dot{\gamma}$ is enormous.

Remark 3.15 | Energy and the second law

The one-signedness of Φ is not an accident of the Newtonian model; it is demanded by the entropy inequality. For a purely viscous fluid of the kind considered here, thermodynamic admissibility requires $\boldsymbol{\tau} : \boldsymbol{D} \geq 0$, so that the mechanical energy spent on deformation is never returned. The statement is restricted to this class. A viscoelastic material stores part of the work done on it and can give it back later, and for such materials the requirement applies to the entropy production of the whole model rather than to $\boldsymbol{\tau} : \boldsymbol{D}$ alone. As noted in Sec. 3.6, the Newtonian law with $\mu > 0$ and $\eta \geq 0$ is the simplest relation consistent with this requirement.

3.9 Boundary Conditions

The balance laws of this chapter, closed by the Newtonian constitutive relation, give the Navier–Stokes equations, a system of partial differential equations for $\boldsymbol{v}$ and p. A differential equation is not a problem until it is fitted with side conditions, and it is the boundary conditions that turn the universal field equations into the specific flow in a specific geometry. They also carry most of the physics that distinguishes one problem from another: whether a wall is wetted or slippery, whether a surface is rigid or free to deform, whether a fluid is confined or unbounded. This section collects the conditions we shall use repeatedly.[8]

The conditions come in two kinds. *Kinematic* conditions constrain how the fluid and the boundaries *move*, normal velocities and the motion of material surfaces; they follow from geometry and from the material nature of the fluid. *Dynamic* conditions constrain the *forces* at a boundary, tractions and their jumps, and therefore involve the stress and its constitutive law. We label each condition below as one or the other.

Let us count how many conditions are needed. The viscous momentum equations are second order in $\boldsymbol{v}$, through the term $\mu\nabla^2\boldsymbol{v}$, so three scalar conditions are required at each point of the boundary, corresponding to the three components of the velocity. What those conditions prescribe is a separate question: depending on the physical boundary they may fix the velocity, the traction, or a combination of the two. A rigid wall fixes all three components of the velocity; a free surface fixes the traction instead; a symmetry plane fixes one velocity component and two components of traction. The pressure carries no such requirement: it

[8] The interfacial stress balance is developed carefully in L. G. Leal, *Advanced Transport Phenomena: Fluid Mechanics and Convective Transport Processes* (Cambridge University Press, 2007). The notation and emphasis are our own.

enters only through its gradient, and is fixed only up to an additive constant, to be pinned down by a single reference value or by a stress condition on some surface. We return to this counting in Rem. 3.16.

Important | Sign and notation conventions

Two conventions are fixed here once and used throughout the boundary conditions below.

Pressure. In a compressible flow p is a thermodynamic variable, fixed by an equation of state $p = p(\rho, T)$ in the density and temperature. In strictly incompressible flow there is no such relation: p becomes a Lagrange multiplier that enforces $\boldsymbol{\nabla} \cdot \boldsymbol{v} = 0$, determined by the flow only up to an additive constant. A *modified* pressure may be introduced to absorb a conservative body-force potential (for gravity, $P = p + \rho g z$), but it must be defined explicitly before use, since it shifts the meaning of every "pressure" written afterwards.

Traction and normals. The unit normal $\boldsymbol{n}$ is the *outward* normal from the fluid domain. The traction $\boldsymbol{t}(\boldsymbol{n}) = \boldsymbol{T} \cdot \boldsymbol{n}$ is the force per unit area exerted *on* the fluid inside the domain *by* the material outside it. The force of the fluid on a wall or neighbouring body is the opposite vector, $-\boldsymbol{T} \cdot \boldsymbol{n}$, by Newton's third law. Where we deliberately point $\boldsymbol{n}$ from a solid into the fluid, as at the interface of Sec. 3.9.2, we say so and state whether a given traction is wall-on-fluid or fluid-on-wall. A heading such as "force on the wall" must never sit above an equation that is really the wall-on-fluid traction.

3.9.1 Conditions at a Solid Wall

A rigid impermeable wall imposes two requirements, one on each part of the velocity. The first is purely geometric: no fluid may cross an impermeable surface, so the normal component of the fluid velocity must equal that of the wall,

$$\boldsymbol{v} \cdot \boldsymbol{n} = \boldsymbol{v}_{\text{wall}} \cdot \boldsymbol{n} \qquad \text{(no penetration)}. \tag{3.6}$$

For a wall at rest this is simply $\boldsymbol{v} \cdot \boldsymbol{n} = 0$. The condition is kinematic and holds regardless of the fluid's constitution. It applies equally to a viscous and an inviscid fluid.

The second requirement concerns the *tangential* motion and is a statement about the fluid, not merely the geometry. Experience shows that a viscous fluid does not slide along a solid: the layer of fluid in contact with the wall is carried along with it, so that the *tangential* velocity of fluid and wall coincide as well. Combining this with Eq. (3.6) gives the celebrated no-slip condition.

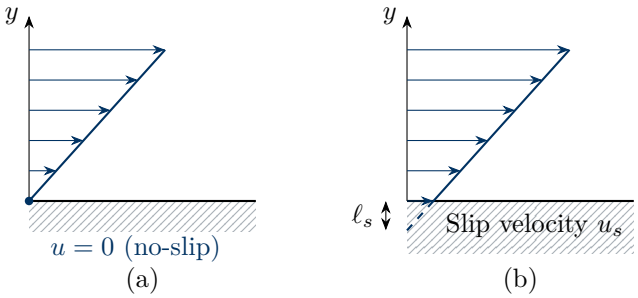


Figure 3.3: Velocity near a solid wall. **(a)** At an ordinary wall the tangential velocity of the fluid vanishes at the surface: the profile is anchored to zero (no-slip). **(b)** The NAVIER slip condition Eq. (3.8) permits a finite wall velocity u_s; its linear extrapolation reaches zero a distance ℓ_s, the slip length, below the surface. No-slip is the limit $\ell_s \to 0$.

Important | No-slip at a solid wall

On the surface of a rigid wall moving with velocity $\boldsymbol{v}_{\text{wall}}$, a viscous fluid takes on the full velocity of the wall,

$$\boldsymbol{v} = \boldsymbol{v}_{\text{wall}} \qquad \text{on the wall.} \tag{3.7}$$

The normal part expresses impermeability; the tangential part is the no-slip condition proper.

Unlike no-penetration, which is kinematic, no-slip is an empirical interfacial *constitutive* condition, a statement about how a given fluid interacts with a given solid, not a consequence of geometry, and hence an empirical law rather than a theorem. It was contested through the nineteenth century, C.-L. NAVIER himself proposed a partial-slip boundary condition in which the tangential velocity is proportional to the wall shear rate,

$$\boldsymbol{v}_{\parallel} = \ell_s \left. \frac{\partial \boldsymbol{v}_{\parallel}}{\partial n} \right|_{\text{wall}}, \tag{3.8}$$

with a *slip length* ℓ_s measuring how far into the wall the velocity profile would have to be extrapolated to vanish. The no-slip limit is $\ell_s \to 0$; a perfectly shear-free wall is $\ell_s \to \infty$. For ordinary liquids on wettable solids ℓ_s is of molecular size, a few tenths of a nanometre, so no-slip is an excellent approximation at every scale of engineering interest, and we adopt it throughout. Slip becomes measurable only in special circumstances: rarefied gases (large mean free path), molecularly smooth or strongly non-wetting surfaces, and micro- and nano-fluidic channels, where ℓ_s can reach microns (Fig. 3.3).

3.9.2 Conditions at a Fluid Interface

When the boundary is not a solid but the surface separating two immiscible fluids (a liquid and a gas, or two liquids), its location is itself unknown, part of the solution, and two *kinds* of condition are required: a *kinematic* condition governing how the surface moves, and a *dynamic* condition balancing the forces acting across it.

The kinematic condition. An interface between immiscible fluids is a *material surface*: the same fluid particles remain on it for all time, since none can cross. If the surface is described implicitly by $F(\boldsymbol{x}, t) = 0$, this material character is expressed by requiring F to be constant following the flow,

$$\frac{\mathrm{D}F}{\mathrm{D}t} = \frac{\partial F}{\partial t} + \boldsymbol{v} \cdot \boldsymbol{\nabla} F = 0 \qquad \text{on } F = 0. \tag{3.9}$$

For a nearly flat interface written as a height $y = h(x, t)$, so that $F = y - h(x, t)$, Eq. (3.9) reduces to the intuitive statement

$$v = \frac{\partial h}{\partial t} + u\,\frac{\partial h}{\partial x} = \frac{\mathrm{D}h}{\mathrm{D}t}, \tag{3.10}$$

the normal velocity of the fluid equals the velocity with which the surface itself rises. Across the interface the velocity is continuous, neither fluid penetrates ($\boldsymbol{v}^{(1)} \cdot \boldsymbol{n} = \boldsymbol{v}^{(2)} \cdot \boldsymbol{n}$), and for a clean interface we ordinarily impose continuity of the tangential velocity as well ($\boldsymbol{v}^{(1)} = \boldsymbol{v}^{(2)}$); interfacial slip would require an additional interfacial constitutive model.

The dynamic condition. An interface has no mass, so the net force on any element of it must vanish. Two contributions compete: the tractions exerted by the fluids on either side, and the line tension, *surface tension* γ, acting along the surface, the free-energy cost per unit area of the interface. A force balance on a massless surface element gives the interfacial stress balance (Fig. 3.4).

Important | Stress balance at an interface

Let $\boldsymbol{n}$ point from fluid 1 into fluid 2. With surface tension γ the only interfacial force, the traction jump across the interface is

$$(\boldsymbol{T}^{(2)} - \boldsymbol{T}^{(1)}) \cdot \boldsymbol{n} = \gamma\,\kappa\,\boldsymbol{n} \;-\; \nabla_s \gamma, \qquad \kappa \equiv \nabla_s \cdot \boldsymbol{n} = \frac{1}{R_1} + \frac{1}{R_2}, \tag{3.11}$$

where κ is the total curvature (R_1, R_2 the principal radii, reckoned positive when the surface is convex toward fluid 2) and ∇_s is the gradient taken within the surface.

The two projections of Eq. (3.11) carry distinct physics. Along the normal, contracting

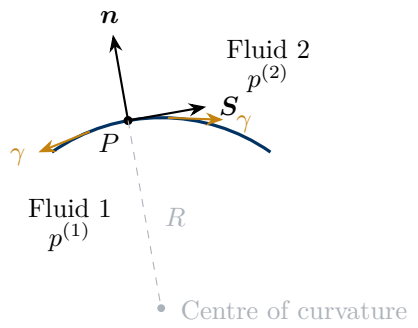


Figure 3.4: The dynamic condition at a curved interface. At a surface point P the unit normal $\boldsymbol{n}$ points from fluid 1 into fluid 2 and $\boldsymbol{s}$ is a unit tangent; the surface has principal radius R. Surface tension γ pulls tangentially along the interface (gold). The normal balance is the Young–Laplace law: the pressure is higher on the concave side (fluid 1). A tangential *gradient* of γ would leave the two gold arrows unbalanced, driving a Marangoni flow.

with $\boldsymbol{n}$ gives the jump in normal stress,

$$\boldsymbol{n}\cdot\boldsymbol{T}^{(2)}\cdot\boldsymbol{n}-\boldsymbol{n}\cdot\boldsymbol{T}^{(1)}\cdot\boldsymbol{n}=\gamma\,\kappa. \tag{3.12}$$

For two fluids at rest, $\boldsymbol{T}=-p\,\boldsymbol{I}$ and this is the Young–Laplace equation

$$p^{(1)}-p^{(2)}=\gamma\,\kappa=\gamma\left(\frac{1}{R_1}+\frac{1}{R_2}\right): \tag{3.13}$$

the pressure is higher on the concave side, by an amount set by the curvature. A spherical drop of radius R (fluid 1 inside) carries an excess pressure $\Delta p=2\gamma/R$, the reason small bubbles are stiffer than large ones.

Along a tangent $\boldsymbol{s}$ ($\boldsymbol{s}\cdot\boldsymbol{n}=0$), contracting Eq. (3.11) gives the jump in *shear* stress,

$$\boldsymbol{s}\cdot\left(\boldsymbol{T}^{(2)}-\boldsymbol{T}^{(1)}\right)\cdot\boldsymbol{n}=-\,\boldsymbol{s}\cdot\nabla_s\gamma. \tag{3.14}$$

When γ is uniform the tangential traction is continuous across the interface. When it is not, because temperature or the concentration of a surfactant varies along the surface, the gradient $\nabla_s\gamma$ must be balanced by a jump in viscous shear, and the interface drags the adjacent fluid from regions of low tension toward regions of high tension. This is the Marangoni effect, the engine behind tears of wine and the drift of small particles in temperature gradients.

The free surface. A common and important special case is a liquid (1) bounded by a gas (2) so light and inviscid that it transmits only a uniform pressure p_{gas} and no appreciable shear: $\boldsymbol{T}^{(2)}=-p_{\text{gas}}\boldsymbol{I}$. The stress balance then constrains the liquid alone. The tangential

condition Eq. (3.14) (uniform γ) becomes

$$\boldsymbol{s}\cdot\boldsymbol{T}^{(1)}\cdot\boldsymbol{n}=0 \qquad \text{(shear-free surface)}, \tag{3.15}$$

while the normal condition fixes the liquid's normal stress against the gas pressure and curvature. For a *flat* free surface ($\kappa=0$) the normal condition reads

$$-p+\tau_{nn}=-p_{\mathrm{gas}}, \qquad \text{that is} \qquad p=p_{\mathrm{gas}}+\tau_{nn},$$

where $\tau_{nn}=\boldsymbol{n}\cdot\boldsymbol{\tau}\cdot\boldsymbol{n}$ is the normal viscous stress in the liquid. The familiar statement $p=p_{\mathrm{gas}}$ is the special case $\tau_{nn}=0$. It is not automatic, but it does hold in the unidirectional films of Ch. 4: there $\boldsymbol{v}=u(y)\,\boldsymbol{e}_x$ with $\boldsymbol{n}=\boldsymbol{e}_y$, so $D_{yy}=\partial u_y/\partial y=0$ and the normal viscous stress vanishes. Together with the shear-free condition Eq. (3.15), that is what we shall impose at the top of the falling film.

3.9.3 Far Field, Inflow, Outflow, and Symmetry

Several further conditions recur in unbounded or truncated domains. Unlike the wall and interface conditions, these are *external* conditions: they do not describe a physical surface but serve to complete an unbounded or artificially truncated domain. *Far-field* conditions apply when the fluid extends to infinity: one requires the flow to approach a prescribed state, most often a uniform stream $\boldsymbol{v}\to\boldsymbol{v}_\infty$ and $p\to p_\infty$ as $|\boldsymbol{x}|\to\infty$, as in the flow past a sphere or an aerofoil. At an *inflow* boundary one prescribes the incoming velocity profile. An *outflow* boundary is different in kind, since it is an artificial edge introduced by truncating the domain rather than a physical surface. No single condition is correct in general. Fully developed ($\partial\boldsymbol{v}/\partial n=\boldsymbol{0}$), traction-free, and convective conditions are all in common use, and the right choice depends on the problem and on how far downstream the boundary has been placed. A plane of *symmetry* behaves like a shear-free wall: no flux through it and no shear across it,

$$\boldsymbol{v}\cdot\boldsymbol{n}=0, \qquad \boldsymbol{s}\cdot\boldsymbol{T}\cdot\boldsymbol{n}=0 \qquad \text{on a symmetry plane}, \tag{3.16}$$

which halves the computational domain of any symmetric problem.

Remark 3.16 | Counting conditions: viscous versus inviscid

The order of the governing equation dictates how many conditions a wall can carry, and this has a famous consequence. The viscous term $\mu\nabla^2\boldsymbol{v}$ makes Navier–Stokes second order in $\boldsymbol{v}$, so we may, and must, impose the full vector condition Eq. (3.7): both no-penetration and no-slip. Drop the viscosity and the Euler equations of an ideal fluid are of lower order; they can enforce only the single scalar condition Eq. (3.6). An inviscid fluid is therefore permitted to *slip* along a wall. The reconciliation of these

two pictures, an outer, nearly inviscid flow that slips, matched to a thin wall layer in which no-slip is restored, is the subject of boundary-layer theory in **??**.

3.10 A Note on the Millennium Problem

The Navier–Stokes equations will serve us, for the rest of this book, as the trusted description of every viscous flow. One question about them is a very deep and fundamental one in mathematics, and it is worth stating precisely, because it is often stated loosely.

The question concerns *regularity* of the solutions to the equations. Given smooth initial data of finite energy, do the three-dimensional incompressible Navier-Stokes equaations possess a unique smooth solution for *all* time, or can the velocity field spontaneously sharpen into a *singularity*, an infinite velocity or velocity gradient, after a finite interval? This scenario is called *finite-time blow-up*. In two dimensions the matter was long ago settled in favour of eternal smoothness. ~~In three dimensions it remains wide open.~~ In three dimensions it remained open for ninety years, and its status as of this writing is described below. The question is one of global existence and smoothness, not of whether the equations are usable. J. Leray proved in 1934 that global *weak* solutions always exist, but could neither guarantee that they stay smooth nor rule out singular "turbulent" solutions of his own imagining.[9]

So central and so stubborn is the problem that the Clay Mathematics Institute made it one of its seven *Millennium Prize Problems* in 2000, each carrying a one-million-dollar award.[10] In the official statement C. L. Fefferman concedes that "standard methods from PDE appear inadequate to settle the problem. Instead, we probably need some deep, new ideas." The difficulty is of a piece with turbulence itself, a phenomenon that has humbled the finest physical minds: Three statements should be kept apart. Global weak solutions are known to exist, by Leray's construction. Whether they are smooth and unique for all time in three dimensions is unresolved. Neither question bears on the standing of the equations as a physical model.

> "I am an old man now, and when I die and go to heaven there are two matters on which I hope for enlightenment. One is quantum electrodynamics, and the other is the turbulent motion of fluids. And about the former I am rather optimistic.", attributed to H. Lamb

And yet the practising engineer computes with these equations every day. Whether a smooth solution can blow up in finite time bears almost not at all on the design of a

[9] J. Leray, "Sur le mouvement d'un liquide visqueux emplissant l'espace," *Acta Mathematica* **63** (1934), 193–248, the paper that introduced weak solutions to the subject.

[10] C. L. Fefferman, "Existence and smoothness of the Navier–Stokes equation," official problem description, Clay Mathematics Institute (2000).

pump, a pipeline, or an aircraft. The open question is one of *regularity* for an idealized partial differential equation, and it is logically separate from how well the continuum model describes a physical flow within its range of validity. Finite-time blow-up, the onset of turbulence, the failure of the continuum description, and the divergence of a numerical calculation are four distinct things, and none of them implies any of the others.

That ninety-year stalemate broke, noisily, in September 2026. On 8 September OpenAI announced that a system of coordinated language-model agents had produced a proof of finite-time blow-up for the three-dimensional equations, formally checked in the proof assistant Lean, and claimed the Millennium Problem as resolved.[11] Three cautions belong with that announcement, and they are the reason this section has not been rewritten around it. First, the Clay Institute has recognized no solution: its president, M. Bridson, has said that the prize requires peer-reviewed publication, and that the Institute's evaluation will not be hurried. Second, the precise statement proved, in particular whether the forcing is smooth and whether the result concerns the unforced equations of the official problem, was still being clarified in public at the time of writing; a blow-up theorem for a forced or modified system is a different theorem from the one Fefferman states. Third, priority is disputed. On the previous day T. Buckmaster and L. Alpöge had released finite-time blow-up results with smooth forcing for the incompressible porous-medium, Boussinesq, and three-dimensional incompressible Euler equations, Lean-verified on 22 August, and Buckmaster has alleged that knowledge of their progress reached OpenAI beforehand.[12] Both groups credit the underlying program to D. Córdoba and L. Martínez-Zoroa, who had spent several years constructing forced blow-ups. On 11 September twenty-five Fields Medalists, among them T. Tao, published a declaration objecting less to the mathematics than to the manner of its release: results announced at a pace that leaves no time for a proper writeup, for isolating the new ideas, or for citing the earlier work they rest on.[13]

For our purposes the episode changes nothing and illustrates something. For mathematicians, the Navier Stokes regularity problem has been a "north-star" whose solution may expose something deeply fundamental about mathematical structures of PDEs, and the insights gained in solving it may, in the future, impact other fields of mathematics or open entirely new ones. Yet, on the other hand, it changes nothing because the standing of the Navier–Stokes equations as a practical tool never depended on the regularity question, and

[11] OpenAI, "On the Navier–Stokes Millennium Prize Problem," 8 September 2026. Contemporary reporting, with the caveats that attended it, in *Quanta Magazine*, 8 September 2026, and *Nature*, 8 September 2026.

[12] T. Buckmaster, statement, Courant Institute, 7 September 2026. He writes there that "the credit for the basic idea of this program goes to Diego Córdoba and Luis Martínez-Zoroa," and that the work with Alpöge was "a purely personal collaboration, free of any institutional agreements." Furthermore, in a statement of pure integrity and collegiality Buckmaster has explicitly acknowledged that in his view Martínez-Zoroa deserves the fields medal.

[13] "A Severe Misalignment of AI in Mathematics," mathandai.org, 11 September 2026, signed by twenty-five Fields Medalists. The declaration objects that such results are "announced in a rush, leaving no time for a proper writeup, the isolation of new methods and ideas, and citing relevant previous work of others."

does not depend on its answer. Readers who want the current status should consult the Clay Institute directly rather than this paragraph, which is dated 13 September 2026. The only thing I am certain of is that these views are likely to age badly, *very badly*.

This completes the machinery of the chapter. A transport problem is fully *posed*, and ready to be solved, once four ingredients are assembled:

balance laws + constitutive relations + material properties + initial and boundary conditions = a formulated transport problem.

Ch. 4 puts this framework to work, applying it to a sequence of canonical unidirectional flows in which each ingredient can be watched doing its job.

Appendix 3.A The Cauchy Tetrahedron and Noll's Theorem

We owe two debts from the main text: the existence and linearity of the stress tensor (Cauchy's stress theorem, Thm. 3.8), and the legitimacy of Cauchy's hypothesis itself (Def. 3.7).

3.A.1 The Tetrahedron Argument

We wish to show that there exists a tensor field $\boldsymbol{T}(\boldsymbol{x},t)$ such that $\boldsymbol{t}(\boldsymbol{x},t;\boldsymbol{n}) = \boldsymbol{T}(\boldsymbol{x},t)\,\boldsymbol{n}$. The argument is Cauchy's, and it repays close reading: linearity in $\boldsymbol{n}$ is assumed nowhere, and the only tools used are the balance of linear momentum and the mean value theorem for integrals.

Let $\mathcal{P}$ be the tetrahedron of Fig. 3.5. Write σ for the oblique face, of outward unit normal $\boldsymbol{n}$ and area A, and $\sigma_1,\sigma_2,\sigma_3$ for the three orthogonal faces, of outward normals $-\boldsymbol{e}_1,-\boldsymbol{e}_2,-\boldsymbol{e}_3$ and areas A_1,A_2,A_3. These areas are not independent of A. Resolving the oriented oblique face along the coordinate directions,

$$\boldsymbol{n}A = n_i\boldsymbol{e}_i A = A\,n_i\,\boldsymbol{e}_i = A_i\,\boldsymbol{e}_i, \qquad \text{so} \qquad A_i = A\,n_i.$$

Apply the balance of linear momentum to $\mathcal{P}$, using the transport theorem on the left to bring the time derivative inside,

$$\int_{\mathcal{P}} \rho\left(\frac{\mathrm{D}\boldsymbol{v}}{\mathrm{D}t} - \boldsymbol{b}\right)\mathrm{d}V = \int_{\sigma} \boldsymbol{t}(\boldsymbol{n})\,\mathrm{d}A + \sum_{i=1}^{3}\int_{\sigma_i} \boldsymbol{t}(-\boldsymbol{e}_i)\,\mathrm{d}A.$$

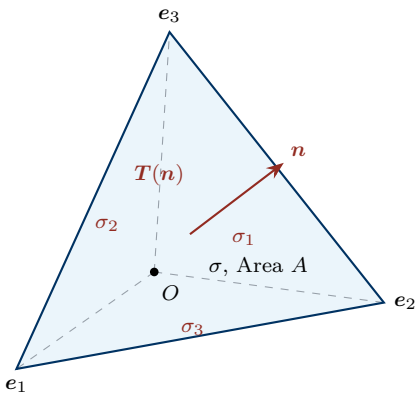


Figure 3.5: The Cauchy tetrahedron at a point O. Three faces lie in the coordinate planes, with outward normals $-\boldsymbol{e}_1$, $-\boldsymbol{e}_2$, $-\boldsymbol{e}_3$ and areas A_1, A_2, A_3; the oblique face has outward unit normal $\boldsymbol{n}$ and area A, so that $A_j = A\, n_j$. The oblique face is written σ and the three coordinate faces $\sigma_1, \sigma_2, \sigma_3$. Bounding the volume terms and applying the mean value theorem to each face integral, then letting the height $h \to 0$, forces the traction to depend linearly on $\boldsymbol{n}$ and gives $t_i = T_{ij} n_j$.

Cauchy's lemma Eq. (3.3) replaces $\boldsymbol{t}(-\boldsymbol{e}_i)$ by $-\boldsymbol{t}(\boldsymbol{e}_i)$, so that

$$\int_{\mathcal{P}} \rho \left(\frac{\mathrm{D}\boldsymbol{v}}{\mathrm{D}t} - \boldsymbol{b} \right) \mathrm{d}V = \int_{\sigma} \boldsymbol{t}(\boldsymbol{n}) \, \mathrm{d}A - \sum_{i=1}^{3} \int_{\sigma_i} \boldsymbol{t}(\boldsymbol{e}_i) \, \mathrm{d}A.$$

Nothing has been approximated yet. The content of the argument is that the two sides scale differently as the tetrahedron shrinks, and to expose this we bound the left side. Set $k(\boldsymbol{x}, t) := \left| \rho \left(\mathrm{D}\boldsymbol{v}/\mathrm{D}t - \boldsymbol{b} \right) \right|$, so that

$$\left| \int_{\mathcal{P}} \rho \left(\frac{\mathrm{D}\boldsymbol{v}}{\mathrm{D}t} - \boldsymbol{b} \right) \mathrm{d}V \right| \;\leq\; \int_{\mathcal{P}} k(\boldsymbol{x}, t) \, \mathrm{d}V \;=\; k^* V,$$

the last equality being the mean value theorem for integrals, with $k^* = k(\boldsymbol{x}^*, t)$ at some interior point $\boldsymbol{x}^*$ of $\mathcal{P}$. Writing h for the height of the tetrahedron above the oblique face gives $V = \frac{1}{3} h A$, and therefore

$$\tfrac{1}{3} h \, k^* A \;\geq\; \left| \int_{\sigma} \boldsymbol{t}(\boldsymbol{n}) \, \mathrm{d}A - \sum_{i=1}^{3} \int_{\sigma_i} \boldsymbol{t}(\boldsymbol{e}_i) \, \mathrm{d}A \right|.$$

Apply the mean value theorem once more, now to each of the four surface integrals: there are points on σ and on each σ_i at which

$$\int_{\sigma} \boldsymbol{t}(\boldsymbol{n}) \, \mathrm{d}A = \boldsymbol{t}^*_{(\boldsymbol{n})} A, \qquad \int_{\sigma_i} \boldsymbol{t}(\boldsymbol{e}_i) \, \mathrm{d}A = \boldsymbol{t}^*_{(\boldsymbol{e}_i)} A_i,$$

where a star marks evaluation at the corresponding interior point. Substituting these and

then using $A_i = A\,n_i$,

$$\tfrac{1}{3}h\,k^*A \;\geq\; \left|\boldsymbol{t}^*_{(\boldsymbol{n})}A - \sum_{i=1}^{3}\boldsymbol{t}^*_{(\boldsymbol{e}_i)}A_i\right| = \left|\boldsymbol{t}^*_{(\boldsymbol{n})} - \sum_{i=1}^{3}\boldsymbol{t}^*_{(\boldsymbol{e}_i)}n_i\right|A.$$

The area A now cancels from both sides, which is precisely what writing $A_i = A\,n_i$ was for, and we are left with

$$\left|\boldsymbol{t}^*_{(\boldsymbol{n})} - \sum_{i=1}^{3}\boldsymbol{t}^*_{(\boldsymbol{e}_i)}n_i\right| \;\leq\; \tfrac{1}{3}k^*h.$$

Let $h \to 0$. The tetrahedron collapses onto the point $\boldsymbol{x}$, so each interior point supplied by the mean value theorem must approach $\boldsymbol{x}$, and continuity of the traction gives $\boldsymbol{t}^*_{(\boldsymbol{n})} \to \boldsymbol{t}_{(\boldsymbol{n})}(\boldsymbol{x},t)$ and $\boldsymbol{t}^*_{(\boldsymbol{e}_i)} \to \boldsymbol{t}_{(\boldsymbol{e}_i)}(\boldsymbol{x},t)$. The right-hand side vanishes with h, provided k^* remains bounded, which is what the smoothness assumed of ρ, $\boldsymbol{v}$, and $\boldsymbol{b}$ provides. Hence

$$\boldsymbol{t}_{(\boldsymbol{n})} = \sum_{i=1}^{3}\boldsymbol{t}_{(\boldsymbol{e}_i)}\,n_i = \left(\sum_{i=1}^{3}\boldsymbol{t}_{(\boldsymbol{e}_i)}\otimes\boldsymbol{e}_i\right)\boldsymbol{n}.$$

The quantity in parentheses carries no dependence on $\boldsymbol{n}$. Defining

$$\boldsymbol{T}(\boldsymbol{x},t) := \sum_{i=1}^{3}\boldsymbol{t}_{(\boldsymbol{e}_i)}(\boldsymbol{x},t)\otimes\boldsymbol{e}_i$$

therefore yields Cauchy's stress theorem (Thm. 3.8), $\boldsymbol{t}_{(\boldsymbol{n})}(\boldsymbol{x},t) = \boldsymbol{T}(\boldsymbol{x},t)\,\boldsymbol{n}$. Linearity in $\boldsymbol{n}$ was never postulated: it is forced by the balance of linear momentum together with the mismatch in scaling between volume and surface terms.

Extracting the components is worth doing, because it is where the physical reading of T_{ij} comes from. Write $\boldsymbol{T} = T_{ij}\,\boldsymbol{e}_i\otimes\boldsymbol{e}_j$ and compare with the definition just given,

$$T_{ij}\,\boldsymbol{e}_i\otimes\boldsymbol{e}_j = \boldsymbol{t}_{(\boldsymbol{e}_i)}\otimes\boldsymbol{e}_i.$$

Applying both sides to $\boldsymbol{e}_k$ and using $\boldsymbol{e}_j\cdot\boldsymbol{e}_k = \delta_{jk}$ gives $T_{ik}\,\boldsymbol{e}_i = \boldsymbol{t}_{(\boldsymbol{e}_k)}$; dotting with $\boldsymbol{e}_j$ then leaves

$$T_{ij} = \boldsymbol{e}_i\cdot\boldsymbol{t}_{(\boldsymbol{e}_j)}.$$

So T_{ij} is the i-component of the traction acting on the face whose outward normal is $\boldsymbol{e}_j$. A stress is a force per unit area on a face, resolved along a direction, and both indices are needed: one names the face, the other the direction. This is the honest meaning of the stress tensor, and all of it has emerged from the balance of linear momentum.

3.A.2 Noll's Theorem

This subsection is foundational rather than practical, and may be passed over on a first reading; nothing later in these notes depends on it. The tetrahedron argument takes Cauchy's hypothesis (Def. 3.7), that traction depends on the surface only through its normal $\boldsymbol{n}$, and not, say, on its curvature, as a starting assumption, and further presumes enough regularity to pass to the limit. Placing this on a rigorous footing was accomplished by W. Noll, who showed that Cauchy's hypothesis is not an independent postulate but a *consequence* of the balance laws themselves.[14]

Theorem 3.17 | Noll's theorem, informal statement

Assume the balance of linear momentum holds for *every* subbody, that the body force is bounded, and that the traction $\boldsymbol{t}(\boldsymbol{x};\boldsymbol{n})$ depends continuously on $\boldsymbol{x}$ for each fixed $\boldsymbol{n}$. Then the traction necessarily depends on the surface only through its unit normal $\boldsymbol{n}$, i.e. Cauchy's hypothesis holds, and, by the tetrahedron argument, $\boldsymbol{t}(\boldsymbol{n}) = \boldsymbol{T}\cdot\boldsymbol{n}$ for a unique stress tensor $\boldsymbol{T}$.

The content of Noll's result is that surface interactions in a continuum, if they are to be consistent with the balance of momentum on arbitrary subregions, cannot be more exotic than a linear map of the normal. What began with A.-L. Cauchy as a plausible hypothesis became, through W. Noll, a theorem, an instructive example of how the foundations of a classical subject can be reexamined and secured long after its equations are in routine use.

Appendix 3.B Change of Observer and the Objectivity of the Rate of Strain

This appendix derives the transformation laws quoted in Sec. 3.6: under a change of observer the rate of strain is objective, $\boldsymbol{D}^{*} = \boldsymbol{Q}\boldsymbol{D}\boldsymbol{Q}^{\top}$, whereas the rotation tensor is not, $\boldsymbol{W}^{*} = \boldsymbol{Q}\boldsymbol{W}\boldsymbol{Q}^{\top} + \dot{\boldsymbol{Q}}\boldsymbol{Q}^{\top}$.

Setup. Two observers relate the same event by

$$\boldsymbol{x}^{*} = \boldsymbol{Q}(t)\,\boldsymbol{x} + \boldsymbol{c}(t), \qquad \boldsymbol{Q}^{\top}\boldsymbol{Q} = \boldsymbol{I},$$

with $\boldsymbol{Q}(t)$ a rotation and $\boldsymbol{c}(t)$ a translation, both possibly time-dependent.

[14] W. Noll, "The foundations of classical mechanics in the light of recent advances in continuum mechanics," in *The Axiomatic Method, with Special Reference to Geometry and Physics* (Proceedings of the 1957 Berkeley Symposium), North-Holland, 1959, pp. 266–281. See also C. Truesdell and W. Noll, *The Non-Linear Field Theories of Mechanics* (Springer, 1965).

Velocity. Follow one material particle and differentiate in time. With $\boldsymbol{v} = \dot{\boldsymbol{x}}$ and $\boldsymbol{v}^* = \dot{\boldsymbol{x}}^*$,

$$\boldsymbol{v}^* = \dot{\boldsymbol{Q}}\,\boldsymbol{x} + \boldsymbol{Q}\,\boldsymbol{v} + \dot{\boldsymbol{c}}.$$

The two observers therefore disagree even about the velocity: the terms $\dot{\boldsymbol{Q}}\,\boldsymbol{x}$ and $\dot{\boldsymbol{c}}$ are the rigid motion of one frame relative to the other.

Velocity gradient. The velocity gradient is $\boldsymbol{L} = \boldsymbol{\nabla}\boldsymbol{v}$, with $L_{ij} = \partial v_i/\partial x_j$; the starred observer forms $L^*_{ij} = \partial v^*_i/\partial x^*_j$. Since $\boldsymbol{Q}$ and $\boldsymbol{c}$ depend on t alone, differentiating $v^*_i = \dot{Q}_{ik}x_k + Q_{ik}v_k + \dot{c}_i$ gives $\partial v^*_i/\partial x_m = \dot{Q}_{im} + Q_{ik}L_{km}$. Using $\partial x_m/\partial x^*_j = Q_{jm}$ (from $\boldsymbol{x} = \boldsymbol{Q}^\top(\boldsymbol{x}^* - \boldsymbol{c})$), the chain rule yields $L^*_{ij} = (\dot{Q}_{im} + Q_{ik}L_{km})\,Q_{jm}$, that is,

$$\boldsymbol{L}^* = \boldsymbol{Q}\,\boldsymbol{L}\,\boldsymbol{Q}^\top + \dot{\boldsymbol{Q}}\,\boldsymbol{Q}^\top.$$

The extra term is antisymmetric. Differentiating $\boldsymbol{Q}\boldsymbol{Q}^\top = \boldsymbol{I}$ gives $\dot{\boldsymbol{Q}}\boldsymbol{Q}^\top + \boldsymbol{Q}\dot{\boldsymbol{Q}}^\top = \boldsymbol{0}$, hence $\dot{\boldsymbol{Q}}\boldsymbol{Q}^\top = -(\dot{\boldsymbol{Q}}\boldsymbol{Q}^\top)^\top$: the tensor $\dot{\boldsymbol{Q}}\boldsymbol{Q}^\top$ is antisymmetric. (It is the angular velocity of the starred frame.)

Reading off $\boldsymbol{D}$ and $\boldsymbol{W}$. Split $\boldsymbol{L} = \boldsymbol{D} + \boldsymbol{W}$ into symmetric and antisymmetric parts:

$$\boldsymbol{L}^* = \underbrace{\boldsymbol{Q}\,\boldsymbol{D}\,\boldsymbol{Q}^\top}_{\text{symmetric}} + \underbrace{\boldsymbol{Q}\,\boldsymbol{W}\,\boldsymbol{Q}^\top + \dot{\boldsymbol{Q}}\,\boldsymbol{Q}^\top}_{\text{antisymmetric}}.$$

Here $\boldsymbol{Q}\boldsymbol{D}\boldsymbol{Q}^\top$ is symmetric because $\boldsymbol{D}$ is, while $\boldsymbol{Q}\boldsymbol{W}\boldsymbol{Q}^\top$ and $\dot{\boldsymbol{Q}}\boldsymbol{Q}^\top$ are each antisymmetric. Since the symmetric part of $\boldsymbol{L}^*$ is $\boldsymbol{D}^*$ and the antisymmetric part is $\boldsymbol{W}^*$, we read off

$$\boldsymbol{D}^* = \boldsymbol{Q}\,\boldsymbol{D}\,\boldsymbol{Q}^\top, \qquad \boldsymbol{W}^* = \boldsymbol{Q}\,\boldsymbol{W}\,\boldsymbol{Q}^\top + \dot{\boldsymbol{Q}}\,\boldsymbol{Q}^\top.$$

The rate of strain merely co-rotates with the observer and is therefore objective; the rotation tensor carries the extra term $\dot{\boldsymbol{Q}}\boldsymbol{Q}^\top$, the observer's own spin, and is not. This is why the simple Newtonian fluid of Sec. 3.6 is built on $\boldsymbol{D}$: a relation in which $\boldsymbol{W}$ appeared alone would assign different stresses to the same material state for two different observers. The conclusion is specific to that class of models. More general constitutive relations may involve $\boldsymbol{W}$, provided it enters in combinations from which the observer's spin cancels, as in the corotational derivatives used for viscoelastic fluids.

Appendix 3.C The Navier–Stokes Equations in Cartesian, Cylindrical, and Spherical Coordinates

The incompressible Navier–Stokes equations of Sec. 3.7 are written out here component by component, for a fluid of constant density ρ and constant shear viscosity μ, with $\boldsymbol{b}$ the body force per unit mass. Solving a concrete problem means selecting the coordinate system in which the boundaries are simple and then striking out the terms that the geometry kills, so these are the forms actually used in Ch. 4 and after.[15]

Cartesian Coordinates (x, y, z)

Continuity,

$$\frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z} = 0, \tag{3.17}$$

and the three components of momentum,

$$\rho\left(\frac{\partial v_x}{\partial t} + v_x\frac{\partial v_x}{\partial x} + v_y\frac{\partial v_x}{\partial y} + v_z\frac{\partial v_x}{\partial z}\right) = -\frac{\partial p}{\partial x} + \mu\,\nabla^2 v_x + \rho\, b_x, \tag{3.18}$$

$$\rho\left(\frac{\partial v_y}{\partial t} + v_x\frac{\partial v_y}{\partial x} + v_y\frac{\partial v_y}{\partial y} + v_z\frac{\partial v_y}{\partial z}\right) = -\frac{\partial p}{\partial y} + \mu\,\nabla^2 v_y + \rho\, b_y, \tag{3.19}$$

$$\rho\left(\frac{\partial v_z}{\partial t} + v_x\frac{\partial v_z}{\partial x} + v_y\frac{\partial v_z}{\partial y} + v_z\frac{\partial v_z}{\partial z}\right) = -\frac{\partial p}{\partial z} + \mu\,\nabla^2 v_z + \rho\, b_z, \tag{3.20}$$

with the scalar Laplacian

$$\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}.$$

Here, and only here, the vector Laplacian acts componentwise: because the Cartesian basis is fixed, $(\nabla^2\boldsymbol{v})_i = \nabla^2 v_i$. In the two curvilinear systems that is false, and the difference is the origin of every term below that carries no derivative.

Cylindrical Coordinates (r, θ, z)

Continuity,

$$\frac{1}{r}\frac{\partial (r v_r)}{\partial r} + \frac{1}{r}\frac{\partial v_\theta}{\partial \theta} + \frac{\partial v_z}{\partial z} = 0, \tag{3.21}$$

[15] The same components, with the transposed index convention for $\boldsymbol{\nabla v}$ and in a different notation, are tabulated by R. B. Bird, W. E. Stewart, and E. N. Lightfoot, *Transport Phenomena*, 2nd ed. (Wiley, New York, 2002), Appendix B.

and momentum,

$$\rho\left(\frac{\partial v_r}{\partial t}+v_r\frac{\partial v_r}{\partial r}+\frac{v_\theta}{r}\frac{\partial v_r}{\partial \theta}+v_z\frac{\partial v_r}{\partial z}-\frac{v_\theta^2}{r}\right) = -\frac{\partial p}{\partial r}+\mu\left[\nabla^2 v_r-\frac{v_r}{r^2}-\frac{2}{r^2}\frac{\partial v_\theta}{\partial \theta}\right]+\rho\, b_r, \tag{3.22}$$

$$\rho\left(\frac{\partial v_\theta}{\partial t}+v_r\frac{\partial v_\theta}{\partial r}+\frac{v_\theta}{r}\frac{\partial v_\theta}{\partial \theta}+v_z\frac{\partial v_\theta}{\partial z}+\frac{v_r v_\theta}{r}\right) = -\frac{1}{r}\frac{\partial p}{\partial \theta}+\mu\left[\nabla^2 v_\theta-\frac{v_\theta}{r^2}+\frac{2}{r^2}\frac{\partial v_r}{\partial \theta}\right]+\rho\, b_\theta, \tag{3.23}$$

$$\rho\left(\frac{\partial v_z}{\partial t}+v_r\frac{\partial v_z}{\partial r}+\frac{v_\theta}{r}\frac{\partial v_z}{\partial \theta}+v_z\frac{\partial v_z}{\partial z}\right) = -\frac{\partial p}{\partial z}+\mu\,\nabla^2 v_z+\rho\, b_z, \tag{3.24}$$

with the scalar Laplacian

$$\nabla^2=\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial}{\partial r}\right)+\frac{1}{r^2}\frac{\partial^2}{\partial \theta^2}+\frac{\partial^2}{\partial z^2}.$$

The term $-v_\theta^2/r$ in Eq. (3.22) is the centripetal acceleration and the term $+v_r v_\theta/r$ in Eq. (3.23) is the Coriolis-like acceleration; both belong to the material derivative, not to the viscous stress, and neither is present when the flow is unidirectional in the Cartesian sense. The pairs $-v_r/r^2$ with $-(2/r^2)\frac{\partial v_\theta}{\partial \theta}$, and $-v_\theta/r^2$ with $+(2/r^2)\frac{\partial v_r}{\partial \theta}$, are what separates the vector Laplacian from the scalar one. Two equivalent groupings appear often enough to be worth recording,

$$\nabla^2 v_r-\frac{v_r}{r^2}=\frac{\partial}{\partial r}\left(\frac{1}{r}\frac{\partial (r v_r)}{\partial r}\right)+\frac{1}{r^2}\frac{\partial^2 v_r}{\partial \theta^2}+\frac{\partial^2 v_r}{\partial z^2},$$

$$\nabla^2 v_\theta-\frac{v_\theta}{r^2}=\frac{\partial}{\partial r}\left(\frac{1}{r}\frac{\partial (r v_\theta)}{\partial r}\right)+\frac{1}{r^2}\frac{\partial^2 v_\theta}{\partial \theta^2}+\frac{\partial^2 v_\theta}{\partial z^2},$$

which is the form in which the annular and swirling flows of Ch. 4 are most easily integrated.

Spherical Coordinates (r,θ,ϕ)

With θ the polar angle and ϕ the azimuthal angle, continuity reads

$$\frac{1}{r^2}\frac{\partial (r^2 v_r)}{\partial r}+\frac{1}{r\sin\theta}\frac{\partial (\sin\theta\, v_\theta)}{\partial \theta}+\frac{1}{r\sin\theta}\frac{\partial v_\phi}{\partial \phi}=0, \tag{3.25}$$

and the momentum components are

$$\rho\left(\frac{\partial v_r}{\partial t}+v_r\frac{\partial v_r}{\partial r}+\frac{v_\theta}{r}\frac{\partial v_r}{\partial \theta}+\frac{v_\phi}{r\sin\theta}\frac{\partial v_r}{\partial \phi}-\frac{v_\theta^2+v_\phi^2}{r}\right)$$
$$=-\frac{\partial p}{\partial r}+\mu\left[\nabla^2 v_r-\frac{2v_r}{r^2}-\frac{2}{r^2}\frac{\partial v_\theta}{\partial \theta}-\frac{2v_\theta\cot\theta}{r^2}-\frac{2}{r^2\sin\theta}\frac{\partial v_\phi}{\partial \phi}\right]+\rho\, b_r, \tag{3.26}$$

$$\rho\left(\frac{\partial v_\theta}{\partial t}+v_r\frac{\partial v_\theta}{\partial r}+\frac{v_\theta}{r}\frac{\partial v_\theta}{\partial \theta}+\frac{v_\phi}{r\sin\theta}\frac{\partial v_\theta}{\partial \phi}+\frac{v_r v_\theta}{r}-\frac{v_\phi^2\cot\theta}{r}\right)$$
$$=-\frac{1}{r}\frac{\partial p}{\partial \theta}+\mu\left[\nabla^2 v_\theta+\frac{2}{r^2}\frac{\partial v_r}{\partial \theta}-\frac{v_\theta}{r^2\sin^2\theta}-\frac{2\cos\theta}{r^2\sin^2\theta}\frac{\partial v_\phi}{\partial \phi}\right]+\rho\, b_\theta, \tag{3.27}$$

$$\rho\left(\frac{\partial v_\phi}{\partial t}+v_r\frac{\partial v_\phi}{\partial r}+\frac{v_\theta}{r}\frac{\partial v_\phi}{\partial \theta}+\frac{v_\phi}{r\sin\theta}\frac{\partial v_\phi}{\partial \phi}+\frac{v_r v_\phi}{r}+\frac{v_\theta v_\phi\cot\theta}{r}\right)$$
$$=-\frac{1}{r\sin\theta}\frac{\partial p}{\partial \phi}+\mu\left[\nabla^2 v_\phi-\frac{v_\phi}{r^2\sin^2\theta}+\frac{2}{r^2\sin\theta}\frac{\partial v_r}{\partial \phi}+\frac{2\cos\theta}{r^2\sin^2\theta}\frac{\partial v_\theta}{\partial \phi}\right]+\rho\, b_\phi, \tag{3.28}$$

with the scalar Laplacian

$$\nabla^2=\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\right)+\frac{1}{r^2\sin\theta}\frac{\partial}{\partial \theta}\left(\sin\theta\,\frac{\partial}{\partial \theta}\right)+\frac{1}{r^2\sin^2\theta}\frac{\partial^2}{\partial \phi^2}.$$

These are the equations behind the creeping flow past a sphere of **??**. A useful check on any of the three sets: put $\boldsymbol{v}$ equal to a rigid rotation and every viscous bracket must vanish, since a rigid motion deforms nothing.

Exercises

Repeated indices are summed and fields are as smooth as needed. Parts marked [Practice] are for self-study.

1. **(Per-unit-mass transport theorem)** Starting from the Reynolds transport theorem (Thm. 3.2) with $\phi=\rho\psi$ and using the continuity equation (Thm. 3.6), prove $\dfrac{d}{dt}\displaystyle\int_{V(t)}\rho\psi\,\mathrm{d}V=\int_{V(t)}\rho\,\frac{\mathrm{D}\psi}{\mathrm{D}t}\,\mathrm{d}V$. Identify precisely where continuity is used.

2. **(Continuity in disguise)**

   (a) Write the continuity equation in conservative and material forms and show they are equivalent.

   (b) [Practice] Expand $\boldsymbol{\nabla}\cdot(\rho\boldsymbol{v})$ in cylindrical coordinates (r,θ,z) and state the continuity equation there.

3. **(Momentum in conservative form)** The conservative form

$$\frac{\partial}{\partial t}(\rho v_i) + \partial_j(\rho v_i v_j - T_{ij}) = \rho\, b_i$$

was derived in Sec. 3.4.

   (a) Verify the derivation by identifying, step by step, exactly where continuity is used and confirming that no term is dropped.

   (b) Recover Cauchy's first law $\rho\, \mathrm{D}v_i/\mathrm{D}t = \partial_j T_{ij} + \rho b_i$ from the conservative form by reversing the manipulation, and state precisely which terms cancel.

   (c) [Practice] Write the conservative form in the symbolic notation $\frac{\partial(\rho \boldsymbol{v})}{\partial t} + \boldsymbol{\nabla}\cdot(\rho\, \boldsymbol{v}\otimes\boldsymbol{v} - \boldsymbol{T}) = \rho\boldsymbol{b}$, and check that the two forms agree component by component. Interpret $\rho v_i v_j$ as a convective momentum flux and $-T_{ij}$ as a molecular momentum flux.

4. **(Symmetry of the stress)** Fill in the angular-momentum derivation of Sec. 3.5, and show that $\epsilon_{ijk}T_{kj} = 0$ is equivalent to $T_{ij} = T_{ji}$. How many independent components does $\boldsymbol{T}$ then have? What physical situation (a polar or Cosserat medium) would spoil the symmetry?

5. **(Reading the stress tensor)** At a point the stress tensor is T_{ij} (given, symmetric). For a surface with unit normal $\boldsymbol{n}$:

   (a) Find the traction $\boldsymbol{t} = \boldsymbol{T}\cdot\boldsymbol{n}$, its normal component $t_n = \boldsymbol{n}\cdot\boldsymbol{T}\cdot\boldsymbol{n}$, and the shear magnitude.

   (b) Explain why the eigenvalues of $\boldsymbol{T}$ are the principal stresses and its eigenvectors the directions of zero shear.

6. **(Cauchy's lemma)** Apply the balance of linear momentum to a thin disk-shaped pillbox of thickness h straddling a surface, and take $h \to 0$ to prove $\boldsymbol{t}(-\boldsymbol{n}) = -\boldsymbol{t}(\boldsymbol{n})$.

7. **(From Newtonian stress to the Laplacian.)** For an incompressible Newtonian fluid ($\boldsymbol{T} = -p\boldsymbol{I} + 2\mu\boldsymbol{D}$, constant μ), show in indices that $\partial_j T_{ij} = -\partial_i p + \mu\,\partial_j\partial_j v_i$, i.e. $\boldsymbol{\nabla}\cdot\boldsymbol{T} = -\boldsymbol{\nabla}p + \mu\nabla^2\boldsymbol{v}$. Hence complete the derivation of the Navier–Stokes equations.

8. **(Hydrostatics.)** For a fluid at rest ($\boldsymbol{v} = \boldsymbol{0}$) under gravity $\boldsymbol{b} = \boldsymbol{g}$, reduce the momentum balance to $\boldsymbol{\nabla}p = \rho\boldsymbol{g}$. Integrate to obtain the barometric pressure distribution for constant ρ, and comment on the compressible-atmosphere case.

9. **[practice] (Bernoulli, revisited.)** For steady inviscid incompressible flow, combine

the Euler equation with the convective-acceleration identity of Ch. 1 (Exercise 7 there) to show that $\tfrac{1}{2}|\boldsymbol{v}|^2 + p/\rho + gz$ is constant along streamlines.

10. (Mechanical energy from momentum.) Starting from $\rho\,\mathrm{D}\boldsymbol{v}/\mathrm{D}t = \boldsymbol{\nabla}\cdot\boldsymbol{T} + \rho\boldsymbol{b}$, form the dot product with $\boldsymbol{v}$ and derive the local mechanical energy equation of Sec. 3.8. Show explicitly that $T_{ij}\partial_j v_i = \boldsymbol{T} : \boldsymbol{D}$ requires the symmetry of $\boldsymbol{T}$.

11. (Dissipation and decay.) For an incompressible flow in a fixed domain with $\boldsymbol{v} = \boldsymbol{0}$ on the boundary, integrate the mechanical energy equation to show that the total kinetic energy can only decrease: $\dfrac{d}{dt}\displaystyle\int_V \tfrac{1}{2}\rho|\boldsymbol{v}|^2\,\mathrm{d}V = -\int_V 2\mu\,\boldsymbol{D}:\boldsymbol{D}\,\mathrm{d}V \le 0$. Interpret.

12. [practice] (Dissipation in a pipe.) For fully developed Hagen–Poiseuille flow, evaluate the dissipation Φ and integrate over the pipe to show that the total power dissipated equals the pumping power $Q\,\Delta p$, with Q the volumetric flow rate and Δp the pressure drop.

13. (Material volume versus control volume.) A balloon drifts on the wind while slowly inflating.

(a) Taking the balloon's interior as a *material* volume, state how its bounding surface moves and whether any air crosses it. What does the Reynolds surface term represent if no air crosses the boundary?

(b) Taking instead a *fixed* spherical control volume of the same instantaneous size and position, explain why air now crosses the boundary, and write the rate of change of the enclosed mass as a surface flux.

(c) For a control volume whose surface moves with velocity $\boldsymbol{w}$, explain why the boundary flux involves the relative velocity $(\boldsymbol{v} - \boldsymbol{w})\cdot\boldsymbol{n}$, and recover the material case ($\boldsymbol{w} = \boldsymbol{v}$) and the fixed case ($\boldsymbol{w} = \boldsymbol{0}$) as limits.

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4
Unidirectional and Nearly-Unidirectional Flows

When can we solve the Navier–Stokes equations without neglecting their nonlinear term? One useful case is flow along straight, parallel paths, with no variation in the direction of motion. A particle then encounters no change in velocity as it travels, so its convective acceleration vanishes. The remaining equation is linear, even though we began with the full equations of motion. These are the *unidirectional* (or *fully developed*) flows, and they furnish one of the few families of exact solutions available in closed form.

We will use these flows to connect velocity profiles to forces and transport: how a moving wall transmits shear, how a pressure difference sets a flow rate, and how viscosity spreads motion through a fluid. Circular flow provides a useful contrast, because a particle accelerates even when its speed is constant. Thin gaps introduce a different simplification, in which small terms are estimated and neglected rather than vanishing exactly.

Their importance reaches beyond the specific geometries they describe. Each solution highlights a distinct physical mechanism: the linear velocity profile of Couette flow reflects the uniform transmission of shear stress across a viscous gap; the parabolic profile of Poiseuille flow encodes the balance between a driving pressure gradient and viscous resistance; the transient solutions to start-up problems reveal the diffusive character of momentum transport and the time scales on which viscosity acts. These flows also form the basis for experimental measurement techniques—the Couette viscometer, capillary rheometry, the rotating-spindle rheometer—making them as practically relevant as they are theoretically transparent.

In this chapter we develop a systematic approach to unidirectional flow problems. We begin with the Navier–Stokes equations in the form used throughout, then identify the structure that makes unidirectional flows tractable. We proceed through the canonical steady flows: plane Couette flow, two-layer Couette flow with interfacial matching, Hagen–Poiseuille flow in a circular pipe, and the rotating-spindle viscometer. Next, we examine the nearly unidirectional thin-film flows of lubrication theory, and finally two transient problems: the start-up of Couette flow (solved via Sturm–Liouville theory) and the diffusion of vorticity from an impulsively started plate (the Rayleigh problem, solved by similarity). A few more problems are included as exercises for the reader. Each problem follows the same logical

template, and working through them builds both analytical technique and physical intuition for viscous flows.

4.1 The Navier–Stokes Equations

We consider a viscous, incompressible, Newtonian fluid with constant density ρ and shear viscosity μ. The fluid state is described by the velocity field $\boldsymbol{v}(\boldsymbol{x}, t)$ and the pressure $p(\boldsymbol{x}, t)$.

Important

Incompressible Navier–Stokes Equationsnavier-stokes Conservation of mass and linear momentum for an incompressible Newtonian fluid yield

$$\boldsymbol{\nabla} \cdot \boldsymbol{v} = 0\,, \tag{4.1}$$

$$\rho\left(\frac{\partial \boldsymbol{v}}{\partial t} + (\boldsymbol{\nabla}\boldsymbol{v})\,\boldsymbol{v}\right) = -\boldsymbol{\nabla} p + \mu\,\nabla^2 \boldsymbol{v} + \rho\,\boldsymbol{g}\,, \tag{4.2}$$

where $\boldsymbol{g}$ is the gravitational acceleration and the left-hand side is the inertial (material) acceleration of the fluid. Equations Eqs. (4.1) and (4.2) form a closed system for $(\boldsymbol{v}, p)$ given suitable initial and boundary conditions.

The term $\rho\,(\boldsymbol{\nabla}\boldsymbol{v})\,\boldsymbol{v}$ is the *convective acceleration*: it is quadratic in $\boldsymbol{v}$ and is the source of the fundamental nonlinearity of the Navier–Stokes equations. For all flows in this chapter this term will vanish identically.

Solving a problem means choosing the coordinate system in which the boundaries are simple and then striking out the terms that the geometry kills. The component forms of Eqs. (4.1) and (4.2) in Cartesian, cylindrical, and spherical coordinates are collected in Appendix 3.C, and the components of the velocity gradient $\boldsymbol{\nabla}\boldsymbol{v}$, from which the stresses and the vorticity are built, in Appendix 2.A. We refer to those two tables at each reduction below rather than rederiving them.

Remark 4.1 | Modified pressure

When gravity acts in a fixed direction, it is often convenient to absorb the hydrostatic contribution into the pressure by defining the *modified* (or, *dynamic*) pressure $\mathcal{P} = p + \rho\,\phi_g$, where $\boldsymbol{g} = -\boldsymbol{\nabla}\phi_g$. The momentum equation then takes the form $\rho\,\mathrm{D}\boldsymbol{v}/\mathrm{D}t = -\boldsymbol{\nabla}\mathcal{P} + \mu\nabla^2\boldsymbol{v}$, with gravity absent explicitly. We will switch between p and $\mathcal{P}$ as convenient, always stating clearly which is in use.

Remark 4.2 | No-slip condition

At every solid boundary, a viscous fluid satisfies the *no-slip condition*: the fluid velocity equals the velocity of the boundary. This is the only boundary condition needed for the velocity field in the problems below. An additional condition on p (or $\mathcal{P}$) is not required at solid walls because pressure in an incompressible flow is determined globally by the divergence-free constraint, not locally by a wall condition.

4.2 Structure of Unidirectional Flows

A flow is called *unidirectional* if, in some Cartesian coordinate system (x, y, z), the velocity takes the form

$$\boldsymbol{v}(\boldsymbol{x}, t) \;=\; u(y, z, t)\, \boldsymbol{e}_x \,. \tag{4.3}$$

The fluid moves solely in the x-direction; the speed u varies in the transverse directions (y, z) and in time, but not along the flow direction x (full development).

Effect on the continuity equation. Substituting Eq. (4.3) into Eq. (4.1),

$$\boldsymbol{\nabla} \cdot \boldsymbol{v} = \frac{\partial u}{\partial x} = 0 \,,$$

which is satisfied automatically since u is independent of x by assumption. The incompressibility constraint imposes no additional restriction on u.

Effect on the momentum equation. The convective acceleration evaluates to

$$(\boldsymbol{\nabla} \boldsymbol{v})\, \boldsymbol{v} = u\, \frac{\partial}{\partial x}(u\, \boldsymbol{e}_x) = u\, \frac{\partial u}{\partial x}\, \boldsymbol{e}_x = \boldsymbol{0} \,,$$

using $\partial u / \partial x = 0$. The nonlinear term disappears. The three scalar components of Eq. (4.2) in Cartesian coordinates, Eqs. (3.18)–(3.20), reduce to

$$\rho\, \frac{\partial u}{\partial t} = -\frac{\partial p}{\partial x} + \mu \left(\frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} \right) + \rho\, g_x \,, \tag{4.4}$$

$$0 = -\frac{\partial p}{\partial y} + \rho\, g_y \,, \tag{4.5}$$

$$0 = -\frac{\partial p}{\partial z} + \rho\, g_z \,. \tag{4.6}$$

Equations Eqs. (4.5) and (4.6) express hydrostatic equilibrium in the transverse directions and can be integrated immediately to find p as a function of (y, z). Equation Eq. (4.4) is then a *linear* scalar PDE for u, driven by the streamwise pressure gradient and body force.

Important | Systematic procedure for unidirectional flows

Each problem in this chapter is solved by the following steps.

(i) **Postulate** the velocity form $\boldsymbol{v} = u(y, z, t)\, \boldsymbol{e}_x$ based on the geometry and symmetry.

(ii) **Verify** that continuity is satisfied automatically.

(iii) **Reduce** the Navier–Stokes equations, in the component form of Appendix 3.C for the coordinate system that fits the geometry: ~~drop the convective term~~evaluate each component of the acceleration $\mathrm{D}\boldsymbol{v}/\mathrm{D}t$ for the postulated field, and keep whatever survives, use the transverse momentum equations to determine p where appropriate, and extract the scalar PDE (or ODE) for u.

(iv) **Apply** ~~boundary conditions (no-slip at walls; symmetry conditions where appropriate)~~all the data the particular model needs: wall, interface, end, far-field, and, for a transient, initial conditions.

(v) **Solve** for u and, where needed, p.

(vi) **Interpret**: compute the shear stress, net flow rate, and forces on boundaries; identify the controlling dimensionless parameters.

~~The ansatz defines the first and simplest of three classes of flow treated in this chapter, in which the convective acceleration vanishes identically. Two later problems relax this.~~ The ansatz Eq. (4.3) is only the simplest of four situations treated in this chapter. They differ in what becomes of inertia, and the distinction is worth absorbing before we start.

Class	Example	What happens to inertia?
Straight parallel flow	Plane Couette, pipe flow	Convective acceleration is exactly zero.
Unsteady parallel flow	Start-up Couette, Rayleigh	Convective term zero; local term retained.
Circular flow	Rotating spindle	Azimuthal part vanishes; centripetal part remains.
Thin-gap flow	Slider and squeeze films	Neglected under explicit scale restrictions.

Only the first two rows have an acceleration that vanishes identically. In the rotating spindle the azimuthal balance is linear, but the radial balance retains the centripetal term v_θ^2/r, which fixes the pressure rather than the velocity; a particle there accelerates even though its speed never changes. In the thin-film flows inertia does not vanish at all, but is small under conditions we shall have to state explicitly (Sec. 4.7).

An exact velocity field is not automatically the flow realized in an experiment. The solution must also be compatible with the entrance conditions, the end effects, and whatever disturbances are present. The parallel-flow reductions do *not* require a small Reynolds number, and nothing in this chapter assumes one. A measured flow can nonetheless depart from these solutions when disturbances grow or when the assumed geometry is a poor approximation.

4.3 Plane Couette Flow

4.3.1 Problem Statement

The simplest unidirectional flow is generated by the relative motion of two parallel plates with a viscous fluid filling the gap between them. Consider two infinite, horizontal plates separated by a distance H (Fig. 4.1). The lower plate at $y = 0$ is held stationary; the upper plate at $y = H$ is driven at a constant velocity $U > 0$ in the x-direction. There is no externally imposed pressure gradient. Gravity acts in the $-y$ direction; working with the modified pressure $\mathcal{P} = p + \rho g y$ removes it from the equations.

Definition 4.3 | Plane Couette Flow

An incompressible Newtonian fluid occupies the region $0 \le y \le H$ between two infinite parallel plates. The boundary conditions on the velocity are

$$u\big|_{y=0} = 0\,, \qquad u\big|_{y=H} = U\,, \tag{4.7}$$

and the modified pressure satisfies $\partial \mathcal{P}/\partial x = 0$ (no imposed pressure gradient).

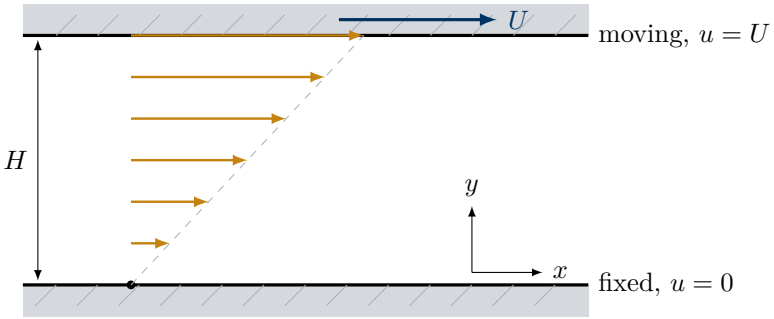


Figure 4.1: Plane Couette flow geometry. The lower plate is stationary; the upper plate moves at speed U in the x-direction. Gold arrows show the linear velocity profile $u(y) = Uy/H$; their tips lie on the dashed line.

4.3.2 Reduction of the Governing Equations

Following Step 1 of the general procedure, the flow is steady, symmetric in z, and driven by the boundary alone. We therefore postulate

$$\boldsymbol{v} = u(y)\,\boldsymbol{e}_x\,. \tag{4.8}$$

Continuity is satisfied identically. With $\partial\mathcal{P}/\partial x = 0$, $\rho\, g_x = 0$, and $\partial/\partial t = 0$, the streamwise component Eq. (4.4), which is the Cartesian x-equation Eq. (3.18) stripped of everything the geometry removes, reduces to the ordinary differential equation

$$\mu\,\frac{d^2u}{dy^2} = 0\,. \tag{4.9}$$

This states that the shear stress $\tau = \mu\, du/dy$ is constant across the gap: in steady flow with no pressure gradient, the viscous force on every fluid layer must vanish, so momentum is transmitted uniformly from the moving plate to the stationary one.

4.3.3 Solution

Integrating Eq. (4.9) twice yields the general solution

$$u(y) = C_1\, y + C_2\,.$$

Applying the boundary conditions Eq. (4.7): $u(0) = 0$ gives $C_2 = 0$, and $u(H) = U$ gives $C_1 = U/H$.

Theorem 4.4 | Couette Flow Velocity Profile

The unique solution to plane Couette flow is the linear profile

$$\boxed{u(y) \;=\; U\,\frac{y}{H}\,.} \tag{4.10}$$

The velocity varies linearly from zero at the lower plate to U at the upper plate.

4.3.4 Vorticity

The vorticity $\boldsymbol{\omega} = \boldsymbol{\nabla}\times\boldsymbol{v}$ measures the local rotation rate of fluid elements. For $\boldsymbol{v} = (Uy/H)\,\boldsymbol{e}_x$, the three components are

$$\begin{aligned}\omega_x &= \frac{\partial v_z}{\partial y} - \frac{\partial v_y}{\partial z} = 0\,,\\ \omega_y &= \frac{\partial v_x}{\partial z} - \frac{\partial v_z}{\partial x} = 0\,,\\ \omega_z &= \frac{\partial v_y}{\partial x} - \frac{\partial v_x}{\partial y} = -\frac{du}{dy} = -\frac{U}{H}\,.\end{aligned}$$

Therefore

$$\boldsymbol{\omega} = -\frac{U}{H}\,\boldsymbol{e}_z\,. \tag{4.11}$$

The vorticity is *spatially uniform*: every fluid parcel rotates at the same angular rate $U/(2H)$ about the $-z$ axis, a direct consequence of the constant velocity gradient.

4.3.5 The Full Stress Tensor and Forces on the Plates

Rate-of-deformation tensor. For a Newtonian fluid, $\boldsymbol{T} = -p\,\boldsymbol{I} + 2\mu\,\boldsymbol{D}$ where $\boldsymbol{D} = \frac{1}{2}(\boldsymbol{\nabla v} + (\boldsymbol{\nabla v})^{\top})$. The velocity gradient has components $(\boldsymbol{\nabla v})_{ij} = \partial v_i/\partial x_j$; for our flow the only nonzero entry is $(\boldsymbol{\nabla v})_{12} = U/H$. Thus

$$\boldsymbol{D} = \frac{U}{2H}\begin{pmatrix}0 & 1 & 0\\ 1 & 0 & 0\\ 0 & 0 & 0\end{pmatrix},$$

representing pure shear with no normal straining or dilatation.

Full stress tensor. Substituting into $\boldsymbol{T} = -p\,\boldsymbol{I} + 2\mu\,\boldsymbol{D}$:

$$\boldsymbol{T} = \begin{pmatrix}-p & \mu U/H & 0\\ \mu U/H & -p & 0\\ 0 & 0 & -p\end{pmatrix}. \tag{4.12}$$

The diagonal entries are the isotropic pressure; the off-diagonal xy and yx entries are the viscous shear stress $\tau_{yx} = \mu U/H$, which is constant throughout the fluid.

Traction on the lower plate ($y = 0$). The traction (force per unit area exerted by the fluid on a surface) is $\boldsymbol{t} = \boldsymbol{T}\cdot\boldsymbol{n}$, where $\boldsymbol{n}$ is the outward unit normal from the plate into the

fluid. For the lower plate, $\boldsymbol{n} = +\boldsymbol{e}_y$:

$$\boldsymbol{t}\big|_{y=0} = \boldsymbol{T}\,\boldsymbol{e}_y = \begin{pmatrix} -p & \mu U/H & 0 \\ \mu U/H & -p & 0 \\ 0 & 0 & -p \end{pmatrix} \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} \mu U/H \\ -p \\ 0 \end{pmatrix}. \tag{4.13}$$

The x-component $\mu U/H$ is a viscous drag in the direction of plate motion: the fluid would pull the lower plate in the $+x$ direction if it were free. The y-component $-p$ is the compressive normal stress.

Traction on the upper plate ($y = H$). The upper plate's outward normal (pointing down into the fluid) is $\boldsymbol{n} = -\boldsymbol{e}_y$:

$$\boldsymbol{t}\big|_{y=H} = \boldsymbol{T}\,(-\boldsymbol{e}_y) = \begin{pmatrix} -p & \mu U/H & 0 \\ \mu U/H & -p & 0 \\ 0 & 0 & -p \end{pmatrix} \begin{pmatrix} 0 \\ -1 \\ 0 \end{pmatrix} = \begin{pmatrix} -\mu U/H \\ p \\ 0 \end{pmatrix}. \tag{4.14}$$

The fluid resists the plate motion with drag $\mu U/H$ in the $-x$ direction. To sustain velocity U, an external force of $\mu U/H$ per unit area in the $+x$ direction must be applied—this is the operating principle of the Couette viscometer.

Example 4.5 | Couette viscometer

A pair of parallel plates separated by $H = 1\,\mathrm{mm}$ is used to characterize a polymer solution. The upper plate is driven at $U = 0.5\,\mathrm{m\,s^{-1}}$ and the shear stress measured on the lower plate is $\tau = 25\,\mathrm{Pa}$. Determine the shear viscosity μ.

From Eq. (4.13), the tangential traction on the lower plate equals $\mu U/H$. Therefore

$$\mu = \frac{\tau H}{U} = \frac{25 \times 10^{-3}}{0.5} = 0.05\ \mathrm{Pa \cdot s},$$

about 50 times the viscosity of water at 20°C—consistent with a moderately concentrated polymer solution.

Remark 4.6 | Linear profile from the force balance

In steady flow with no pressure gradient, the net viscous force on any fluid slab is zero, requiring $d\tau_{yx}/dy = \mu\, d^2u/dy^2 = 0$ and hence a constant velocity gradient. The unique function with constant du/dy satisfying $u(0) = 0$ and $u(H) = U$ is the linear profile Eq. (4.10); the ODE Eq. (4.9) encodes exactly this argument.

4.4 Two-Layer Couette Flow and Interfacial Matching

Plane Couette flow transmits a uniform shear stress across a single viscous gap. The next canonical problem shows what happens when *two* immiscible fluids share that gap: the shear stress is still uniform, but the velocity gradient jumps at the interface, and the matching of velocity and traction there is a direct application of the fluid-interface conditions of Sec. 3.9.2.

Geometry and problem statement. Two immiscible, incompressible Newtonian fluids fill the gap between parallel plates (Fig. 4.2). The lower plate at $y = 0$ is stationary; the upper plate at $y = H$ moves with velocity $U\,\boldsymbol{e}_x$. A flat, stationary interface at $y = h$ separates fluid 1 ($0 \le y \le h$, viscosity μ_1) below from fluid 2 ($h \le y \le H$, viscosity μ_2) above. There is no imposed streamwise pressure gradient, surface tension is uniform (so there is no Marangoni stress), and the interface remains flat. We seek the steady, fully developed field

$$\boldsymbol{v}^{(1)} = u_1(y)\,\boldsymbol{e}_x, \qquad \boldsymbol{v}^{(2)} = u_2(y)\,\boldsymbol{e}_x. \tag{4.15}$$

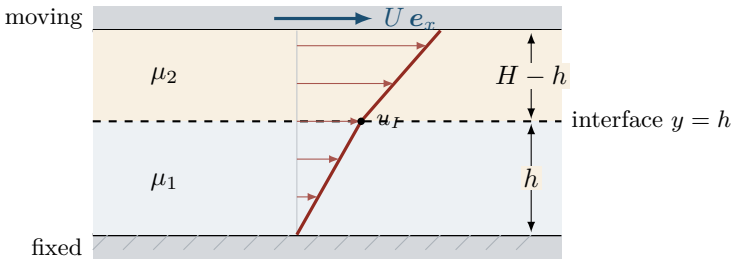


Figure 4.2: Two-layer Couette flow. The upper plate at $y = H$ moves at U; the lower plate is fixed. Fluids of viscosity μ_1 (lower) and μ_2 (upper) meet at a flat interface $y = h$. The velocity profile (red) is piecewise linear: its slope changes at the interface because the viscosities differ, while the shear stress τ is continuous. The interface moves at u_I.

Governing equations. Each layer is an exact unidirectional flow, so the convective term vanishes and, with no pressure gradient, the streamwise momentum equation Eq. (3.18) reduces within each fluid to

$$\mu_1 \frac{d^2 u_1}{dy^2} = 0, \qquad \mu_2 \frac{d^2 u_2}{dy^2} = 0. \tag{4.16}$$

The velocity is therefore linear within each fluid; the two slopes need not agree.

Wall and interface conditions. No-slip at the two plates gives

$$u_1(0) = 0, \qquad u_2(H) = U. \tag{4.17}$$

At the clean interface $y = h$, the conditions of Sec. 3.9.2 require continuity of velocity and of tangential traction,

$$u_1(h) = u_2(h), \qquad \mu_1\, u_1'(h) = \mu_2\, u_2'(h). \tag{4.18}$$

The first expresses that neither fluid penetrates the interface and that a clean interface supports no slip between the fluids; the second is continuity of tangential traction, which holds when surface tension is uniform. The central lesson deserves to be stated plainly:

Important | Velocity and traction matching

Velocity is continuous at a clean interface, and tangential traction is continuous when surface tension is uniform; the *velocity gradient* is generally discontinuous when the viscosities differ.

Solution through the common shear stress. Since $\mu_k u_k'' = 0$ in each layer and the tangential traction is continuous across the interface, the shear stress

$$\tau = \mu_1 \frac{du_1}{dy} = \mu_2 \frac{du_2}{dy} \tag{4.19}$$

is one and the same constant throughout both fluids. Integrating with $u_1(0) = 0$, integrating with $u_2(H) = U$, and matching the velocities at $y = h$ fixes its value:

$$\boxed{\tau = \frac{U}{\dfrac{h}{\mu_1} + \dfrac{H-h}{\mu_2}}} \tag{4.20}$$

The profiles are then piecewise linear,

$$u_1(y) = \frac{\tau}{\mu_1}\, y \quad (0 \le y \le h), \qquad u_2(y) = \frac{\tau h}{\mu_1} + \frac{\tau}{\mu_2}(y - h) \quad (h \le y \le H), \tag{4.21}$$

and the interface moves at

$$u_I = u_1(h) = u_2(h) = \frac{\tau h}{\mu_1} = U\, \frac{h/\mu_1}{h/\mu_1 + (H-h)/\mu_2}. \tag{4.22}$$

Resistance in series. Rearranging Eq. (4.20) as

$$U = \tau \left(\frac{h}{\mu_1} + \frac{H-h}{\mu_2} \right) \tag{4.23}$$

displays the plate-speed difference as the sum of two velocity drops driven by the *same* shear stress τ. Each term—h/μ_1 for the lower layer and $(H-h)/\mu_2$ for the upper—is the velocity drop per unit shear stress carried by that layer, a compliance (an inverse shear resistance).

Just as springs sharing a common force add their compliances in series, the two layers add theirs: the softer, less viscous layer accommodates the larger share of the velocity drop, and the total compliance is the sum of the two.

Dimensionless ratios and limits. Introduce the viscosity ratio and thickness fraction

$$m = \frac{\mu_2}{\mu_1}, \qquad \alpha = \frac{h}{H}. \tag{4.24}$$

The interface velocity Eq. (4.22) becomes

$$\frac{u_I}{U} = \frac{m\alpha}{m\alpha + (1-\alpha)}, \tag{4.25}$$

while the piecewise slopes are $u_1' = \tau/\mu_1$ and $u_2' = \tau/\mu_2$, in ratio $u_2'/u_1' = 1/m$. The limits are instructive:

- $m = 1$: the two fluids behave as one, $u_I/U = \alpha$, and the profile is a single straight line.
- $m \gg 1$ (viscous fluid on top): $u_I/U \to 1$; the upper fluid shears little and moves nearly as a plug with the upper plate, so almost the entire velocity drop occurs in the less viscous lower fluid.
- $m \ll 1$ (less viscous fluid on top): $u_I/U \to 0$; the interface velocity approaches that of the lower plate and almost the entire drop occurs in the upper fluid.
- $\alpha \to 0$ or $\alpha \to 1$: one layer fills the gap and the single-fluid Couette profile of Sec. 4.3 is recovered.

Vorticity, strain, and dissipation. Within each layer the vorticity and rate of strain are constant,

$$\omega_z^{(k)} = -u_k'(y) = -\frac{\tau}{\mu_k}, \qquad D_{xy}^{(k)} = D_{yx}^{(k)} = \tfrac{1}{2}u_k'(y) = \frac{\tau}{2\mu_k}, \tag{4.26}$$

so vorticity and shear rate are piecewise constant and jump across the interface whenever $\mu_1 \neq \mu_2$, even though the shear stress τ passes through continuously. The viscous dissipation per unit plate area is

$$\mathcal{D} = \int_0^h \mu_1 \,(u_1')^2 \, dy + \int_h^H \mu_2 \,(u_2')^2 \, dy = \tau^2\left(\frac{h}{\mu_1} + \frac{H-h}{\mu_2}\right) = \tau U, \tag{4.27}$$

which is exactly the mechanical power τU delivered by the moving plate per unit area—the steady mechanical-energy balance of Sec. 3.8 with dissipation Sec. 3.8.2.

Remark 4.7 | Prescribed Marangoni stress

If surface tension varies along the interface, the continuity of tangential traction in Eq. (4.18) is replaced by the interfacial stress jump of Sec. 3.9.2: the jump in tangential traction balances the surface-tension gradient $\partial\gamma/\partial x$. A prescribed $\partial\gamma/\partial x$ can then drive motion even when *both* plates are stationary—the mechanism of Marangoni flows. Solving the coupled thermal or surfactant transport that sets $\gamma(x)$ is beyond our scope here.

4.5 Hagen–Poiseuille Flow

Problem Statement. We consider a viscous fluid driven through a straight circular pipe of radius R by a constant imposed pressure gradient. We adopt cylindrical coordinates (r, θ, z) with z along the pipe axis and define

$$G \;=\; -\frac{dp}{dz} > 0 \tag{4.28}$$

as the pressure drop per unit length, so the fluid is driven in the $+z$ direction.

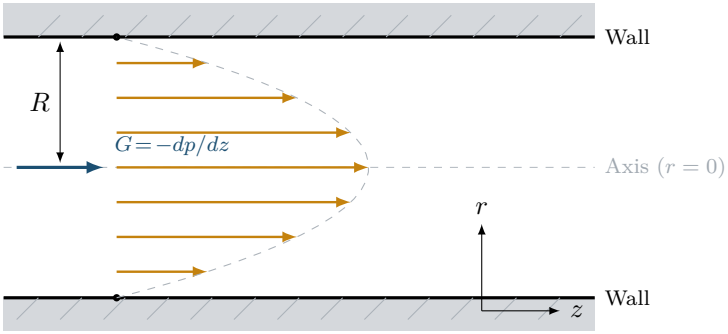


Figure 4.3: Longitudinal cross-section of Hagen–Poiseuille flow in a circular pipe of radius R. The axis of symmetry is the dashed centreline. Gold arrows show the paraboloidal profile $v_z = v_{\max}(1 - r^2/R^2)$; the pressure gradient G drives the flow to the right.

Definition 4.8 | Hagen–Poiseuille Flow

An incompressible Newtonian fluid occupies a circular pipe $0 \le r \le R$. The boundary conditions are

$$v_z\big|_{r=R} = 0 \quad \text{(no-slip)}\,, \qquad \left.\frac{dv_z}{dr}\right|_{r=0} = 0 \quad \text{(axial symmetry)}\,. \tag{4.29}$$

Reduction of the Governing Equations. By axisymmetry and translational invariance in z, we postulate $\boldsymbol{v} = v_z(r)\,\boldsymbol{e}_z$. Continuity ($\partial v_z/\partial z = 0$) is satisfied identically, and the

convective term vanishes. The z-momentum equation in cylindrical coordinates, Eq. (3.24), reduces for steady axisymmetric fully developed flow to

$$\frac{\mu}{r}\frac{d}{dr}\left(r\frac{dv_z}{dr}\right) = -G\,. \tag{4.30}$$

This is the cylindrical analogue of the constant-coefficient ODEs in the previous sections: the operator $(1/r)\,d(r\,d/dr)/dr$ is the cylindrical Laplacian, which accounts for the fact that larger-radius shells contain proportionally more fluid.

Solution. Multiplying Eq. (4.30) by r/μ and integrating once,

$$r\,\frac{dv_z}{dr} = -\frac{G}{2\mu}\,r^2 + C_1\,.$$

For $C_1 \neq 0$, $dv_z/dr \sim C_1/r$ diverges at $r = 0$; the symmetry condition requires $C_1 = 0$. Integrating again and applying no-slip:

Theorem 4.9 | Hagen–Poiseuille Profile

The velocity field is the paraboloid

$$\boxed{v_z(r) = \frac{G}{4\mu}\left(R^2 - r^2\right) = v_{\max}\left[1 - \left(\frac{r}{R}\right)^2\right]}\,, \tag{4.31}$$

with centreline velocity $v_{\max} = GR^2/(4\mu)$.

The volume flow rate through the cross-section is

$$Q = \int_0^R v_z(r)\,2\pi r\,dr = \frac{\pi G}{2\mu}\int_0^R (R^2 - r^2)\,r\,dr = \frac{\pi G R^4}{8\mu}\,. \tag{4.32}$$

This is the *Hagen–Poiseuille law*: Q grows as the *fourth power* of R. The cross-sectionally averaged velocity is $\langle v_z\rangle = Q/(\pi R^2) = GR^2/(8\mu) = v_{\max}/2$.

Vorticity. For $\boldsymbol{v} = v_z(r)\,\boldsymbol{e}_z$, the cylindrical velocity gradient Eq. (2.14) leaves a single non-zero vorticity component,

$$\omega_\theta = -\frac{dv_z}{dr} = \frac{Gr}{2\mu}\,, \qquad \boldsymbol{\omega} = \frac{Gr}{2\mu}\,\boldsymbol{e}_\theta\,. \tag{4.33}$$

The vorticity is azimuthal and increases *linearly* from zero on the axis to $GR/(2\mu)$ at the wall.

The Full Stress Tensor and Force on the Wall. For $\boldsymbol{v} = v_z(r)\,\boldsymbol{e}_z$, the symmetric part of Eq. (2.14) has a single off-diagonal component: $D_{rz} = D_{zr} = \tfrac{1}{2}(dv_z/dr) = -Gr/(4\mu)$. With cylindrical components ordered (r, θ, z):

$$\boldsymbol{T} = \begin{pmatrix} -p & 0 & -Gr/2 \\ 0 & -p & 0 \\ -Gr/2 & 0 & -p \end{pmatrix}. \tag{4.34}$$

Traction at the wall ($r = R$). Fix the convention once, since it recurs in every example of this chapter. If $\boldsymbol{n}_f$ points outward from the *fluid*, then $\boldsymbol{T}\cdot\boldsymbol{n}_f$ is the traction the *wall exerts on the fluid*; the fluid exerts the opposite traction $-\boldsymbol{T}\cdot\boldsymbol{n}_f$ on the wall. Here the outward normal from the fluid into the wall is $\boldsymbol{n} = +\boldsymbol{e}_r$, so the display below is the traction *on the fluid*:

$$\boldsymbol{t}\big|_{r=R} = \boldsymbol{T}\,\boldsymbol{e}_r\big|_{r=R} = \begin{pmatrix} -p & 0 & -GR/2 \\ 0 & -p & 0 \\ -GR/2 & 0 & -p \end{pmatrix}\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} -p \\ 0 \\ -GR/2 \end{pmatrix}. \tag{4.35}$$

~~The z-component $-GR/2$ is the viscous drag per unit area opposing the flow.~~ Written out, the two tractions are

$$\boldsymbol{t}_{\text{wall}\to\text{fluid}} = -p\,\boldsymbol{e}_r - \frac{GR}{2}\,\boldsymbol{e}_z, \qquad \boldsymbol{t}_{\text{fluid}\to\text{wall}} = p\,\boldsymbol{e}_r + \frac{GR}{2}\,\boldsymbol{e}_z.$$

The signs say what one expects: the wall retards the fluid, and the fluid pulls the pipe downstream. On a length L the axial wall force on the fluid is $-\pi GR^2 L\,\boldsymbol{e}_z$, which balances the driving force $+\Delta P\,\pi R^2\boldsymbol{e}_z$ when $G = \Delta P/L$ and there is no axial gravity. This is the axial balance only; it should not be identified with the integral of all wall tractions when transverse gravity or other forces are retained.

Example 4.10 | Pipe flow viscometry

~~Blood flows through a capillary of radius 4 micrometres and length 1 mm under a pressure drop of 1 Pa. Estimate the flow rate.~~

A Newtonian calibration liquid has viscosity $\mu = 10^{-3}\,\text{Pa}\cdot\text{s}$ and density $\rho = 10^{3}\,\text{kg}\,\text{m}^{-3}$. It flows through a rigid circular tube of radius $R = 50\,\mu\text{m}$ and length $L = 1\,\text{cm}$ under a pressure drop $\Delta P = 100\,\text{Pa}$. Assuming fully developed flow, estimate the flow rate.

With $G = \Delta P/L$,

$$Q = \frac{\pi R^4\,\Delta P}{8\mu L} \simeq 2.45\times10^{-11}\ \text{m}^3\,\text{s}^{-1} = 1.47\ \mu\text{L}\,\text{min}^{-1}.$$

As a consistency check on the model rather than a condition for it, the mean speed is $Q/\pi R^2 \simeq 3.13\,\mathrm{mm\,s^{-1}}$ and the diameter-based Reynolds number is $\rho\bar{u}(2R)/\mu \simeq 0.31$. Blood in a capillary of radius $4\,\mu$m is deliberately *not* the example used here. The same formula would give a number, but red cells are not small compared with such a channel, so the homogeneous Newtonian model that produced Eq. (4.31) does not apply and the number should not be read as a physiological prediction. Asking which assumption fails first is a better exercise than evaluating the formula.

Remark 4.11 | R^4 sensitivity

A 10% reduction in pipe radius reduces Q by $1-(0.9)^4 \approx 34\%$. This fourth-power sensitivity explains the dramatic effect of even partial occlusion in blood vessels or microfluidic channels.

4.6 The Rotating Spindle Viscometer

Problem Statement. We consider a Newtonian fluid filling the annular gap between two coaxial cylinders of radii $R_1 < R_2$. The inner cylinder (the spindle) rotates at constant angular velocity Ω; the outer cylinder is fixed. Measuring the torque required to maintain the rotation at known Ω yields μ directly.

We use cylindrical coordinates (r, θ, z) with z along the common axis.

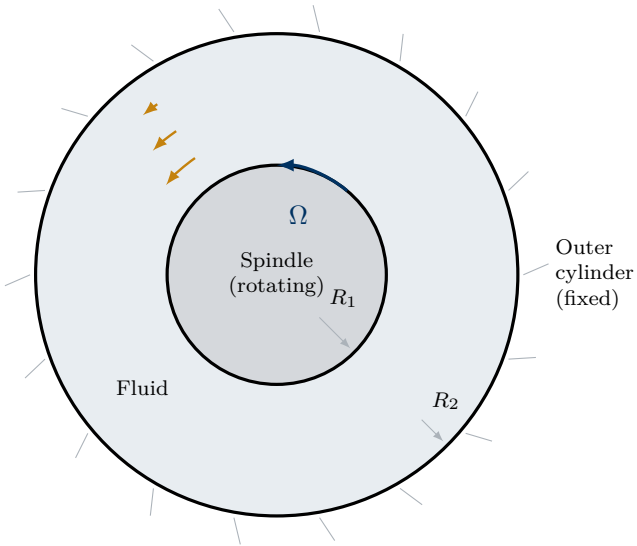


Figure 4.4: Cross-section (plan view) of the rotating spindle viscometer. The inner cylinder of radius R_1 rotates at angular velocity Ω; the outer cylinder of radius R_2 is stationary. Gold arcs show the azimuthal velocity in the gap, largest near the inner cylinder.

Definition 4.12 | Rotating Spindle Flow

An incompressible Newtonian fluid occupies the annulus $R_1 \le r \le R_2$. The boundary conditions are

$$v_\theta\big|_{r=R_1} = \Omega R_1 \quad \text{(inner cylinder, rotating)}\,, \tag{4.36}$$

$$v_\theta\big|_{r=R_2} = 0 \quad \text{(outer cylinder, fixed)}. \tag{4.37}$$

Reduction of the Governing Equations. By axisymmetry and z-invariance we postulate $\boldsymbol{v} = v_\theta(r)\,\boldsymbol{e}_\theta$. Continuity is satisfied automatically. The r-momentum equation Eq. (3.22) gives $\partial p/\partial r = \rho v_\theta^2/r$ (centripetal balance), which determines the pressure but not the velocity field: the only surviving term on its left is the $-v_\theta^2/r$ carried by the material derivative. The θ-momentum equation Eq. (3.23), for steady, axisymmetric, purely azimuthal flow, keeps only the viscous group $\nabla^2 v_\theta - v_\theta/r^2$ and reduces to

$$\frac{d}{dr}\left[\frac{1}{r}\frac{d(r\,v_\theta)}{dr}\right] = 0\,. \tag{4.38}$$

Integrating once gives $(1/r)\,d(rv_\theta)/dr = 2A$ (constant), and integrating again yields

$$v_\theta(r) = Ar + \frac{B}{r}\,. \tag{4.39}$$

The two terms are physically distinct: Ar is rigid-body rotation ($\boldsymbol{\omega} = 2A\,\boldsymbol{e}_z$, nonzero vorticity) and B/r is an irrotational vortex (zero vorticity).

Solution. Applying the boundary conditions Eq. (4.37):

$$\Omega R_1 = AR_1 + B/R_1, \qquad 0 = AR_2 + B/R_2.$$

From the second equation $B = -AR_2^2$; substituting into the first, $A(R_1^2 - R_2^2) = \Omega R_1^2$, giving

$$A = -\frac{\Omega R_1^2}{R_2^2 - R_1^2}\,, \qquad B = \frac{\Omega R_1^2 R_2^2}{R_2^2 - R_1^2}\,.$$

Theorem 4.13 | Rotating Spindle Velocity Profile

The azimuthal velocity in the gap is

$$\boxed{v_\theta(r) = \frac{\Omega R_1^2}{R_2^2 - R_1^2}\left(\frac{R_2^2}{r} - r\right).} \tag{4.40}$$

At $r = R_1$ this gives $v_\theta = \Omega R_1$; at $r = R_2$ it gives zero, as required. For $R_2 \to \infty$ the profile approaches $\Omega R_1^2/r$, a pure irrotational vortex.

Vorticity. The z-vorticity in cylindrical coordinates is $\omega_z = (1/r)\, d(rv_\theta)/dr = 2A$, a constant. Therefore

$$\boldsymbol{\omega} = -\frac{2\Omega R_1^2}{R_2^2 - R_1^2}\, \boldsymbol{e}_z\,. \tag{4.41}$$

The vorticity is *spatially uniform* even though the velocity profile Eq. (4.40) is not linear. This is because only the solid-body component Ar contributes to ω_z; the irrotational vortex B/r contributes nothing.

The Full Stress Tensor and Torque on the Spindle. The relevant component of the rate-of-deformation tensor, the symmetric part of Eq. (2.14), is

$$D_{r\theta} = D_{\theta r} = \frac{r}{2}\frac{d}{dr}\left(\frac{v_\theta}{r}\right) = -\frac{\Omega R_1^2 R_2^2}{(R_2^2 - R_1^2)\, r^2}\,,$$

and the shear stress is $\tau_{r\theta} = 2\mu D_{r\theta}$. With cylindrical components ordered (r, θ, z):

$$\boldsymbol{T} = \begin{pmatrix} -p & \tau_{r\theta}(r) & 0 \\ \tau_{r\theta}(r) & -p & 0 \\ 0 & 0 & -p \end{pmatrix}, \quad \tau_{r\theta}(r) = -\frac{2\mu\Omega R_1^2 R_2^2}{(R_2^2 - R_1^2)\, r^2}\,. \tag{4.42}$$

The magnitude $|\tau_{r\theta}|$ decays as r^{-2}: the stress is largest near the spindle and smallest near the outer cylinder.

Traction on the spindle ($r = R_1$). The outward normal from the spindle into the fluid is $\boldsymbol{n} = +\boldsymbol{e}_r$:

$$\boldsymbol{t}\big|_{r=R_1} = \boldsymbol{T}\,\boldsymbol{e}_r\big|_{r=R_1} = \begin{pmatrix} -p & \tau_{r\theta}(R_1) & 0 \\ \tau_{r\theta}(R_1) & -p & 0 \\ 0 & 0 & -p \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} -p \\ \tau_{r\theta}(R_1) \\ 0 \end{pmatrix}, \tag{4.43}$$

where

$$\tau_{r\theta}(R_1) = -\frac{2\mu\Omega R_2^2}{R_2^2 - R_1^2}$$

is negative (opposing the rotation direction).

Torque and viscometry. The torque the fluid exerts on a spindle of length L is

$$\mathcal{T} = |\tau_{r\theta}(R_1)| \times 2\pi R_1 L \times R_1 = \frac{4\pi\mu\Omega R_1^2 R_2^2 L}{R_2^2 - R_1^2}\,. \tag{4.44}$$

Rearranging gives the *viscometer equation*:

$$\boxed{\mu = \frac{\mathcal{T}(R_2^2 - R_1^2)}{4\pi\Omega R_1^2 R_2^2 L}\,.} \tag{4.45}$$

Remark 4.14 | r^{-2} stress and uniform torque

Because $\tau_{r\theta} \propto r^{-2}$, the torque transmitted through any cylindrical shell at radius r in the gap, $2\pi r\,|\tau_{r\theta}(r)| \cdot r = 4\pi\mu\Omega R_1^2 R_2^2/(R_2^2 - R_1^2)$, is independent of r. This is the angular-momentum analogue of the constant shear-stress result in Couette flow: without inertia or body torques, the torque transmitted through each shell must be the same.

4.7 Thin-Film Flows and Lubrication Theory

The exact solutions above were all strictly unidirectional. Many important flows are only *nearly* so: they occupy thin gaps whose walls converge or move slowly, so that the velocity points almost, but not quite, in a single direction. The oil film in a bearing, the air layer under a hard-disk read head, the tear film on the eye, the mucus lining an airway, the melt between rolls in a coating line all share this geometry, in which one dimension is far smaller than the others. That disparity of scales collapses the Navier–Stokes equations into something almost as simple as the unidirectional flows just studied, yet rich enough to explain how a wedge of fluid can support an enormous load. This is *lubrication theory*, the systematic exploitation of thinness.

4.7.1 The Lubrication Approximation

Consider flow in a gap of thickness $h(x)$ that varies slowly over a streamwise length L, so that the aspect ratio

$$\varepsilon = \frac{h_0}{L} \ll 1,$$

where h_0 is a representative gap. Take the lower surface to move at speed U in the x-direction; the flow is $\boldsymbol{v} = (u, v)$ with $u \sim U$.[1] Continuity $\partial_x u + \partial_y v = 0$ then forces the transverse velocity to be small, $v \sim \varepsilon U$: the fluid moves *along* the gap far faster than across it.

[1] The symbol $\sim$ is used here in the sense of *is of the order of*, and it is worth being explicit about what such an estimate claims, since the rest of this section rests on a string of them. Writing $u \sim U$ says that U is

Weigh the terms of the Cartesian x-momentum equation Eq. (3.18). The two viscous contributions scale as

$$\mu \frac{\partial^2 u}{\partial x^2} \sim \frac{\mu U}{L^2}, \qquad \mu \frac{\partial^2 u}{\partial y^2} \sim \frac{\mu U}{h_0^2},$$

so the cross-gap term is larger by $1/\varepsilon^2$ and dominates.

Now for inertia, and here the choice of what to compare it against is the whole point. The convective acceleration scales as

$$\rho\, u \frac{\partial u}{\partial x} \sim \frac{\rho U^2}{L},$$

and the question is whether it can compete with viscosity. The ordinary Reynolds number answers a slightly different question: $\mathrm{Re} = \rho U L/\mu$ is inertia divided by the *streamwise* viscous term $\mu U/L^2$, and in a thin gap that is the weakest viscous mechanism present, the one we have just discarded. Judging a lubrication film by Re would therefore overstate the importance of inertia badly. The honest comparison is against the viscous term that actually survives, $\mu U/h_0^2$, and it gives

$$\frac{\text{inertia}}{\text{dominant viscous term}} \sim \frac{\rho U^2/L}{\mu U/h_0^2} = \frac{\rho U h_0^2}{\mu L} \equiv \mathrm{Re}^* = \mathrm{Re}\,\varepsilon^2,$$

the *reduced Reynolds number* (the Reynolds number is developed fully in **??**). The factor ε^2 is not a refinement of Re; it is what appears when inertia is measured against the correct viscous term, and it is what makes the approximation work. Because ε^2 is so small, Re^* can be tiny even when Re is of order one or very much larger: *thinness makes a flow effectively creeping*. A journal bearing with an oil film $h_0 \approx 50\,\mu\mathrm{m}$ thick over a bearing length $L \approx 5\,\mathrm{cm}$ has $\varepsilon \approx 10^{-3}$, so a shaft fast enough to give $\mathrm{Re} \approx 10^3$ still leaves $\mathrm{Re}^* \approx 10^{-3}$. The film creeps even though the machine does not. Dropping inertia and the small streamwise diffusion, and examining the y-momentum equation Eq. (3.19) (whose every term is smaller still), the Navier–Stokes equations reduce to

a *scale* for u: the ratio u/U is of order unity over most of the region, so that factors of 2 or π are deliberately thrown away. Derivatives are estimated the same way, by dividing the change in a quantity by the distance over which that change occurs. Here u varies by about U over a streamwise distance L but by the same amount U across a gap of only h_0, so $\frac{\partial u}{\partial x} \sim U/L$ while $\frac{\partial u}{\partial y} \sim U/h_0$, and the second is larger by $1/\varepsilon$. That asymmetry, the same velocity difference spread over two very different lengths, is the whole content of the argument that follows. Estimates of this kind are not proofs; they tell us which terms are large and which are small, and the conclusion is confirmed afterwards by nondimensionalizing the equations properly (the scaling exercise at the end of the chapter), where the same groups reappear as the coefficients of the scaled terms.

Important | Lubrication equations

In a thin gap ($\varepsilon \ll 1$, $\mathrm{Re}^* \ll 1$),

$$\frac{\partial p}{\partial x} = \mu \frac{\partial^2 u}{\partial y^2}, \qquad \frac{\partial p}{\partial y} = 0.$$

The second equation says the pressure does not vary across the gap: $p = p(x)$ only. The first is then, at each station x, exactly the equation of plane Poiseuille–Couette flow driven by the local pressure gradient, the one-dimensional problem of Sec. 4.3 and Sec. 4.5, solved anew at every x.

4.7.2 The Reynolds Equation

Integrate $\mu\, \partial_{yy} u = \frac{\mathrm{d}p}{\mathrm{d}x}$ across the gap with the no-slip conditions $u(0) = U$ (moving lower surface) and $u(h) = 0$ (stationary upper surface). The result is a Couette shear plus a Poiseuille parabola,

$$u(x, y) = U\left(1 - \frac{y}{h}\right) - \frac{1}{2\mu}\frac{\mathrm{d}p}{\mathrm{d}x}\, y\,(h - y),$$

and the volumetric flux per unit width through the gap is

$$q(x) = \int_0^h u \,\mathrm{d}y = \frac{Uh}{2} - \frac{h^3}{12\mu}\frac{\mathrm{d}p}{\mathrm{d}x}. \tag{4.46}$$

For a steady flow, mass conservation requires the flux to be the same at every station, $\frac{\mathrm{d}q}{\mathrm{d}x} = 0$. Differentiating Eq. (4.46) gives the governing equation of lubrication theory.

Theorem 4.15 | Reynolds lubrication equation

The pressure in a steady thin film satisfies

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(h^3 \frac{\mathrm{d}p}{\mathrm{d}x}\right) = 6\mu U \frac{\mathrm{d}h}{\mathrm{d}x}.$$

The right-hand side is the source of everything interesting: pressure is generated only where the gap *varies* ($\mathrm{d}h/\mathrm{d}x \neq 0$). A gap of constant thickness sustains no pressure and carries no load; a *converging* gap, as we now see, generates a large positive pressure that can support a heavy weight on a vanishingly thin film.[2]

[2]The lubrication equations and the theory of the bearing are due to O. Reynolds, "On the theory of lubrication…," *Philosophical Transactions of the Royal Society* **177** (1886), 157–234, explaining the bearing experiments of Beauchamp Tower.

4.7.3 The Slider Bearing

A planar slider of length L rides at a small tilt above a flat surface that moves beneath it at speed U (Fig. 4.5). The gap decreases linearly from h_1 at the inlet to $h_2 < h_1$ at the outlet,

$$h(x) = h_1 - (h_1 - h_2)\frac{x}{L},$$

and the film is open to ambient pressure at both ends, $p(0) = p(L) = 0$.

Integrating Eq. (4.46) once (the flux q is a constant to be fixed by the end conditions) gives

$$\frac{\mathrm{d}p}{\mathrm{d}x} = 6\mu\,\frac{Uh - 2q}{h^3} = 6\mu U\,\frac{h - h^*}{h^3}, \qquad h^* \equiv \frac{2q}{U}.$$

Requiring p to return to zero at the outlet fixes the constant flux; a short calculation (Exercise 3) gives

$$q = \frac{U\,h_1 h_2}{h_1 + h_2}, \qquad \text{equivalently} \qquad h^* = \frac{2h_1 h_2}{h_1 + h_2},$$

so h^* is the harmonic-mean gap, lying between h_2 and h_1. The structure of the pressure follows at once from the sign of $\mathrm{d}p/\mathrm{d}x$:

$$\frac{\mathrm{d}p}{\mathrm{d}x} > 0 \text{ where } h > h^* \text{ (inlet)}, \qquad \frac{\mathrm{d}p}{\mathrm{d}x} < 0 \text{ where } h < h^* \text{ (outlet)}.$$

The pressure therefore *rises* from zero at the inlet, reaches a maximum at the station where $h = h^*$, and *falls* back to zero at the outlet, and it is **positive throughout** (Fig. 4.5). Integrated over the slider, this pressure exerts a net upward force

$$W = \int_0^L p\,\mathrm{d}x > 0,$$

the *load-bearing capacity*. A tilted pad dragged over a moving surface is lifted clear of it on a film of fluid: this is *hydrodynamic lubrication*, the reason a bearing can carry tonnes with metal surfaces never touching.

Remark 4.16 | Why convergence is essential

Everything hinges on the gap converging in the direction of motion. Reverse the tilt (a diverging gap) and the same analysis gives $p < 0$ everywhere, the film would have to sustain tension, it cavitates instead, and the load vanishes. A *parallel* gap generates no pressure at all. The load capacity is a pure consequence of geometry coupled to shear: no converging wedge, no lift.

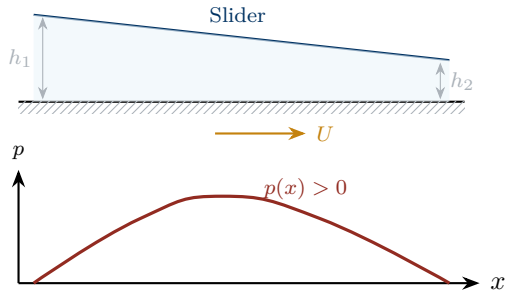


Figure 4.5: The slider bearing. A tilted pad rides above a surface moving at U; the gap converges from h_1 to h_2. By the Reynolds equation, the converging gap generates a pressure $p(x)$ that is positive throughout and integrates to a net upward load W, lifting the pad off the surface on a thin fluid film.

4.7.4 The Squeeze Film

A second archetype replaces sliding with approach. Two parallel circular discs of radius R, separated by a gap $h(t)$, are pushed together so that $\mathrm{d}h/\mathrm{d}t < 0$, squeezing the fluid out radially. Now the gap does not vary in space but does in time, and continuity must account for the shrinking volume. Balancing the outflow against the rate of volume loss and using the lubrication form of the cylindrical radial momentum equation Eq. (3.22) gives the pressure (Exercise 5)

$$p(r) = \frac{3\mu}{h^3}\left(-\frac{\mathrm{d}h}{\mathrm{d}t}\right)(R^2 - r^2),$$

a paraboloid, positive everywhere since the plates approach ($\mathrm{d}h/\mathrm{d}t < 0$). Integrating over the disc yields the force resisting the approach,

$$W = \int_0^R p\, 2\pi r\, \mathrm{d}r = \frac{3\pi\mu R^4}{2h^3}\left(-\frac{\mathrm{d}h}{\mathrm{d}t}\right). \tag{4.47}$$

Important | The tenacity of a thin film

The squeeze-film force grows as h^{-3}: as the gap closes, the resistance diverges. Expelling the last thin layer of a viscous fluid from between two surfaces is exceedingly slow, formally, they never touch in finite time under a constant load.

This h^{-3} law explains the stickiness of two wet glass plates, the damping in a squeeze-film shock absorber, the adhesion in a smear of honey, and the difficulty of draining the film that keeps two coalescing droplets briefly apart. Like the slider bearing, it shows how a vanishingly thin film mediates forces out of all proportion to its size, the recurring lesson of lubrication theory.

Table 4.1: The exact steady unidirectional flows of this chapter at a glance.

Flow	Driving	Velocity profile	Shear stress & output	Key lesson
Plane Couette	moving wall	linear, $u = Uy/H$	$\tau = \mu U/H$ (uniform); force/area $\mu U/H$	shear transmitted uniformly; viscometry
Two-layer Couette	moving wall	piecewise linear, slope $\propto 1/\mu_k$	common τ; interface speed u_I	velocity and τ continuous, gradient jumps
Hagen–Poiseuille (pipe)	pressure gradient G	parabolic, $u \propto R^2 - r^2$	$\tau_w = GR/2$; $Q = \pi G R^4/8\mu$	$Q \propto R^4$
Rotating spindle	rotating cylinder	azimuthal $v_\theta(r)$	torque/length $\propto \mu\Omega$	constant angular-momentum flux; viscometry

4.8 Start-Up of Couette Flow

All solutions so far have been *steady*: the velocity field is time-independent from the outset. Real flows, however, begin from some initial state and approach the steady solution only after a transient. The start-up of Couette flow is the simplest problem in which both a spatial boundary-value structure and a temporal initial-value structure are present simultaneously. It reveals two fundamental ideas: (i) momentum diffuses from boundaries into the fluid interior on a timescale set by the kinematic viscosity, and (ii) the transient can be resolved by expanding the solution in a complete set of eigenfunctions—the subject of Sturm–Liouville theory.

4.8.1 Problem Statement and Governing Equation

Consider the Couette geometry of Sec. 4.3: two infinite parallel plates separated by a gap L, lower plate stationary at $y = 0$, upper plate at $y = L$. At time $t < 0$ both plates and the fluid are at rest. At $t = 0$ the upper plate is impulsively set into motion with velocity V in the x-direction and held there for all $t > 0$. We seek the evolving velocity field $v_x(y, t)$.

By the same symmetry argument as before, $\boldsymbol{v} = v_x(y, t)\, \boldsymbol{e}_x$. With no imposed pressure gradient, the Cartesian x-momentum equation Eq. (3.18) reduces to the *diffusion equation* for v_x:

Theorem 4.17 | Diffusion Equation for Start-Up Flow

The velocity field satisfies

$$\frac{\partial v_x}{\partial t} = \nu \frac{\partial^2 v_x}{\partial y^2}, \qquad \nu \equiv \frac{\mu}{\rho}, \tag{4.48}$$

where ν is the *kinematic viscosity* (units $\mathrm{m^2\,s^{-1}}$), subject to

$$\text{IC:}\quad v_x(y,0) = 0 \qquad (0 < y < L)\,, \tag{4.49}$$

$$\text{BCs:}\quad v_x(0,t) = 0\,, \quad v_x(L,t) = V \qquad (t > 0)\,. \tag{4.50}$$

Eq. (4.48) is first order in time and second order in space, requiring one initial condition and two boundary conditions—exactly what is supplied. The kinematic viscosity ν is the natural diffusivity for momentum: it has dimensions $[L^2/T]$, so the only timescale formed from ν and the gap L is

$$t_{\mathrm{char}} \sim \frac{L^2}{\nu}\,. \tag{4.51}$$

We will recover this estimate—with the precise prefactor—from the analytical solution.

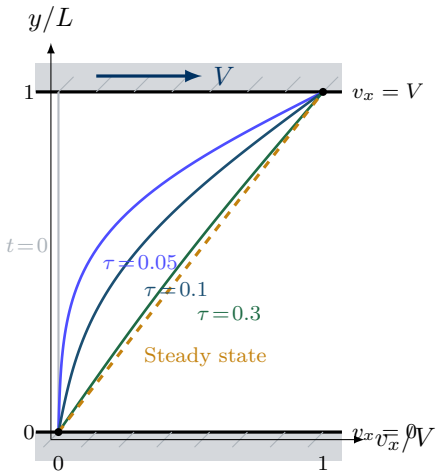


Figure 4.6: Velocity profiles during start-up of Couette flow. At $t = 0$ the fluid is at rest; the upper plate is then impulsively set to velocity V. Profiles are shown at dimensionless times $\tau = \nu t/L^2 = 0.05$, 0.1, 0.3 and at steady state (dashed gold line). Momentum diffuses downward from the moving upper plate into the gap, and the profile approaches the linear Couette solution.

4.8.2 Decomposition into Steady State and Transient

The boundary conditions Eq. (4.50) are *inhomogeneous*: $v_x = V \neq 0$ at $y = L$. Direct separation of variables fails for inhomogeneous BCs. The remedy is to decompose the solution as

$$v_x(y,t) = U(y) + W(y,t)\,, \tag{4.52}$$

where $U(y)$ is the *steady-state* solution (satisfying the full BCs) and $W(y,t)$ is a transient correction with *homogeneous* BCs.

Steady-state part. $U(y)$ satisfies $d^2U/dy^2 = 0$ with $U(0) = 0$ and $U(L) = V$, giving

$$U(y) = V\,\frac{y}{L}\,, \tag{4.53}$$

the Couette profile of Thm. 4.4.

Transient part. Substituting Eq. (4.52) into Eq. (4.48) and using $d^2U/dy^2 = 0$, the transient W satisfies the same diffusion equation with *homogeneous* BCs and a modified initial condition:

$$\frac{\partial W}{\partial t} = \nu\,\frac{\partial^2 W}{\partial y^2}\,, \tag{4.54}$$

$$W(0,t) = 0\,, \quad W(L,t) = 0\,, \tag{4.55}$$

$$W(y,0) = -U(y) = -\frac{V y}{L}\,. \tag{4.56}$$

The homogeneous BCs make this problem accessible to separation of variables and eigenfunction expansion—but first we need the mathematical machinery.

4.8.3 Sturm–Liouville Theory

The transient problem Eq. (4.54)–Eq. (4.56) is an instance of a broader class of problems studied by Sturm and Liouville in the 1830s. Their theory provides a systematic framework for expanding functions in terms of the eigenfunctions of self-adjoint differential operators. We develop it here in a form suitable for all the problems that appear in this chapter.

Definition 4.18 | Sturm–Liouville Equation

A second-order ODE of the form

$$\frac{d}{dx}\left[p(x)\,\frac{dy}{dx}\right] + q(x)\,y + \lambda\,r(x)\,y = 0\,, \qquad x \in (a,b)\,, \tag{4.57}$$

with real, continuous coefficients satisfying $p(x) > 0$ and $r(x) > 0$ on $[a,b]$, is called a *Sturm–Liouville equation*. The real number λ is the *eigenvalue* to be determined; the corresponding nontrivial solution $y_\lambda(x)$ is the *eigenfunction*.

Definition 4.19 | Regular Sturm–Liouville Problem

A *regular Sturm–Liouville problem* (SLP) pairs Eq. (4.57) with boundary conditions of the form

$$\alpha_1\,y(a) - \alpha_2\,y'(a) = 0\,, \qquad \beta_1\,y(b) + \beta_2\,y'(b) = 0\,, \tag{4.58}$$

where $\alpha_1^2 + \alpha_2^2 > 0$ and $\beta_1^2 + \beta_2^2 > 0$. Special cases include Dirichlet BCs ($\alpha_2 = \beta_2 = 0$) and Neumann BCs ($\alpha_1 = \beta_1 = 0$).

The power of the Sturm–Liouville framework lies in its eigenvalue theorem:

Theorem 4.20 | Sturm–Liouville Theorem

A regular Sturm–Liouville problem has the following properties.

(i) **Real, ordered eigenvalues.** There exist countably many eigenvalues $\lambda_1 < \lambda_2 < \lambda_3 < \cdots$ with $\lambda_n \to +\infty$; all are real.

(ii) **Oscillation.** For each λ_n there is a unique (up to scaling) eigenfunction $\phi_n(x)$ that has exactly $n-1$ zeros in the open interval (a, b).

(iii) **Orthogonality.** Eigenfunctions corresponding to distinct eigenvalues are orthogonal with respect to the weight $r(x)$:

$$\langle \phi_n, \phi_m \rangle_r := \int_a^b r(x)\, \phi_n(x)\, \phi_m(x)\, dx = \begin{cases} 0 & n \neq m, \\ \|\phi_n\|_r^2 & n = m. \end{cases} \tag{4.59}$$

(iv) **Completeness.** The eigenfunctions $\{\phi_n\}$ form a complete basis: every piecewise-smooth function f on $[a, b]$ can be represented as

$$f(x) = \sum_{n=1}^{\infty} A_n\, \phi_n(x)\,, \tag{4.60}$$

with coefficients obtained by projecting f onto each eigenfunction:

$$A_n = \frac{\langle f, \phi_n \rangle_r}{\|\phi_n\|_r^2} = \frac{\displaystyle\int_a^b r(x)\, f(x)\, \phi_n(x)\, dx}{\displaystyle\int_a^b r(x)\, \phi_n^2(x)\, dx}\,. \tag{4.61}$$

Remark 4.21 | Geometric interpretation

Thm. 4.20 is the infinite-dimensional analogue of decomposing a vector in $\mathbb{R}^n$ into orthogonal components. The weight $r(x)$ defines the inner product; orthogonality means each eigenfunction "sees" only its own component of f; completeness means nothing is missed. Formula Eq. (4.61) is the projection formula: A_n is the r-weighted correlation of f with ϕ_n, normalized by the squared norm of ϕ_n.

4.8.4 Eigenvalue Problem for the Transient

We seek solutions to Eq. (4.54) of the separable form $W(y,t) = T(t)\,Y(y)$. Substituting and separating variables,

$$\frac{1}{\nu T}\frac{dT}{dt} = \frac{1}{Y}\frac{d^2Y}{dy^2} = -\lambda^2 \qquad (\lambda > 0)\,,$$

where the separation constant must be negative for $T(t)$ to remain bounded as $t \to \infty$. This yields two ODEs:

$$\frac{dT}{dt} = -\lambda^2 \nu\, T \quad \Longrightarrow \quad T(t) = e^{-\lambda^2 \nu t}\,, \tag{4.62}$$

$$\frac{d^2Y}{dy^2} = -\lambda^2 Y\,, \quad Y(0) = 0\,, \quad Y(L) = 0\,. \tag{4.63}$$

The spatial problem Eq. (4.63) is a regular SLP with $p = r = 1$, $q = 0$ on $[0, L]$ and Dirichlet BCs. By Thm. 4.20, the eigenfunctions and eigenvalues are

Theorem 4.22 | Eigenfunctions for the Start-Up Problem

The eigenvalues and eigenfunctions of Eq. (4.63) are

$$\lambda_n = \frac{n\pi}{L}\,, \qquad \phi_n(y) = \sin\!\left(\frac{n\pi y}{L}\right), \qquad n = 1, 2, 3, \ldots \tag{4.64}$$

The orthogonality relation Eq. (4.59) with $r = 1$ reads

$$\int_0^L \sin\!\left(\frac{m\pi y}{L}\right)\sin\!\left(\frac{n\pi y}{L}\right) dy = \frac{L}{2}\,\delta_{mn}\,. \tag{4.65}$$

4.8.5 Full Solution and Timescales

By completeness (Thm. 4.20), the general solution for W is

$$W(y,t) = \sum_{n=1}^{\infty} A_n \sin\!\left(\frac{n\pi y}{L}\right) e^{-\lambda_n^2 \nu t}\,. \tag{4.66}$$

Applying the initial condition Eq. (4.56) at $t = 0$:

$$\sum_{n=1}^{\infty} A_n \sin\!\left(\frac{n\pi y}{L}\right) = -\frac{Vy}{L}\,.$$

Using Eqs. (4.61) and (4.65):

$$A_n = \frac{2}{L}\int_0^L \left(-\frac{Vy}{L}\right)\sin\left(\frac{n\pi y}{L}\right)dy = -\frac{2V}{L^2}\cdot\frac{L^2\,(-1)^{n+1}}{n\pi} = \frac{2V(-1)^n}{n\pi}\,.$$

(The integral follows from integration by parts: $\int_0^L y\sin(n\pi y/L)\,dy = L^2(-1)^{n+1}/(n\pi)$.) Assembling the full solution $v_x = U + W$:

Theorem 4.23 | Start-Up of Couette Flow

The velocity field during start-up is

$$\boxed{v_x(y,t) = V\left[\frac{y}{L} + \frac{2}{\pi}\sum_{n=1}^{\infty}\frac{(-1)^n}{n}\sin\left(\frac{n\pi y}{L}\right)e^{-n^2\pi^2\nu t/L^2}\right].} \tag{4.67}$$

At $t = 0$ the series reconstructs $-y/L$, giving $v_x = 0$; as $t \to \infty$ all terms decay and $v_x \to Vy/L$ (the Couette profile).

Characteristic timescales. Each mode n decays with its own rate:

$$\alpha_n = \lambda_n^2\,\nu = \frac{n^2\pi^2\nu}{L^2}\,. \tag{4.68}$$

The characteristic decay time of mode n is $\tau_n = 1/\alpha_n = L^2/(n^2\pi^2\nu)$. The slowest (longest-lived) mode is $n = 1$, which sets the time to reach steady state:

$$t_{\text{char}} = \frac{L^2}{\pi^2\nu}\,. \tag{4.69}$$

This is the precise version of the dimensional estimate Eq. (4.51): the prefactor is $1/\pi^2 \approx 0.101$. Modes with $n \geq 2$ decay at least four times faster, so for $t \gtrsim t_{\text{char}}$ the solution is well approximated by the $n = 1$ term alone:

$$v_x(y,t) \approx V\left[\frac{y}{L} - \frac{2}{\pi}\sin\left(\frac{\pi y}{L}\right)e^{-\pi^2\nu t/L^2}\right] \qquad (t \gg t_{\text{char}}/4)\,. \tag{4.70}$$

Remark 4.24 | Why the Fourier series works here

The key step that made the eigenfunction expansion possible was the decomposition $v_x = U + W$: by subtracting the steady state, we obtained homogeneous BCs on W, and the eigenfunctions $\sin(n\pi y/L)$ vanish at both walls. Any attempt to expand the full v_x (with $v_x = V$ at $y = L$) in sine functions would fail because sine series on $[0, L]$ with $\sin(n\pi \cdot L/L) = 0$ can never represent a nonzero value at $y = L$.

4.9 The Rayleigh Problem and Similarity Solutions

Our final problem is the purest instance of viscous diffusion. An infinite flat plate lies beneath a semi-infinite fluid, both at rest. At $t = 0$ the plate is suddenly jerked into motion at constant speed U in its own plane and held there. There is no pressure gradient and no second wall, nothing but the plate, the fluid, and viscosity. How far, and how fast, does the motion penetrate? This is the *Rayleigh problem*, also called Stokes' first problem.[3]

Theorem 4.25 | The Rayleigh problem

A semi-infinite fluid ($y > 0$) of kinematic viscosity $\nu = \mu/\rho$ rests above a plane wall at $y = 0$; for $t > 0$ the wall moves in its own plane at constant speed U. With $\boldsymbol{v} = v_x(y,t)\,\boldsymbol{e}_x$ and no applied pressure gradient, the Cartesian x-momentum equation Eq. (3.18) reduces to the diffusion equation

$$\frac{\partial v_x}{\partial t} = \nu \frac{\partial^2 v_x}{\partial y^2}, \tag{4.71}$$

subject to the initial and boundary conditions

$$v_x(y,0) = 0, \qquad v_x(0,t) = U \quad (t > 0), \qquad v_x(\infty,t) = 0.$$

Notice what is *absent* from this data: there is no length. The wall is infinite, the fluid unbounded, and the forcing a scale-free step. The only quantity with the dimension of length that can be built from the parameters ν and the elapsed time t is $\sqrt{\nu t}$, and it *grows* as the flow develops. Because there is no fixed length against which to measure y, the velocity profile cannot depend on y and t separately; it can depend only on their dimensionless combination. This absence of an imposed length scale is the hallmark of a *self-similar* problem.

4.9.1 The Philosophy of Similarity

A solution is *self-similar* when its shape is preserved in time: the profile at one instant is a stretched copy of the profile at any other, the only change being the scale of the stretch. If we rescale the height y by the growing diffusion length $\sqrt{\nu t}$ and the velocity by U, the family of profiles $v_x(y,t)$ at different times must all fall on a single master curve. Writing this expectation as

$$\frac{v_x(y,t)}{U} = f(\eta), \qquad \eta \propto \frac{y}{\sqrt{\nu t}},$$

[3] G. G. Stokes, "On the effect of the internal friction of fluids on the motion of pendulums," *Transactions of the Cambridge Philosophical Society* **9** (1851), 8–106; the impulsively-started-plate version is usually credited to Lord Rayleigh, *Philosophical Magazine* **21** (1911), 697–711.

collapses the two independent variables (y, t) into the one *similarity variable* η, and, if the guess is consistent, turns the partial differential equation Eq. (4.71) into an ordinary differential equation for f. That reduction is the entire payoff of the method.

4.9.2 Finding the Similarity Variable: Dilatational Transforms

Where does η come from? Dimensional reasoning suggested it, but the systematic and general route, the one that works even when dimensional analysis is inconclusive, is to demand that the problem be invariant under a one-parameter group of *dilatational* (stretching) transformations. This is the group-theoretic view of similarity: *a scaling symmetry shared by the governing equation and its auxiliary data provides a systematic route to a group-invariant similarity solution, with the similarity variable the invariant of that group.* The converse is not claimed: not every useful similarity reduction arises from the simple dilatational group considered here, and other symmetries (translations, spiral groups) can generate similarity variables of their own.

Introduce the stretching transformation, with parameter $\lambda > 0$ and exponents to be determined,

$$t \mapsto \lambda^a\, t, \qquad y \mapsto \lambda^b\, y, \qquad v_x \mapsto \lambda^c\, v_x.$$

Under it the two sides of Eq. (4.71) scale as

$$\frac{\partial v_x}{\partial t} \mapsto \lambda^{c-a}\,\frac{\partial v_x}{\partial t}, \qquad \nu\,\frac{\partial^2 v_x}{\partial y^2} \mapsto \lambda^{c-2b}\,\nu\,\frac{\partial^2 v_x}{\partial y^2}.$$

The equation is invariant only if the two exponents agree,

$$c - a = c - 2b \quad \Longrightarrow \quad a = 2b.$$

The boundary condition $v_x(0, t) = U$ fixes the remaining freedom: since U is a constant, $\lambda^c U = U$ for all λ forces $c = 0$, so the velocity itself is invariant. Choosing $b = 1$ (hence $a = 2$), the group is

$$(t,\, y,\, v_x) \;\mapsto\; (\lambda^2 t,\; \lambda y,\; v_x).$$

Its invariants, the quantities unchanged by the stretch, are v_x and the ratio $y/\sqrt{t}$, since $y \mapsto \lambda y$ and $\sqrt{t} \mapsto \lambda\sqrt{t}$ scale identically. The solution can therefore depend only on these: $v_x/U = f\left(y/\sqrt{\nu t}\right)$, exactly the self-similar form anticipated above. The group fixes η only up to a constant multiple; we take

$$\eta = \frac{y}{2\sqrt{\nu t}} \tag{4.72}$$

for later algebraic convenience.

Theorem 4.26 | Rayleigh similarity solution

The solution of the Rayleigh problem posed above is

$$\frac{v_x(y,t)}{U} = \operatorname{erfc}(\eta) = 1 - \frac{2}{\sqrt{\pi}} \int_0^{\eta} e^{-s^2}\, \mathrm{d}s, \qquad \eta = \frac{y}{2\sqrt{\nu t}}, \tag{4.73}$$

where erfc is the complementary error function.

Proof. Substitute $v_x = U f(\eta)$ with η from Eq. (4.72). Since $\eta \propto t^{-1/2}$, the chain rule gives $\partial\eta/\partial t = -\eta/(2t)$ and $\partial\eta/\partial y = 1/(2\sqrt{\nu t})$, so

$$\frac{\partial v_x}{\partial t} = U f'(\eta)\left(-\frac{\eta}{2t}\right), \qquad \frac{\partial^2 v_x}{\partial y^2} = U f''(\eta)\,\frac{1}{4\nu t}.$$

Inserting these into Eq. (4.71) and cancelling $U/(4t)$ leaves an ordinary differential equation in η alone, the hallmark of a successful similarity reduction:

$$f'' + 2\eta\, f' = 0, \qquad f(0) = 1, \quad f(\infty) = 0.$$

The two boundary conditions on v_x (at the wall and at infinity) and the initial condition ($y \to \infty$ and $t \to 0$ both send $\eta \to \infty$) have merged into just two conditions on f, a necessary consistency check. Treating f' as the unknown, $(\ln f')' = -2\eta$ gives $f'(\eta) = f'(0)\, e^{-\eta^2}$, and integrating,

$$f(\eta) = 1 + f'(0) \int_0^{\eta} e^{-s^2}\, \mathrm{d}s.$$

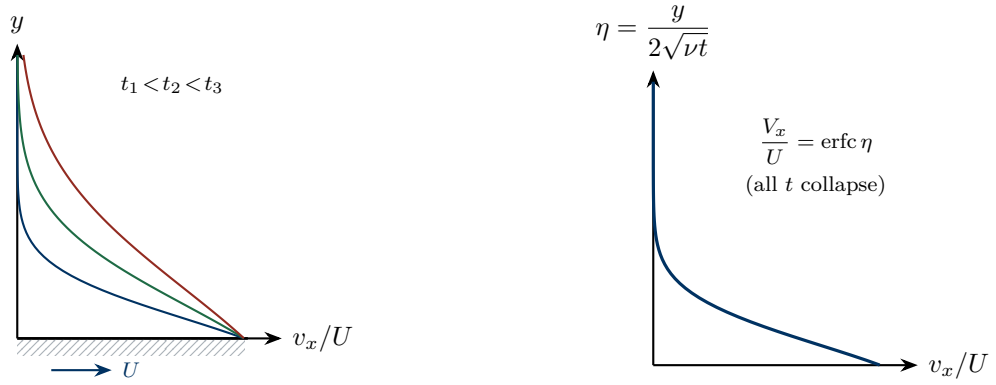
The condition $f(\infty) = 0$ with $\int_0^{\infty} e^{-s^2}\mathrm{d}s = \sqrt{\pi}/2$ fixes $f'(0) = -2/\sqrt{\pi}$, yielding $f(\eta) = \operatorname{erfc}(\eta)$. ∎

Remark 4.27 | Diffusion length and wall stress

The solution Eq. (4.73) makes the physics quantitative (Fig. 4.7). Because $\operatorname{erfc}(\eta)$ falls from 1 to near zero over $\eta \sim 1$, the motion has penetrated to a depth

$$\delta(t) \sim \sqrt{\nu t},$$

the *diffusion length*: viscosity spreads momentum, and with it the vorticity generated at the wall, outward as $\sqrt{t}$. This is a characteristic penetration depth, not a sharp front: the complementary error function is nonzero at every finite y for any $t > 0$. Nor



(a) Profiles thicken as $\sqrt{\nu t}$. **(b)** One universal curve in η.

Figure 4.7: Viscous diffusion from an impulsively started plate. **(a)** The velocity profile $v_x/U = \operatorname{erfc}\big(y/2\sqrt{\nu t}\big)$ at three successive times; the layer of dragged fluid thickens in proportion to $\sqrt{\nu t}$. **(b)** Replotted against the similarity variable $\eta = y/2\sqrt{\nu t}$, the profiles at *all* times collapse onto the single universal curve $\operatorname{erfc}\eta$, the visual signature of self-similarity.

does "never reaching a steady state" mean the fluid never settles at a given place. It means that no steady field satisfies both the moving-wall and far-field conditions; at any fixed y, $u \to U$ as t grows. ~~The traction the plate exerts on the fluid is~~ The shear stress component at the wall is

$$T_{xy}(0,t) = \mu \left.\frac{\partial v_x}{\partial y}\right|_{y=0} = \frac{\mu U}{2\sqrt{\nu t}}\, f'(0) = -\frac{\mu U}{\sqrt{\pi\, \nu t}},$$

and its sign must be read with care. The fluid occupies $y > 0$, so its outward normal at the plate is $\boldsymbol{n}_f = -\boldsymbol{e}_y$, and

$$t_x^{\text{plate}\to\text{fluid}} = -T_{xy}(0,t) = +\frac{\mu U}{\sqrt{\pi\, \nu t}}, \qquad t_x^{\text{fluid}\to\text{plate}} = T_{xy}(0,t) = -\frac{\mu U}{\sqrt{\pi\, \nu t}}.$$

The plate drags the fluid forward, and the fluid drags back on the plate; the negative number above is the *fluid on the plate*, not the plate on the fluid. The magnitude is infinite at the instant of start-up and decays as $t^{-1/2}$ as the sheared layer thickens. That initial singularity is an artifact of the imposed discontinuous wall speed, and it is integrable in time; a finite acceleration history removes it.

A global check ties the sign back to the balance laws of Ch. 3. The x-momentum of the fluid above unit plate area, and its rate of change, are

$$\frac{P_x}{A} = \rho \int_0^\infty u(y,t)\,\mathrm{d}y = 2\rho U \sqrt{\frac{\nu t}{\pi}}, \qquad \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{P_x}{A}\right) = \frac{\mu U}{\sqrt{\pi\, \nu t}},$$

which is exactly the positive plate-on-fluid traction. The momentum the fluid gains is the momentum the plate supplies. The same $\sqrt{\nu t}$ scaling governs boundary layers, thermal penetration, and mass diffusion, which is why this little problem is a template for so much of transport.

Remark 4.28 | When does similarity work?

The dilatational-group argument is general: a similarity solution exists whenever the governing equation and *all* of the auxiliary data are invariant under a common stretching group. Here the semi-infinite domain and the scale-free step forcing were essential, introduce a second wall at $y = L$, or a fixed length in the initial data, and the group is broken, the length L competes with $\sqrt{\nu t}$, and the profiles no longer collapse. That is exactly the finite-gap start-up problem of Sec. 4.8, which is *not* self-similar and instead required the eigenfunction expansion developed there.

Exercises

These problems span the exact and nearly-exact solutions of this chapter—plane and interfacial Couette flow, gravity-driven and oscillatory flows, duct flow, and the thin-film lubrication material of Sec. 4.7; fields are as smooth as needed. Parts marked [PRACTICE] are for self-study; those marked [ADVANCED] are more demanding.

1. **(Plane Couette–Poiseuille flow.)** An incompressible Newtonian fluid fills the gap between infinite plates at $y = 0$ (stationary) and $y = H$ (moving at speed $U > 0$ along x), with a constant applied pressure gradient. Write $G \equiv -\,dp/dx$.

   (a) Postulate $\boldsymbol{v} = u(y)\boldsymbol{e}_x$ and show $\mu\, u'' = -G$ with $u(0) = 0$, $u(H) = U$.

   (b) Derive the profile $u(y) = U\dfrac{y}{H} + \dfrac{G}{2\mu}\, y(H - y)$, and identify the Couette and plane-Poiseuille contributions.

   (c) With $Y = y/H$ and $\Lambda = GH^2/(2\mu U)$, show $u/U = Y + \Lambda Y(1 - Y)$. Interpret $\Lambda = 0$ (pure Couette), $\Lambda > 0$ (pressure assists the wall), and $\Lambda < 0$ (adverse gradient).

   (d) Show $\tau_{xy}(y) = \mu U/H + G(H/2 - y)$; find the wall shears and the location of zero shear when it lies in the gap.

   (e) Show that reverse flow first appears at the stationary wall when $\Lambda < -1$, i.e. $G < -2\mu U/H^2$, and locate the interior stagnation plane.

   (f) Derive the flux per unit width $q = \int_0^H u\, dy = UH/2 + GH^3/(12\mu)$, show that zero net flux occurs at $\Lambda = -3$ ($G = -6\mu U/H^2$), and explain why zero net flux does

not mean zero velocity everywhere.

(g) [PRACTICE] Compute the pressure work, the moving-wall work, and the viscous dissipation per unit length and span, and verify the steady mechanical-energy balance with a consistent traction sign convention.

2. (Gravity-driven film down an inclined plane.) A Newtonian liquid film of uniform thickness H flows down an infinite plane inclined at angle α to the horizontal; x points downslope and y normal to the plate into the liquid, with the free surface at atmospheric pressure p_0 (Fig. 4.8).

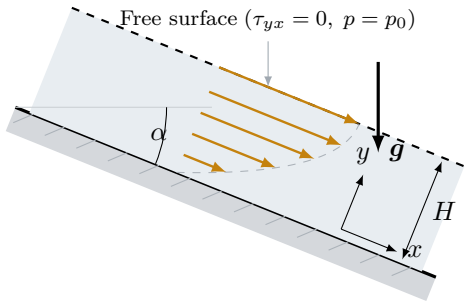


Figure 4.8: Gravity-driven flow down an inclined plane of angle α. The free surface at $y = H$ is stress-free and at atmospheric pressure p_0; the velocity profile is parabolic, maximal at the free surface.

(a) Resolve gravity: $g_x = g\sin\alpha$, $g_y = -g\cos\alpha$.

(b) Postulate $\boldsymbol{v} = u(y)\boldsymbol{e}_x$ and verify continuity.

(c) From transverse momentum and $p(H) = p_0$, derive $p(y) = p_0 + \rho g\cos\alpha\,(H - y)$.

(d) Show $\mu\,u'' = -\rho g\sin\alpha$ with no-slip at $y = 0$ and zero tangential traction at $y = H$.

(e) Obtain $u(y) = \dfrac{\rho g\sin\alpha}{2\mu}(2Hy - y^2)$.

(f) Determine the maximum velocity, the flux per unit width, the wall shear stress, the vorticity distribution, the rate-of-strain components, and the full traction on the plate.

(g) [PRACTICE] Verify by a global momentum balance that the downslope wall shear equals the downslope weight of the film per unit plate area, and explain why the maximum velocity is at the free surface rather than the film centre.

[Advanced] (Oscillatory plate flow: Stokes' second problem.) A semi-infinite Newtonian fluid occupies $y > 0$, at rest far from the wall. The wall at $y = 0$ oscillates tangentially, $u(0,t) = U_0\cos\omega t$. After transients decay the flow is time-periodic.

(a) With $\boldsymbol{v} = u(y,t)\boldsymbol{e}_x$ show $\partial_t u = \nu\,\partial_{yy}u$, with $u(0,t) = U_0\cos\omega t$ and $u \to 0$ as $y \to \infty$.

(b) Seek $u = \mathrm{Re}\{\hat{u}(y)e^{i\omega t}\}$, obtain $i\omega\hat{u} = \nu\hat{u}''$, and choose the decaying root.

(c) Defining $\delta = \sqrt{2\nu/\omega}$, show $u(y,t) = U_0 e^{-y/\delta}\cos(\omega t - y/\delta)$.

(d) Interpret the exponential amplitude decay, the phase lag y/δ, and why higher frequencies give a thinner oscillatory layer.

(e) Compute the vorticity and the wall shear stress; state whether your sign refers to fluid-on-wall or wall-on-fluid traction, and find the amplitude and phase of the wall shear relative to the wall velocity.

(f) Compute the cycle-averaged power per unit area supplied by the plate, show it is positive, and relate it to the viscous dissipation integrated over the fluid.

(g) Compare the penetration depths $\sqrt{\nu t}$ (impulsively started plate, Sec. 4.9) and $\sqrt{2\nu/\omega}$ (oscillating plate), and explain how one is set by elapsed time and the other by forcing frequency.

[Advanced] (Poiseuille flow in a rectangular duct.) A straight duct has cross-section $0 < y < a$, $-b < z < b$, with streamwise coordinate x and constant pressure gradient $G \equiv -\,dp/dx > 0$; take $\boldsymbol{v} = u(y,z)\boldsymbol{e}_x$.

(a) Show continuity is automatic and $\partial_{yy}u + \partial_{zz}u = -G/\mu$, with $u = 0$ on all four walls.

(b) Expand the forcing on $0 < y < a$ using $1 = \dfrac{4}{\pi}\displaystyle\sum_{n\ \mathrm{odd}} \frac{1}{n}\sin\frac{n\pi y}{a}$.

(c) Seek $u = \sum_{n\ \mathrm{odd}} U_n(z)\sin(n\pi y/a)$, derive the ODE for $U_n(z)$, and impose symmetry about $z = 0$ and no slip at $z = \pm b$.

(d) Show $u(y,z) = \dfrac{4Ga^2}{\mu\pi^3}\displaystyle\sum_{n\ \mathrm{odd}} \frac{1}{n^3}\left[1 - \frac{\cosh(n\pi z/a)}{\cosh(n\pi b/a)}\right]\sin\frac{n\pi y}{a}$, and verify all four wall conditions and the Poisson equation.

(e) Integrate to obtain $Q = \dfrac{16Ga^3 b}{\mu\pi^4}\displaystyle\sum_{n\ \mathrm{odd}} \frac{1}{n^4}\left[1 - \frac{a}{n\pi b}\tanh\frac{n\pi b}{a}\right]$.

(f) In the wide-duct limit $b/a \to \infty$, use $\sum_{n\ \mathrm{odd}} n^{-4} = \pi^4/96$ to recover plane Poiseuille flow, $Q \sim Ga^3 b/(6\mu)$, and explain why this is the plane-Poiseuille flux per unit width times the duct width $2b$.

(g) [PRACTICE] For a square duct ($2b = a$), evaluate the first one or two terms to estimate the centreline velocity and Q, and comment on convergence.

(h) Explain why the profile is not a paraboloid and why the corners reduce Q relative to a parallel-plate estimate.

3. **(Two-layer Couette: limits and dissipation.)** Using the results of Sec. 4.4:
   (a) Recover the single-fluid Couette profile as $\alpha \to 0$, $\alpha \to 1$, or $m \to 1$.
   (b) Examine $m = \mu_2/\mu_1 \to 0$ and $\to \infty$ and describe which layer carries most of the velocity drop in each case.
   (c) Verify directly that the total dissipation equals τU.
   (d) [PRACTICE] Add a prescribed surface-tension gradient $\partial\gamma/\partial x$ at the interface, replace tangential-traction continuity by the appropriate jump (Sec. 3.9.2), and show that a Marangoni stress can drive flow with both plates at rest.

4. **(Scaling the equations.)** Nondimensionalize the two-dimensional Navier–Stokes equations Eqs. (3.18) and (3.19) with $x \sim L$, $y \sim h_0$, $u \sim U$, $v \sim \varepsilon U$, and a pressure scale $\mu U L/h_0^2$. Show that the surviving balance is $\partial_x p = \partial_{yy} u$, $\partial_y p = 0$, with corrections of order $\mathrm{Re}^* = \mathrm{Re}\,\varepsilon^2$ and ε^2.

5. **(The velocity profile and flux.)** Solve $\mu\,\partial_{yy} u = \mathrm{d}p/\mathrm{d}x$ with $u(0) = U$, $u(h) = 0$ to recover the Couette–Poiseuille profile, and integrate it to obtain the flux Eq. (4.46).

6. **(Finishing the slider bearing.)** Impose $p(0) = p(L) = 0$ on $\mathrm{d}p/\mathrm{d}x = 6\mu U(h - h^*)/h^3$ to show $q = U h_1 h_2/(h_1 + h_2)$. Then integrate for $p(x)$, verify it is positive throughout, and (for $h_2/h_1 = 1/2$) estimate the load W.

7. **[practice] (Why not a parallel or diverging pad?)** Show from the Reynolds equation that a parallel gap ($h = \mathrm{const}$) supports no load, and that a diverging gap would require $p < 0$. Explain physically what happens instead (cavitation).

8. **(The squeeze film.)** Derive Eq. (4.47) for two approaching discs: write the axisymmetric lubrication balance $\partial_r p = \mu\,\partial_{zz} u_r$, integrate for u_r, impose flux conservation against the volume change $-\pi r^2\,\mathrm{d}h/\mathrm{d}t$, and integrate the resulting pressure over the disc.

9. **[practice] (Viscous heating in a bearing.)** Using the dissipation function of Sec. 3.8.2, estimate the power converted to heat in a slider bearing of gap h_0 and area A operating at surface speed U, and comment on why high-speed bearings require cooling.